Component 2: Independent Design and Make Project 2025-26

Will Cowley

RGS Worcester

Client Introduction

Mr Robert Cowley (Pictured In Figure 1) is a retired engineer and experienced skipper of a small yacht which he sails recreationally, predominately in his local area in the Bristol channel and the southwest of England/Wales for many years.
The sailboat (Figure 2) named 'Libra Lass' is a 31 ft Moody 31 Mk1 with a diesel onboard engine, space to sleep three comfortably and limited cooking facilities. Mr Cowley, is an also very competent in the upkeep of yachts, performing routine maintenance and engineering upgrades himself. These are often done at Cardiff bay Marina or in his home workshop. Mr Cowley is interested in further improvement to the safety, ease of use and comfort of his vessel to suit his changing needs as he ages.



Figure 1



Figure 2

Identifying Design Possibilities and Proposals

In this section I will introduce problems faced by my client and recreational sailors in general. The paragraphs covers initial ideas about the problem identified and start to explore what could be designed to solve them. **Colour Code Key 1:** Text coloured **blue** on this page only gives more specific information on the design of a proposal.

1. Collapsible wind deflector/canopy
While in motion the steering wheel and various lines must be attended to constantly. This involves being forced to stay above deck inside the cockpit area for long periods. Also, while cruising with passengers on board they often voluntarily stay in the open to avoid discomfort caused by seasickness. Passengers are often not well equipped with the performance clothing required to stay warm and dry.
The cockpit is designed with low seating and a canvas sprayhood which provide limited shelter; limiting windchill, splash and spray traveling over the bow along the length of the boat. Problems arise because it is not possible to sail directly towards the wind, so a proportion of the water/wind can always get to the passengers over the windward side of the boat or over the stern while travelling downwind.
One solution to this issue would be a collapsible barrier/deflector, deployable when conditions demanded its use. One solution would be to raise the back of the benches in order to cover the torso and head. Modularity, Adjustability and disassembly, would allow it to function in providing shelter, while not compromising visibility while manoeuvring.

2. Mast Platform
Occasionally climbing the mast of a sailboat is necessary to perform routine maintenance and upgrades, as my client does both in the Marina or (rarely) at sea. This could be to inspect the rigging, retrieve a lost halyard or unfoul a jammed block (pulley). Climbing the 40ft mast is performed using a harness and fold out steps. The serious issue with this arrangement is insufficient stability, as it is easy to fall off the steps and swing around as the boat rocks.
Mr R Cowley suggested a small (just big enough to fit two feet on) removeable platform that could slide up the mast and be clamped/secured at the level where work needed to be done. This would improve safety and help with balance and comfort by taking weight off the harness. The platform could slot into the track which the mainsail fits into to keep it secure and stable while having slots or pockets to briefly hold hand tools and small parts such as bolts and shackles .

3. Man Overboard recovery device
If a person falls overboard there are various conventional techniques and devices in order to quickly and reliably attempt to retrieve them from the water. Equipment to do this is often installed - with many offshore cruisers having throw lines, Life jacket, PLB's (Personal Locator Beacons) and a Lifebuoy. The shortfall of these devices is that they all require the MOB(Man Overboard) to be able to assist with their own recovery by grabbing hold of things or climbing up a ladder.
My client noticed a general need to be able to retrieve a MOB casualty who is incapacitated, unconscious or unresponsive. This situation is plausible, - for instance if the MOB has symptoms of hypothermia, trauma during the fall or panic.
A device to solve this problem would need to be able catch, secure and assist with the lifting of the MOB from the water. Yachts/Sailboats such as 'Leebra Lass' have a high deck over a metre above the waterline, so it is not possible to physically reach someone in the water beside the boat without risking falling in yourself. Existing devices such as hooks and slings exist but these compromise on either reach or security.

4. Coiling and rope storage aid
Ropes for mooring, attaching fenders and rigging the Genoa pose a considerable trip hazard and must be stowed and deployed regularly during a journey. A crank handle winder or simply some sort of correctly shaped jig to coil ropes around quickly would be useful. This could be portable or mounted to an accessible section of deck near solar panel.

5. Diesel Bug Prevention
Diesel bug is microbial contamination at the interface between fuel and water in a tank. The bacterium which feeds off on oils can only survive if water has entered the tank though leaks, contaminated fuel or condensation.
Diesel Bug being can be a safety hazard as it creates a black sludge which clogs filters and fuel lines preventing fuel from getting to the engine. This can make engines unreliable in critical situations such as during an emergency where use of the engine is the only way to avoid an obstacle. Constant replacement of filters and redundancy is expensive, and though chemical methods exist to prevent growth cleaning a tank in transit once a colony is established is extremely difficult. A device to clean fuel inside tanks clean either by filtration, straining or siphoning out heavier water from the bottom would be able to prevent this issue. Even just an articulated scrubber to breakup, scoop out and clean tanks while in transit.

6. Orca Attacks – Emergency seal for hole in hull, or steering .
Orca attacks on small boats are rare but dangerous, with about 50-200 reported each year. Orcas are most prevalent in the straight of Gibraltar and along the Iberian Peninsula, and can cause significant damage especially to rudders. This can be extremely dangerous, resulting in vessels getting stuck without steering control or damage to the integrity of the hull where the rudder (which acts as a lever) is attached.
There are several potential solutions to minimise damage or recover from the effects of an attack. Steel and Carbon fibre systems exist, which shield the rudder or improve its rigidity without reducing boat speed/efficiency.

7. Evaporation food/beer cooler
My client's yacht is fitted with a top loading mini fridge capable of storing a few items of food and drink. The electrical consumption is quite high, which is not an issue while hooked up to Marina power. Unfortunately, while sailing under most conditions solar panels cannot keep up with the power demand and he does not want to risk running the battery low which is required to start the engine.
Coolers which use the evaporation of water to provide cooling are already on the market, but none are designed specifically to fit inside a boat. These coolers work by having a revisor or saturated sponge to allow water to slowly evaporate taking in and removing heat.

Further Investigation & Justification of Design Possibilities

Proposal: 1. Collapsible wind deflector/canopy

Design Development/Clarification (8)

An accordion style folding sheet of thin wood with transparent acrylic/polycarbonate windows for visibility would attach to the guardrail behind the seats in the cockpit, be collapsable for storage during calm conditions and be quickly moved to the upwind side.

Research Into Existing Solutions	Relevance to client's requirements (6)	Commercial/Widespread Applicability (3)
Spray dodgers are made from UV stable acrylic canvas and attached to the guard rail. These are commonly seen with the boat name printed on them in white.	Sideways splash protection is only a significant issue rarely in very windy conditions. I would prefer to help with an issue that is more specific to the actual needs of my client.	A wind deflector or canopy would likely need to be specifically shaped to fit my client's boat, making it only marketable to other Moody 31 owners, thus limiting commercial potential.
Achievability (7)		
Creating a barrier could be quite simple, but making it integrate well with the boat and match its traditional aesthetic would be a challenge.		



Figure 3

Client's Thoughts

"Spray can be an issue when I have passengers on board without good waterproof clothing"
"I'm not sure much would be stopped by adding more barriers to the sides as a lot of spray comes down over the top"
"A barrier behind the cockpit seats could possibly deflect some of the spray coming across the boat"

Proposal: 2. Mast Platform


Figure 4


Figure 5


Figure 7


Figure 6

Design Development/Clarification (10)

Modern masts are aluminium extrusions (Figure 6) which have a track for the sail to fit into, the platform would be about 300 x 300mm rectangle which could be attached to a bracket that slotted into this and is pulled up, and held in place before assent with a halliard. The sail track and cross section of a mast are shown

Research Into Existing Solutions	Relevance to user requirements (6)	Commercial/Widespread Applicability (8)
The only comparable product is a Bosuns Chair (Figure 7) but these are not rigidly attached so can still swing. Working seated may make it hard to reach masthead.	Spreader's (labelled Figure 5) can be stood on, but on 'Libra Lass' are too low to stand on while working at the top the mast. Mr R Cowley climbs mast approximately once per year.	No similar existing products for service platforms for small yachts. Simple space saving design could be a useful item to store onboard. Issues could come from differences between mast profiles.
Achievability (7)		
Small size would make cost affordable. Small safety margin required as user always harnessed. Would need to assess if steel or CF would be strong enough to support person limited by mast track width.		

Render general idea of what it might look like


Figure 8

Proposal: 3. Man Overboard recovery device

Design Development/Clarification (8)

Presently, my Client has limited equipment to be able to effectively rescue a MOB if they became unable to physically assist with their own recovery. "Libra Lass" is fitted with throwline and lifebuoy, but these both require the MOB to hold on to be reeled in. An able MOB would have the ability to climb abord with a ladder at the transom (flat surface at rear of boat).

Research Into Existing Solutions	Relevance to client's requirements (9)	Commercial/Widespread Applicability (8)
there are two areas where a device could assist with rescue - Securing and Lifting a MOB. Commercial methods commonly use loops/hooks and slings.	Existing solution is a DIY carabiner style hook welded to a rod. Could attach a rope to the MOB. Not purpose built, main use is catching mooring buoys.	Large market of safety conscious cruisers and racing yachts. Opportunity to improve on existing products effectiveness which are more suited to larger ships.
Achievability (7)		
The required strength and durability to lift the weight of a person could be difficult. A range of mechanism complexities would pe possible to help lifting people.		




Figure 9



Figure 10

Client's Thoughts

"I think a neglected area of safety, its not that much of an unlikely scenario that someone cannot offer any help getting back in"

"Most people have lots of kit for climbing back on board or being pulled"

Proposal: 5. Diesel Bug Prevention

Design Development/Clarification (9)

A device could incorporate a self-starting siphon to drain the bottom layer of a tank with a brush, scraper or other tools allowing a tank to be serviced on the water. The Intake could be made with the correct buoyancy to float at the boundary between the fuel and water where bacteria proliferate. Transparency to allow user to check if contamination is present.

Research Into Existing Solutions

Prevention methods such as chemical fuel additives, or companies provide expensive 'fuel polishing' services where they drain and filter tanks.

Achievability (4)

Difficulty could come from sludge clogging pipes and only removing contaminated layers. Also, filtration with only hand power may not be possible

Relevance to client's requirements (5)

My client has not had issues with diesel bug, due to not needing a full tank of fuel which can stagnate (as reserve for long trips) he generally takes short trips, along with Cardiff's colder climate reducing condensation.

Commercial/Widespread Applicability (6)

Common problem in boating, yachts often moored for several weeks at a time. A low-cost portable fix would be a useful tool. Greater issue due to new biodiesels being more affected.

Alternative Design

A hand crank powered pump and filter circuit to process all the stagnant fuel in a tank either passing it though cheap vacuum cleaner filters or using centrifugal force to separate components based on their density, reducing the number of consumable parts to zero. This could have a drum that throws lighter water and contaminants to the outside allowing fuel to be tapped from the centre. It could be interesting to see if this would work in practice.



Figure 11



Figure 12



Figure 13

Client's Thoughts

"I've never had much of an issue with contamination, but know people who have, I think it depends on how diligent you are with using fuel and engine servicing"

"Probably more money saving"

Proposal: 4. Coiling and rope storage aid

Design Development/Clarification (6)

A compact hand powered geared device to wind ropes into a neat coil more quickly would allow this to be done with less effort and time. A range of complexities of mechanisms could be tested to coil evenly with oval shaped loops. After winding, the coiled rope would be able to slide off one side for storage. A guard to partially enclose moving parts protected and make it simple to use by just feeding a rope in.

Research Into Existing Solutions

Most portable aids are very simple one-part jigs to manually wind around. One exception of note is the "No Rope" which is a spring loaded permanently installed unit to make mooring lines retractable.

Achievability (4)

Designing a system which helps with coiling would be possible but making it convenient and fast enough to justify its use could be challenging.

Relevance to client's requirements (2)

Relevant as multiple ropes are wound and unwound at various points during every trip. May be a solution to a non-existent problem as my client has no particular difficulty tidying ropes

Commercial/Widespread Applicability (8)

Compatibility with different thicknesses and lengths would allow a portable tool to be used on any sailing boat.

Proposal: 6. Orca attack protection

Design Development/Clarification (5)

A welded frame/cage that could be fitted around the rudder to protect it against impacts, while still allowing it to move to steer the boat. This could be easily removable without having to take the boat out of the water by using ratchet straps around the hull to attach without needing permanent modifications.

Research Into Existing Solutions

Advice against attack focuses on strategies to encourage the animals to lose interest in the boat - lowering sails, turning off engine, disabling depth sounders/electronics. Protection cages do exist but are niche

Achievability (3)

Creating a frame hydrodynamic enough not to slow the boat while also being strong enough to withstand strong hits may not be possible.

Relevance to client's requirements (1)

My client has never sailed internationally in his own boat and Orcas are extremely rare in UK waters apart from near northern Scotland, so he is unlikely ever to have issues with them.

Commercial/Widespread Applicability (2)

Might be sailable in countries with colder seas with more orca activity, otherwise the number of orca attacks each year is very small.

7. Evaporation food/beer cooler

Design Development/Clarification (8)

The cooler would be a modern adaption of Zeer pots (Figure 14 & 17) used in ancient Egypt. These work by having a pot inside a larger pot with wet sand sandwiched between. The evaporative cooling effect occurs due to slow evaporation of the water removing heat the same way as sweating. A modern version designed to fit inside the boat may be able to keep food and drink 5-10 ºC below room temperature.

Research Into Existing Solutions

The Fenik No-Ice Cooler (Figure 16) is a Kickstarter project for camping which uses evaporative cooling lowering temperatures by 5-10 ºC. It uses Gortex like materials to increase the rate of evaporation.

Achievability (5)

Effectiveness of cooling could be reduced by humidity and lower ambient temperatures reducing evaporation.

Relevance to client's requirements (6)

This would be useful for my client while staying at Marinas without a power supply. However, he rarely sails far enough to have depleted batteries while in transit. See kitchen area space available in Figure 15.

Commercial/Widespread Applicability (4)

Most yachts designed for long passages are fitted with enough solar or wind generation capability to run a fridge and lights indefinitely and start the engine in an emergency.



Figure 14



Figure 16



Figure 15



Figure 17

Client's Thoughts

"Too much evaporation could cause moisture problems below deck if the hatch was kept closed, unless there was some way of venting it"

"Might be useful as an addition to the fridge for non-perishable items like drinks"

"I'm not really sure that for me I'd get lots of use from it"

Choosing a Design Proposal

I have compared my analysis and research around each design possibility to each other (see table right). The mean score is calculated using a score out of ten for how I felt a design proposal performed in each of my assessment categories. These were: Scope for design development, Achievability (If it would be possible for me to make a working solution), Relevance to client requirements and Likelihood of success commercially. Each subtitle on the previous two pages is a category (score shown in brackets). These give me a quantifiable method of assessing the potential of a design proposal.

Justifying Choice of Design Proposal

Based on the scoring and also both my own and client's preference, the Man Overboard (MOB) rescue device was chosen. This option will give me the greatest ability to experiment with a range of methods of solving the problem. It also is large and potentially complex enough to allow me to incorporate a range of techniques and materials, while the principle of securing and lifting a person is not unachievable. This will challenge me to test and improve my design & manufacturing skills in the available time.
Most importantly, this design possibility aligns with the needs of Mr R Cowley, my client. He feels that insufficient readiness to cope with an incapacitated MOB situation is an important feature lacking in his current safety equipment. He does not posses any purpose-built method of lifting people out of the water. My client also thinks that this is a common area that rarely considered in yachting, so there is a considerable market for devices at various price points.

Design Possibility	Mean Score (out of 10)	My Opinion (ranking of how interested I am in the idea)	Clients Opinion (Ranking based of comments on each)
1. Collapsible wind deflector/canopy	7.75	6th Not enough scope for nuanced/complex development	7th
2. Mast Platform	7.75	2nd Unique solution to the problem of comfort during mast maintenance.	3rd
3. Man Overboard recovery device	8.00	1st Opportunity for many different approaches, important to client.	1st
4. Coiling and rope storage aid	5.00	4th Potential to save time but also could be a gimmick.	4th
5. Diesel Bug Prevention	6.00	5th Potential to be very useful but too complex to build	5th
6. Orca attack protection	3.75	7th Very little relevance to client's needs.	6th
7. Evaporation food/beer cooler	5.75	3rd Interesting concept but may not be effective enough at cooling.	2nd

Assessment of client's Wants, Needs and Values

Introduction

I interviewed my client to gain more in-depth understanding of his priorities and take advantage of his knowledge of the risks of yachting, features of his boat and experience making boat parts.

Client Interview

In your experience what are the main shortfalls of existing similar products?

"The main problem with the fabric type (parbuckle slings) I've seen tested is that if there's any breeze it can be impossible to get them unfurled, fixed and working because they flap all over the place and its hard-to-find which corner goes where"
"On the other hand, anything more rigid, I've seen used commercially just doesn't fit anywhere accessible on smaller boats"
"I haven't seen a product that is good at helping with securing and lifting"

What features would be most important to maximise chance of successful rescue?

"Most of my crew are older friends of mine, so to stand a chance teamwork or mechanical advantage from pulleys must be possible"
"recognisable colours to signal to other vessels and help with setup"
"Should be simple enough to use that I can just tell people where it is and generally what to do"
"The time and difficulty to deploy it must be as little as possible, drifting and hypothermia probably worsen your chances quite fast"
"Being able to get hold of someone is the top priority as you may only get a small window to do it, a large target area to make room for course error and waves would help"
"Often bring my grand children sailing so it would need to work for a range of sizes and weights"

What other aspects do you think I should consider?

"Even though all of my own passengers will be wearing life jackets, it would be useful to have the option to rescue someone I come across without anywhere to hook onto"
"Reliability is the most important factor, we will probably test it once a year, but it needs to be able to withstand storage for long periods"
"Must seem sturdy to encourage trust in it working"
"Visually, matching the style of the boat to not look too out of place"
"If possible, reducing its environmental impact to sort of fit the theme of sailing as a clean method of transport"
"Ergonomics and comfort for the person using it and man overboard, to limit extra injury and make physically training and testing it bearable"

How much demand is there, how much of a common problem?

"Lots of people are safety conscious, but unwilling to spend large amounts on specialised gear for an unlikely scenario"
"Many available products are marketed towards commercial buyers such as fisheries, coastguard and military who can afford to spend more"

How are other items designed to be safe and durable in a marine environment?

"Everything with any weight needs to be securable, so it doesn't move about or rattle, 3D printed clips work well"
"Choosing the correct alloy of aluminium or stainless is important to compromise between performance and cost"

Key takeaways:

Needs

- To be able to rescue any casualty in the water regardless of size and without a lifejacket
- Be capable of reducing its size to be able to fit into storage spaces.
- To be able to be deployed in as little time as possible.
- To be able to get hold of someone as fast as possible.

Wants

- Greater stability and less freedom of movement in all axis than fabric slings and lifting straps.
- Comfortable ergonomics for training.
- Intuitive/Simple to learn/teach to use.
- Matching the style of the boat loosely.
- High visibility to help with locating device and setup.

Values

- Reliability, designed to have a long lifespan with minimal maintenance
- Safety, large safety margins on the strength of materials used.
- Sustainability, not causing avoidable damage to the environment where alternatives are possible.

Software Used for Project

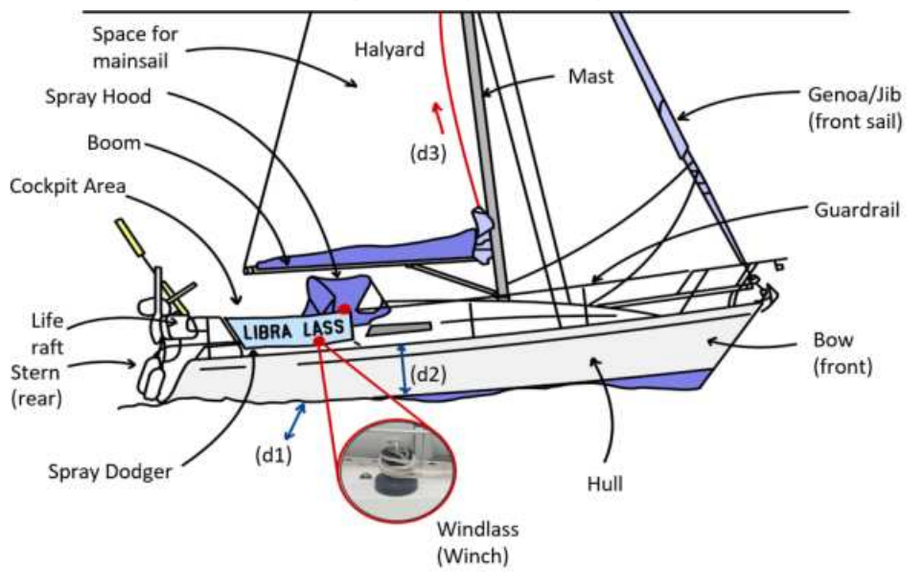
All renders throughout the portfolio were created by 3D modelling using **Blender**, software commonly used In creating 3D animated movies. For some renders, I used a line-art modifier to create cartoon style. **Autodesk Fusion** was used for CAD and to produce drawings. No AI was used in the production of any CAD or images within this project.

Glossary/Abbreviations

To assist those from a non-sailing background understand vocabulary relating to boat parts, dimensions and acronyms.

ABS – Acrylonitrile Butadiene Styrene, a polymer commonly used in 3D printing and injection moulding. The material that Lego is made from.
Buoyancy Aid – buoyant jacket designed to provide assistance with staying afloat in water.
CNC – Computer Numerically Controlled
EPDM – Ethylene Propylene Diene Monomer – a dense rubber material
Halyard – (d3) A rope used for lowering/raising the sail which passes though the top of the mast. This is pulled up using a windlass in the cockpit.
Lifejacket - Sleeveless buoyant or inflatable jacket which is capable of keeping an unconscious person afloat and with their head out of the water.
MOB – Man/Person Overboard
Outboard Distance – (d1) (not a technical term) The phrase I will be using to describe the distance to an MOB from the hull, perpendicular to its surface at the waterline.
PLB – Personal Locator Beacon
PD – Potential Difference
TPU – Thermoplastic Polyurethane, 3D printable polymer that is flexible, shock absorbent.
Topside – (d2)The upper part of a boats side above the waterline. The height of the topside, waterline to deck edge is known as the freeboard.
UV – Ultra Violet
UHMWPE – Ultra High Molecular Weight Polyethylene (band name Dyneema), a polymer commonly used for high strength ropes.
RYA – Royal Yachting Association
Windward Side – the side of the boat which faces into the wind, as opposed to the leeward side which is opposite and sheltered from the wind.
Windlass – Type of cylindrical winch operated by a hand crank which slots into the top, provides mechanical advantage and has a ratchet to lock a rope in position.

Below is a labelled diagram with parts of the yacht for reference.



Colour Code Key 2: For the next 5 pages in the Research section of the portfolio text coloured **blue** are key design needs that I have identified during my research.
All comments made by my client will continue to be coloured **orange**.

Research - Ergonomics and Integration

Having chosen the 'The Man Overboard Rescue Device' proposal, It is important to get an idea of which parts of the boat a design may need to interact with when it is being used. Additionally, ergonomic and anthropometric data influencing dimensions can now be considered. In the section below, I will look at how the device will integrate with and attach to my client's boat while also factoring in human features to create a space envelope to constrain the size of the final design.

Minimum and Maximum sizes justified.

- **Length 1.365 – 1.8m**
  - Limiting factor 1: Maximum possible length of 1.8m because this is the distance between guardrail posts.
  - Limiting factor 2: Minimum possible length 1.365m to ideally support body of 95th percentile MOB measured from feet to chest (Figure 18). Aiming to support an MOB from their ankles to their chest will spread upwards pressure over most of their body.
  - Minimising length will take up less space in storage. Maximising length improves comfort/support.
- **Width (Outboard distance for sling designs) > 0.54m**
  - Limiting factor: A device must have enough width to fit around the shoulders of a 95th percentile MOB (Figure 19)
  - Width a compromise between ease of use in sliding people into/onto the device and material costs.
- **Reach (Furthest diagonal distance MOB can be secured from) 1.5 – 2m**
  - Client suggests being able to secure an MOB from 1.5-2m away is practical to allow for good functionality while being easy to store and move.
- **Deck Edge Rail Dimensions (See diagram and photos)**
  - Slotted holes may be used as attachment points for toggles
  - Length must be a multiple of the distance between holes if toggles to be used (Figure 22)
- **Rope thickness = 12mm (Figure 20)**
  - Ropes available on board capable of carrying the weight of a person have a 12mm diameter.

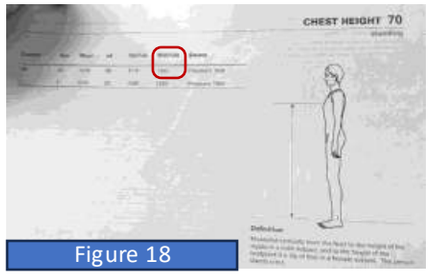


Figure 18

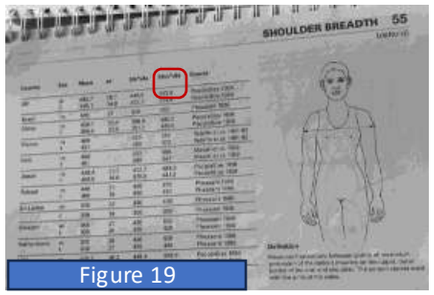


Figure 19

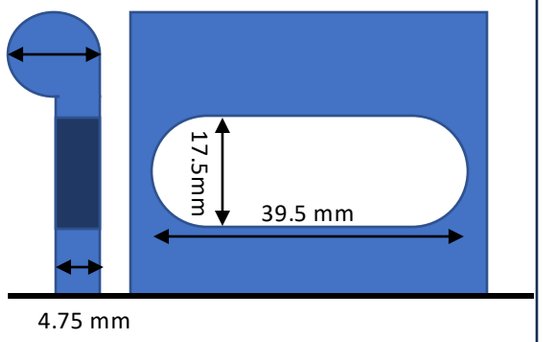


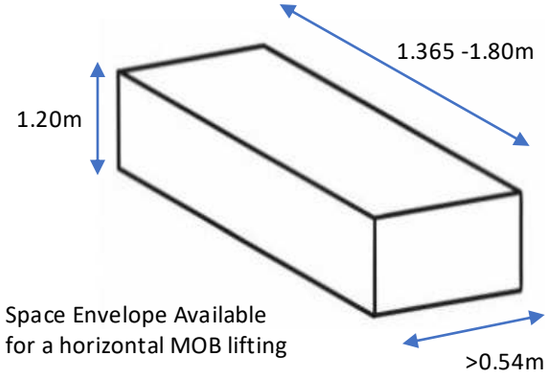

Figure 20



Figure 21



Figure 22



Space Envelope Available for a horizontal MOB lifting

Lifting Assistance

Whatever design chosen, it is likely to make use of built in pulley systems available on my client's boat to provide mechanical advantage - lifting an MOB more manageable .

Windlass winches available provide either 30:1 and 40:1 advantage as they are intended for pulling in large sails. This means that to lift a 100kg person between 2.5 and 3.5 kg of force must be applied. This is enough of a reduction to make lifting easy but exact choice of winch will be a compromise between effort and speed.

Research - Existing Equipment

The design of a device may need to factor in safety equipment already present on my client's boat and commonly installed on others. Equipment assessed are only those which could affect the design of my product - throw-lines, lifebuoys and life-raft would not be useful in an incapacitated MOB situation.

Lifejackets- CO_2 cartridge causes them to inflate upon contact with water, this could increase the space a person needs to fit inside a device considerably. Lifejackets with loops or waist straps could provide a quick method of securing a person.

Guardrail - Reduces the risk of people falling overboard, but could act as an obstacle for lifting people back in.

MSLD (water activated Maritime Survivor Locating Device) – Not deemed necessary by my client, high price means usually only used by racing crews and long-distance solo sailors. An electronic component such as RGB lights could integrate with the software to display different colours based on proximity, if it were adapted to suit these markets.

Windlass- Four vertical winches are present two on each side of the cockpit. Used as winches during sailing provide mechanical advantage to move sails. Windlass could provide upwards lifting force though halyard or more directly.

Research - Medical/Scientific Background

The medical needs of the users (rescuer and MOB) should be factored into the design of a rescue device to maximise their safety and the chance of a successful rescue.

<u>Risk factors affecting rescue success</u>

Time

Reducing the time a person is in the water is critical. Long periods in rough, cold water are detrimental to health and the chance of being able to locate and secure a person due to drifting.

Research by the UK's Marine Accident Investigation Branch confirms the need for rescue/recovery aids, since half of all MOB incidents from pleasure craft result in a fatality. The report also states crew have on average under 11 minutes to recover a crew-mate who has fallen overboard into cold water, and cold incapacitation can occur within 2 to 15 minutes of entering the water. This is when blood flow to muscles and extremities is diverted to preserve heat in the core and vital organs. Include features to allow for: quick unpacking and setup, simple/intuitive to use under pressure, Modular and adaptable to suit any scenario, Easy to locate in storage.

Orientation

According to RYA (Royal Yachting Association) guidelines; when lifting a casualty aboard they should remain in a horizontal position to avoid complications caused by hydrostatic squeeze. Hydrostatic squeeze is the pressure applied by the water to the body, which once released as the MOB is lifted out the water blood can flow from the limbs into the heart causing a heart attack. Lifting vertically also leads to blood pooling in the legs, that can cause low blood pressure that leads to fainting or shock. The RYA suggests that though lifting horizontally can reduce the chance of medical complications, priority should go towards removing a MOB from the water if time or equipment for horizontal lifting is unavailable, as drowning is the biggest risk.

To reduce the risk of "circum-rescue cardiovascular collapse" feature horizontal, sloped, seated or a slightly raised knee position while the MOB is being lifted.

Blunt force trauma

Another source of discomfort and injury to casualties could be inflicted by hard structures present on the boat and rescue device. The chance of an inadvertent collision between the MOB rescue device during rough weather is significant, due to the constantly changing position of the boat and MOB. Though bumps against the MOB only have the potential to cause minor injury, increasing comfort will encourage training and reduce reluctance to use a device.

To reduce the severity of impact against a MOB, add padding to pointed areas, reduce the use hard fittings which protrude, aim for a more rounded shape to reduce sharp corners, minimise the weight of the device.

Blunt force trauma can also be caused by impact with the boat itself, during extreme conditions. Swinging caused by the boat rolling while suspended could be hazardous, banging against topside (outer hull surface above the waterline) could cause major injuries such as fractures and concussion . To reduce this, add some sort of protective barrier or padding between the hull and MOB. Also, make sure it is possible to hold onto and control the device and person being lifted.

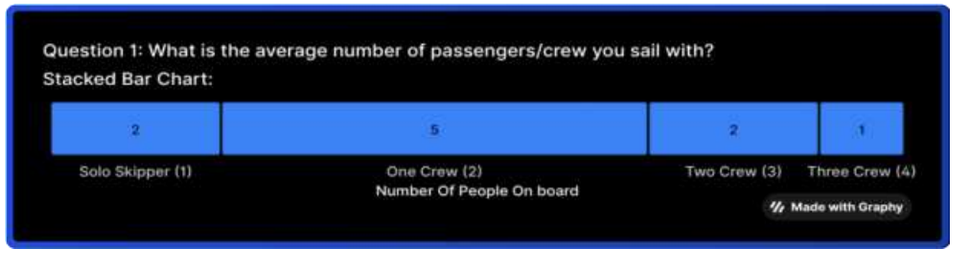
One problem with commonly used lifting slings are that they form a loop very close to the hull which in rough swell can risk the MOB being crushed under the hull, being able to secure a MOB while they are further away could mitigate this.

Drowning

The primary method of preventing drowning is removing the casualty from the water in as little time as possible. Another way is by reducing waves and splash going over the MOB to prevent inhalation. The importance of reducing splash is taken by some life jackets which have built in hoods to cover the MOB's head and face. This could be done by having the outboard side of the device waterproof and have enough rigidity to reduce the strength of a wave to provide some shelter. Also, some sort of deflector or thicker area on the leading edge of any sling designs could help.

Research - Target Market Questionnaire

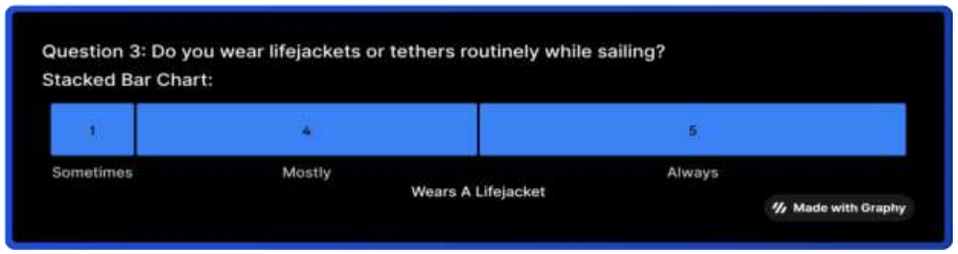
I created a survey completed by other members at Cardiff Bay Yacht Club was made to get a more general sense of the needs of recreational sailors. The questions will give me information about the required cost, crew numbers and experience and maintenance timescales, which will influence the design of the product.



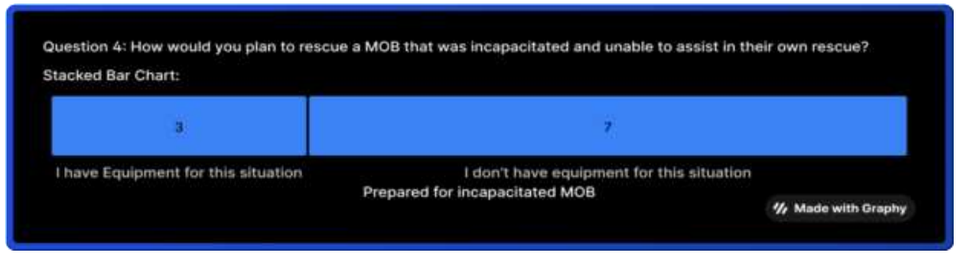
Conclusion: Most sailors have a single other person on-board to rescue them while they are sailing. So, a rescue device must be operable alone.



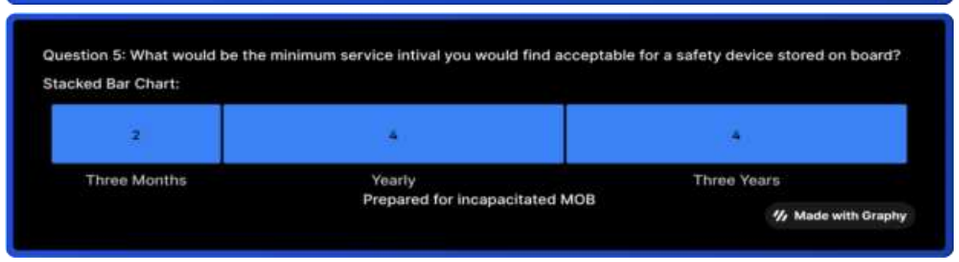
Conclusion: Most of the time passengers have no or limited sailing experience. This means that the device must be very simple to use so they are able to perform a rescue of the helm.



Conclusion: Nearly all the time an MOB will be wearing a lifejacket. The extra volume of an inflated lifejacket could affect the size of a rescue device.



Conclusion: There is demand other than from my client for something to help, in the specific scenario of an incapacitated MOB.



Conclusion: The device must be durable enough to stay in storage for as long as possible without being checked or repaired.

Question 6: What equipment do you have on board to be able to respond to a man overboard situation?
Common Equipment: Life raft, Lifejacket, Tethers, MSLD, Lifebuoy, Hook
Conclusion: Most recreational sailors are very safety conscious and would be keen to buy equipment which offers improved capability.

Question 7: How much would you be willing to spend on safety equipment in total (excluding life raft)?
The average answer was roughly £300. I should have asked directly how much they would spend on a single device as it is difficult to judge how much other safety equipment they buy would cost.

A sample of 10 people from CBYC provided responses, so data collected should be quite representative of recreational sailors.

Research - Choice of Materials

Durability

Common material choices for marine environments and in yachting:
Woods: Marine Ply, Teak, Pressure treated softwoods. Metals: Stainless Steel, Aluminium. Plastics: Nylon, Polyester, fibreglass(GRP)

Resistance To Environmental Factors

A device would be required to withstand complete exposure to the elements intermittently, due to infrequent use during emergencies and training drills. A high degree of toughness would be important from all materials to withstand impacts from being dropped onto hard surfaces or hit against parts of the boat. In the section below I will research how conditions present on boats could affect a device and which material choices to mitigate these effects.

Salt water

- Only corrosion resistant metals to be used such as galvanised steel, stainless steel or aluminium to avoid seizing of mechanisms and weakening of structural components.
- Avoid having areas where water could pool or be trapped by having drainage holes in any hollow spaces.
- Problem: Do not put dis-similar metals in direct contact with each other. This can lead to galvanic corrosion (Figure 23) where the two different metals act like terminals of a battery with salt water as an electrolyte. This causes accelerated corrosion with some combinations of metals. Importantly, this can occur when using Aluminium and Stainless Steel together due to their quite different electrochemical PD's.
- Solutions would be:
    - Use a non-conductive sealant/adhesive such as silicone between parts. Gaskets or washers can also be used to separate metals.
    - Reduce the surface area of contact between aluminium and steel parts.
    - Using fittings (nuts/bolts/rivets) which match the material of a part.
    - Choose alloys with the smallest difference in electrochemical PD.
    - On wooden parts seal edges and faces or apply varnish, use waterproof adhesives.

UV (Sunlight)

- If an item must be stored outside for quick access it will be exposed to UV for prolonged periods. On open water UV is stronger due to reflection of sea surface and lack of shade. Polymer parts, Textiles and Coatings can be affected.
- Problem: Discolouration/Bleaching of high visibility brightly coloured components could be dangerous.
- Problem: Mechanical failure of straps/rope caused by degradation or cracking of brittle plastic would be unsafe.
- UV damage (Figure 25) could decrease the lifespan and worsen the aesthetic of a device.
- Solutions would be:
    - For 3D printed parts avoid the use of PLA, instead use an ABS filament with UV stabilisers, PETG or ASA.
    - Avoid thin plastic parts with sharp edges/corners as these factors can worsen cracking.
    - For ropes or straps made from synthetic fibre, make sure to choose a UV resistant option. Generally Polyester, Polyethylene or Treated Nylon ropes are used for outdoor applications and sailboat rigging.

Temperature

- Device would be stored in uninsulated, unheated storage areas or outdoors so must be able to withstand temperatures from -20C to 60C.
- Temperature fluctuations daily and between seasons can affect certain materials (Figure 24).
- Metal glued to plastics expand and contract at different rates which can be pulled apart slowly.
- Untreated wood can crack and swell if it has absorbed any moisture.
- Direct sunlight can cause surfaces to heat up above the ambient temperature this can affect material and has the potential to cause discomfort or burns. To avoid this, I should use lighter coloured materials and avoid having metal handles or areas that people will touch.

Vibration

- Some storage areas are located close to the diesel engine which is used during mooring, vibrations are moderately strong but over long periods could have an effect.
- Use fastenings which can withstand some vibration, such as lock nuts.
- Must be capable of withstanding being shaken around in a storage compartment significantly due to vessel pitching over waves.

Sustainability

Sustainability is not a high priority for my client given the application of the product, and its bespoke production scale. Generally, I will be prioritising functionality over it being environmentally friendly in areas where performance is critical, though where possible sustainable options should be used.

The durability required for safety has the secondary benefit of increasing lifespan which, prevents the necessity to source a replacement, lowering the total carbon emissions of disposal and remaking. A modular design with easy to disassemble parts will also increase the ease of replacing and recycling components. Reducing the space and weight of the device so it can be stored easily will reduce transport costs and emissions of sourced materials. Hardwoods will be avoided due to being heavy, and since hardwoods are often sourced though deforestation of tropical regions. Metals choice must factor in corrosion resistance, cost and sustainability. Stainless Steel and Aluminium are commonly recycled, though refining is energy intensive and stainless steel is an alloy containing rare earth elements (nickel, cobalt, molybdenum). Textile components such as straps and fabric must be certified to withstand the required loads, and environmental impact will be considered within this constraint. Minimising waste during the manufacturing process by using materials efficiently will save costs and environmental impact.



Figure 23

Galvanic corrosion (bolts different metal to sheet)



Figure 24

Thermoplastic warping and melting when exposed to heat



Figure 25

Brittleness due to UV degradation

Safety

There are numerous organisations providing safety guidance and standards for maritime applications including: BSI (British Standards Institute), IMO(International Maritime Organisation), SOLAS (Safety of Life at Sea), ISO (International standards organisation) and ABS(American Bureau of Shipping). Below I will quote safety standards relevant to a man overboard rescue device.

- **SOLAS Chapter III** – Life Saving Appliances
    - "All life-saving appliances should not be unduly affected by sea-water, oil or fungal attack and be resistant to deterioration from exposure to sunlight"
    - "Must be a highly visible colour" or fitted with "retro-reflective material"
    - Lifting devices – "must support casualty in a safe and ergonomic way in face-up position"
- **ISO 15085** – "Safety arrangements on small craft to prevent going overboard and facilitate reboarding"
    - "Construction and strength requirements – use of load tested materials with defined safety margins"
    - Must be capable of operating in any climate, advisably between -20 C and 60 C
- **IMO FTP (Fire Test Procedures)**
    - Materials tested must meet low limits for flammability, smoke emission, smoke toxicity.
    - Avoid PVC plastics being stored around fuel tanks and engines as this produce toxic fumes.
    - Pick fabrics and foams treated for flame resistance.

The safety and comfort of the man overboard and device operator performing the rescue must be considered. The device must be designed to put the operator in the least risk possible.

Cost

I am aiming to build a device that offers functionality on par with the higher cost products aimed at professional rescuers (priced above £300) but at the cost of some of the more reasonably priced options such as the simple parbuckle slings (priced around £150) marketed towards recreational sailors. To a achieve this I will aim to minimise complexity/number of components but still offer improved functionality (ease of securing and lifting).

Research – Similar Products 1

C-Hero MOB VR-12 Rescue Pole/Recovery Strap (secondary)

Advantages

- Multifunctional (secures and lifts seamlessly) reduces time to gather more equipment.
- Strong closed loop to pull MOB close, detachable lifting strap with looped handles held open with blocks and snaps.
- Padded frame reduces chance of injuring MOB.
- NATO Approved as an "efficient compliant solution".
- Extendable telescopic handle extends from 2.1 – 5.2m improving storage but giving very good reach when extended.
- One person operation.





Price: Approximately £1200

Disadvantages

- Does not lift horizontally, Thin lifting strap could cause discomfort under arms.
- It could be difficult to get hoop/frame positioned correctly around chest, as it must be passed under MOB.
- High sale price mostly restricts to industrial use.
- Little justification for high price as it is not customised for each customer.
- Good at securing an MOB but does not help with lifting them aboard.

Key Features or Analysis

Various states of assembly possible depending on immediate risk or the need for long term storage while moored. Also, very good reach to secure an MOB that has drifted away.

Dacon Fisherman Rescue Frame (secondary)

Advantages

- Robust and durable design with UHMWPE, fiberglass rods and an aluminium pole.
- Floats just below the surface of the water.
- Much greater reach and control than other slings.
- Extremely easy to climb
- Places for multiple people to grip and aid with lifting while being used as a sling.
- Parbuckle lifting provides mechanical advantage.
- Built in wheels reduce friction with side of the boat.







Price:£400 (size dependant)

Disadvantages

- It may be possible to slip though the loop before it is closed due to smooth fibreglass rods.
- Can be rolled up, but not folded lengthways, could make storage more difficult.
- More suited to Rigid Hulled Inflatable Boats with a smaller distance to lift.
- Quite expensive for the materials used.
- The pole used to push it out and hold it open could get in the way of sliding a casualty onto it.

Key Features or Analysis

Ability to be used by multiple people to share load.
Simple, durable and adaptable design

SOS Marine Recovery Ladder (secondary)

Advantages

- Lowest priced sling
- High Vis yellow polyester. Built in stuff pouch storage.
- Option for able crew to climb integral ladder rungs.
- Soft construction avoids causing injury to the MOB if the boat rolls or pitches.
- Mechanical advantage provided by halyard or block and tackle.
- Optional mesh sleeve which is secured on deck's edge and can be grabbed from the water.
- Ladder part can be pulled directly, providing mechanical advantage though parbuckle lifting.





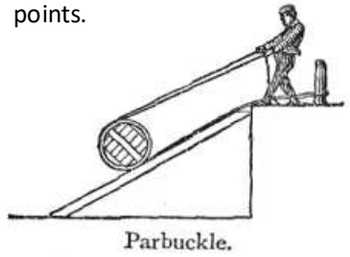




Price: £127

Disadvantages

- Very Little ability to reach out away from hull, meaning MOB would have to be close enough to not be visible by the helm.
- Without quick deploy mesh sleeve it would take a long time to unpack and attach at 3 points.



- Parbuckle Slings half the force required to lift MOB compared to directly lifting them.

Key Features or Analysis

Design to reduce risk of causing injury to MOB
Use of existing features of vessel – attachment points and halyard

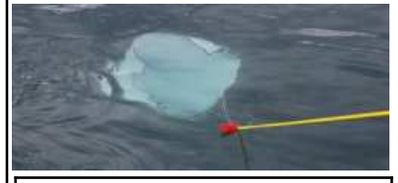
CATCH and LIFT MOB Rescue System (secondary)

Advantages

- Self contained with robust, waterproof case.
- Compact and lightweight (4.9kg)
- Works independently, does not rely main or jib halyard or winch to work.
- Quick/Simple deployment by throwing box into the water.
- No muscle power required to lift MOB. The opposition of a drogue (Water parachute) against the boats forward movement pulls line through block to hoist MOB upwards.







Price: £510

Disadvantages

- High price and complexity.
- Boat must be capable of moving to hoist MOB which could increase danger of them being hit by the propeller.
- May cause discomfort due to unpadded chest loop and does hoist them lying down to reduce hydrostatic shock.
- Method of securing harness around person not included so may not work for an incapacitated MOB.

Key Features or Analysis

Time efficient unpacking, deployment and use.
Use of carabiner/quick fittings/hooks to set up anchor points quickly.

Research – Similar Products 2

Homemade Rescue Sling (Primary)

Advantages

- DIY solution much less expensive than commercial options.
- Mesh material allows prevents filling with water, reducing extra weight (Figure 28)
- Toggles offer quick and secure fastening.
- Rolls MOB onto the deck at the top so no need to lift or drag.
- Makes use of windless or electric winch to provide mechanical advantage.
- Very easy storage, sleeping bag style stuff sack.

Disadvantages

- Time to setup too long, not intuitive so training needed.
- Difficult to orient for setup in a rush.
- Safety fence had to be cut, increasing risk and taking time and tools (Figure 28).
- Sail batons did not provide enough rigidity to prevent scrunching in the water.
- Extremely difficult to do from upwind side, due flapping to it blowing back towards you.
- Toggle holes uncommon on recently built yachts.
- No alternative to halyard for lifting, which would be in use if the sail were raised.

Background Information

A friend of Mr R Cowley, from Cardiff bay marina, was able to lend us an imitation of commercial slings that he had made for his boat. Being able to physically test its design gave me lots of insight into how it attached to the halyard (rope to top of the mast usually used for hoisting sails) and deck. Experiencing it's use also gave me understanding of how difficult it would be to use in rough conditions.



Figure 26



Figure 27



Figure 28

Key Features or Analysis

Toggles (Figure 28) allow it to be secured to the boat quickly.
Too much flexibility, being made from just a sheet of fabric benefits simplicity but has drawbacks of scrunching, not sinking in water and flapping in wind. Adding weight or rigidity to conventional sling designs would improve their functionality by allowing them to be used more easily in harsh conditions.

Latching Recovery Hook (Primary)

Background Information

Mr R Cowley's existing solution to be able to bring MOB's back on board, it consists of a stainless-steel carabiner style gated hook welded to a socket which fits onto a thin plastic pole. The metal hook would have a rope tied though it then act as an anchor point, securing the rope to life jacket straps or a buoyancy aid lifting loop.

Advantages

- Can be stored tied to railing for verry quick access (Figure 35).
- Simple and intuitive to use.
- Good at securing and manovering someone in the water.
- Push-fit of metal hook onto pole comes out when pulled allowing the pole to be discarded once a rope is attached to the person (Figure 33).



Figure 32

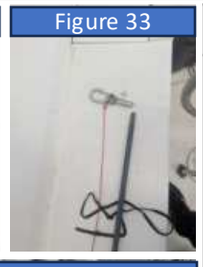


Figure 33



F 34



Figure 35

Disadvantages

- MOB must be wearing a lifejacket to have anywhere to clip onto.
- ABS pole quite short reducing reach and not strong enough to use as a lever for lifting.
- Being lifted by rope offers little spray, trauma protection against hull, or horizontal lifting
- The end of the hook stays attached to the MOB while they are being rescued.

Key Features or Analysis

Spring Loaded Latch ensures MOB stays secured. Durable materials – stainless steel (corrosion resistant) and ABS plastic (UV stable). Simple and intuitive design.

Evaluation of storage capability

Available Storage	Accessibility	Dimensions
Cockpit Under-seat Locker (Figure 30)	Good – Watertight, Could get lost/buried.	Hatch Limits size 0.5m x 1m
Tied to Guardrail (Figure 29)	Very Good – Visible and near waters edge.	Long thin objects only Less than 2m long
Under floor cavity (bilge) (Figure 31)	Poor - Must lift floorboards Potential for damp.	Large Area but small height .
Indoor stowage compartments.	Ok - Open easy to grab, but below deck.	Small volume 0.3 x 0.3 x 0.2m

Client Comments - "The actual space of storage isn't much of an issue, it's more the unusual shapes"
"I think rigid parts must fold or collapse to fit"

I gained a prior understanding of the available spaces to store a recovery device while it is not in use, as this is a factor which will heavily influence their design.
Many spaces are available but there is a compromise between how quickly they could be accessed in an emergency and the space available to store a device.



Figure 29

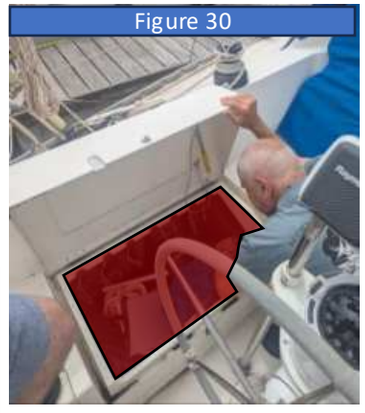


Figure 30



Figure 31

Refined Design Brief

I will design and manufacture a device capable of Securing and Lifting a Man Overboard back into my client's yacht. No assistance from the Man overboard should be necessary, so that an incapacitated MOB could be rescued successfully . Setup and rescue should be as fast as possible to prevent the onset of hypothermia. The device must make lifting the MOB easier by providing mechanical advantage or improved grip and enable the lifting of a casualty in a face up horizontal (or seated knees raised) position to a height of over 1.2m. To ergonomically fit a wide range of MOB's the device must have a width greater than 0.54m and a length if MOB held reclined of 1.365 – 1.80m. The device must be adaptable and modular to be able to suit a variety of situations, number of crew performing the rescue and weather conditions. The device must be able to fit a large range of sizes and weights of casualty wearing an inflated life jacket. It must be possible to fully unpack and ready the device in less than two minutes to avoid the MOB drifting away. I will design the device to be collapsable in some way to fit into the available storage spaces on my client's boat. The device must be durable enough to withstand marine conditions by being highly resistant to Corrosion, UV, Vibration and Temperature with infrequent maintenance/inspections. The device should be custom built to integrate effectively with features of my client's boat. It should not be unnecessarily complex so that it is simple to use in a stressful situation and to make teaching how to use it to crew/passengers quick and easy.

Design Specification Part 1

Area	Specification Point	Justification
Aesthetics	• (A1)The device should have an element which matches the traditional wooden style of my clients boat.	This is an aesthetic preference of the client and would help the device to integrate better visually with the boat when it is being stored.
	• (A2)Should have high visibility bright colours which contrast against each other different areas of the product.	Bright colours allow the device to be seen easily so that it can be found quickly and they would alert other vessels of the emergency situation. Contrasting colours for key parts will allow the device to be oriented and setup fast.
	• (A3)The device should have a modern, robust appearance, that inspires confidence.	Important so that people feel secure while testing or practicing using the device and are inclined to actually use it if it is required.
Manufacturing Process	• (P1)Make use of jigs or templates to produce duplicated parts.	This will increase the accuracy of the shape of parts to the design compared to repeated manual measurements, potentially benefiting reliability and aesthetics.
	• (P2)Plan effectively to produce high quality components within the time available.	Efficient allocation of time key to getting all parts produced to the highest possible standard within the available time.
	• (P3)Prototype designs with 3D printing and use materials efficiently to reduce waste.	Prototyping and an iterative design process will allow for testing of components and improvement without wasting expensive materials.
	• (P4)Produce parts that are dimensionally accurate with a good surface finish.	Good reliability and aesthetics are reliant on the parts being the correct sizes.
Size	• (Si1)The device must fit within one of the available storage spaces on my client's boat.	Collapsible or foldable components are likely to be required to reduce the size that the device takes up while not in use so it can fit inside storage spaces while not in use.
	• (Si2)Must have the width and length to rescue 95% of the population while keeping them comfortably supported.	Anthropometric data for height, shoulder width and weight researched must be considered to produce a device which is ergonomic and functional for the majority people.
	• (Si3)The product must fit features specific to my client's vessel during use, including being less than 1.8m long.	To integrate with or attach to my client's boat dimensions of its features must be accounted for. Especially the less than 1.8m length which is limited by the distance between guardrail posts.
Sustainability	• (Su1)Try to use sustainable materials that can be easily recycled.	Avoids causing pollution or landfill because the product can be disposed of in an environmentally friendly way.
	• (Su2)Avoid the use of adhesives or permanent mechanical fastenings between different materials.	This allows the product to be disassembled to be separated into individual materials for recycling or replacing of individual components.
	• (Su3)The device must have a high level of durability to increase its lifespan.	. A longer lifespan will mean that my client does not have to source a replacement as frequently so that energy and materials for a single device are used.

Design Specification Part 2

Area	Specification Point	Justification
Purpose and Function	• (F1)The Device must be able to safely lift a person weighing over 105 kg to a height of 1.2 m while they are wearing an inflated life jacket.	105kg is a 95% of the population + extra for wet gear, so covers most likely scenario. 1.2m is the distance from the waterline to the deck of my client's boat. Survey showed that nearly all sailors wear auto inflating lifejackets, so the volume they take up and their lifting loop must be factored into the design.
	• (F2)Must be capable of securing a person who is 1.5 meters from the topside without any assistance from the MOB.	Not requiring the MOB to help secure themself is a key requirement will differentiate it from safety equipment already available to my client and most sailors. Being able to rescue an incapacitated MOB will be the main purpose of the product.
	• (F3)It must provide assistance to the rescuer in lifting though mechanical advantage, distributing the load or giving better grip.	Directly lifting the weight of a person 1.2m out of the water in a stressful situation would be extremely difficult and winches are present on the boat so could be used to reduce the strength required. Improved grip or sharing the load between multiple crew is another option available.
	• (F4)Should be simple and intuitive to operate even in extreme waves/wind.	My client often sails with inexperienced people on board so being easy to use would allow them to use it with limited training. Use in extreme waves/wind was a shortfall of the sling I tested as it flapped and became difficult to use in a light breeze (See existing product research section).
Materials	• (M1)Materials must have very good resistance to UV light, with their colour and mechanical properties being unaffected after at least 3 months of 24/7 exposure.	To comply with SOLAS Chapter III guidance for durability of materials in live saving appliances. Likely to be left exposed to sunlight for long periods and material being weakened would compromise safety.
	• (M2)The device must only use materials with excellent corrosion resistance and not place metals with different galvanic potentials in contact with each other.	Salt water and damp storage environment increases likelihood of corrosion of metal components. See materials research for more information on galvanic corrosion.
	• (M3)Temperatures from -20 to 60 degrees Celsius must not affect the performance of the device.	The device must work reliably in the middle of winter and not be damaged by being left in direct sunlight.
	• (M4) Non-flammable or flame-retardant materials to be chosen.	Some possible storage areas are close to the engine, fuel tank or gas hob of the boat which could act as ignition sources. Smoke and toxic gas emissions produced by some polymers burning are also a danger.
	• (M5)Materials which have been certified to withstand required loads are to be used where possible, cost permitting.	Using certified materials is advised in ISO 15085 standard for re-boarding equipment and it will give greater confidence especially if very large safety margins are used.
Safety	• (S1)The device must minimise the risk of injury or falling into the water, for the person performing the rescue.	The device must not create additional risk of causing another MOB, by needing the rescuer to lean overboard, to avoid making a bad situation worse.
	• (S2)Should provide protection against splash and blunt force trauma to the MOB in the water.	These were identified in the medical/scientific background section as being the main risks to the MOB secondary to drowning. Providing some shelter to the person being rescued and reducing sharp edges & corners will also make testing/ practice drills with the device more comfortable.
	• (S3)It must be possible to fully unpack and setup the device for use in less than three minutes.	There is a limited time window to be able to rescue an MOB as they could drift away or symptoms of hypothermia can occur after a few minutes.
	• (S4)The product must lift the MOB face up with their knees level with their chest.	Vertical lifting increases the risk of heart/circulation problems as blood pools in the legs. See the orientation section of medical/scientific background for detail. Horizontal face up lifting is encouraged by the RYA and in SOLAS Chapter III.
Cost	• (Co1)The device should have total material costs of all components less than £150	In order to be affordable for recreational sailors as something that will be used extremely infrequently to supplement safety equipment they already own.
	• (Co2)The device should take less than 20 hours of labour to produce.	To have production completed in a reasonable timeframe and reduce labour costs if the device were to be made commercially.
Client	• (Cl1) The prototype should not cause discomfort to the MOB being rescued.	Being comfortable for the MOB being rescued will be an important feature as my client is likely to be deterred from practicing using the device with new passengers if in a low stake's setting, if it causes any pain.
	• (Cl2) Must be highly reliable and designed to have a long lifespan with minimal maintenance.	One of my client's key values was reliability, he hopes to be able to put it in storage and forget about It, but have it still work 100% of the time it is needed.

Design Idea 1 – Wooden Sling



Figure 36

Boat Interior Design

'Libra Lass' has a classic dark oiled Teak interior (Figure 36), a traditional style that is extremely common in older yachts. Teak is a tropical hardwood with a warm brown tone, historically selected by boat builders for its natural oils which resist moisture.

This design idea would aim to closely match the aesthetic features of my client's boat, primarily by using Teak planks with a similar finish and having rounded corners and slot shaped handles seen inside my client's boat. (A1)

Con: Hardwood construction could make the weight of the device unmanageable for a single person. Alternatively, Teek veneered softwood could be used though this would significantly reduce durability and all edges would need to be covered.

Con: Hardwood construction especially from Teak would be extremely expensive.

The cost of a single 125 mm wide, 20mm thick, 2m long FSC Certified (Su1)Teak board is approximately £40 (Co1). Reclaimed wood from old boats can be cheaper or a synthetic "composite" imitation Teak are other possible alternatives.

Constructed from eight hardwood planks (density greater than water, so that it sinks under MOB). Thin planks to minimise extra weight and conform to shape of the cockpit benches while secured. Durable lacquer or varnish coating for smoothness and good rot resistance. (M3)

Con: Being Immersed in water even with protection measures will likely cause some damage to wood.

Toggles reduce compatibility with other boats that do not have a slotted rail (Figure 42) on their deck edge but are a lightweight fast (S3) method of strongly securing the sling to the side of the boat. These would have high visibility brightly coloured nylon strap going though a milled high strength polymer or aluminium rectangular toggle.



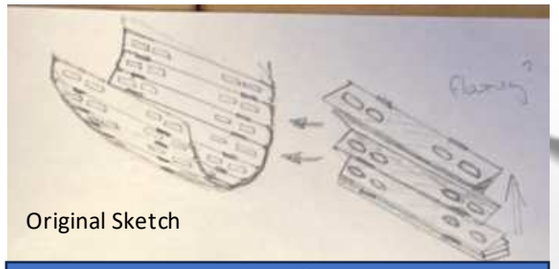
Figure 42

T joint between two perpendicular aluminium pipes TIG welded. Attaches to wood by fitting into round socket cut with Forstner Drill bit.



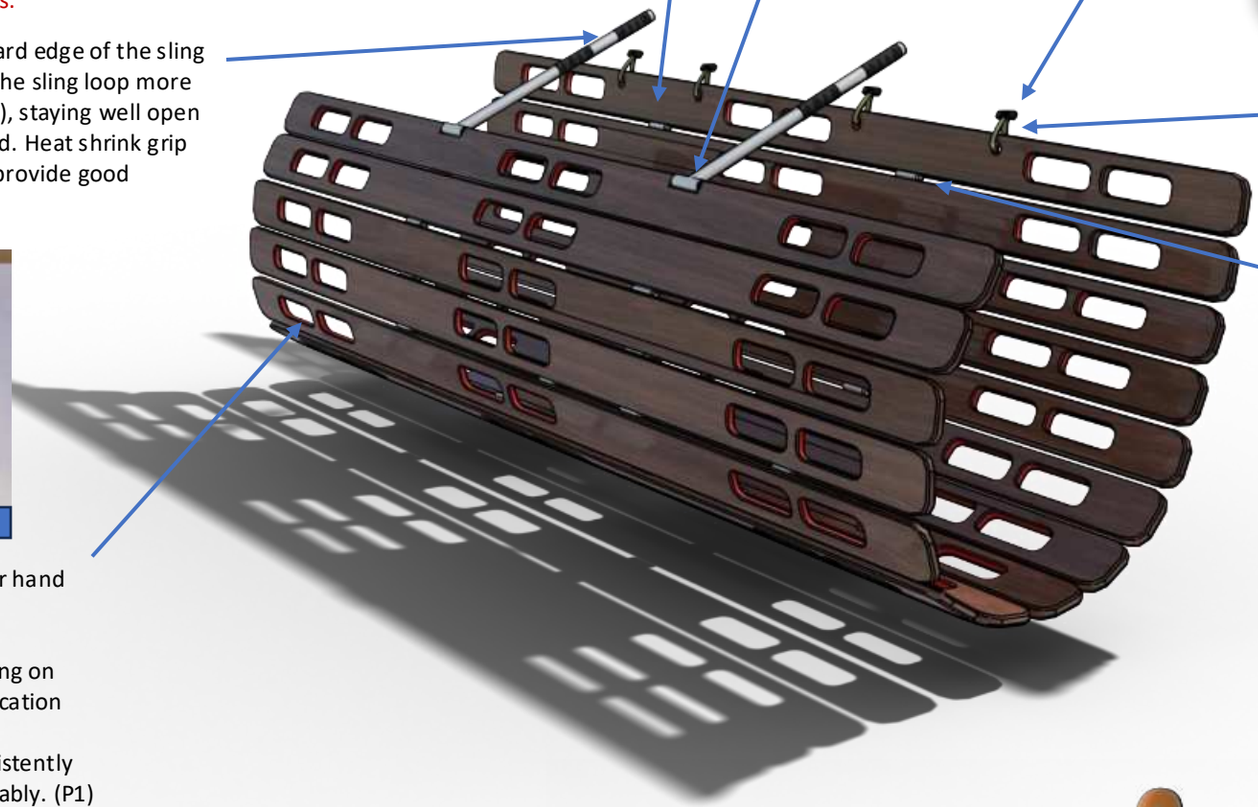
Toggles must be made from a material strong enough to hold weight of a person that does not scratch the deck edge rail. Attached by a polyester or nylon webbing rated to have more than enough strength. (M5)

Aluminium poles attached to the furthest outboard edge of the sling will allow for greater reach and control to open the sling loop more to provide a big target to slide the MOB onto (F2), staying well open was one of the main shortfalls of the sling I tested. Heat shrink grip tube over 50 mm diameter Aluminium pipe will provide good strength and grip to handles.



Original Sketch

Figure 37



Rows of hand shaped cutouts allow for hand over hand "climbing" to gradually lift the MOB aboard.

Three sets of cut out hand holds, allowing for simultaneous use by 1-3 people at once depending on availability to make lifting easier, meeting specification point (F3)(F1) by distributing the load.

A simple router jig could be used to make a consistently shaped handle cutout and chamfer edges repeatably. (P1)

Planks secured together with two Stainless Steel (for corrosion resistance (M2)) hinges (Figure 40) per plank with 360 degrees of movement range. Small low-profile hinges to be used to avoid digging into the MOB (Cl1).



Figure 40



Single Bench Cover

Figure 41



Fitted to act as a permanently on, wooden seating cover on the benches in the cockpit. Secured down with clips or elastic straps.

Aesthetically similar to decking style seat covers installed on some yachts.

Allows for extremely fast accessibility and accommodates long rigid length, because it does not need to fit into storage space.

Separate to the toggles which attach the sling to the deck, toggles or carabiner clips in the middle would allow it to be split into two pieces, one to cover each bench labelled.

Each half of the sling. (8 planks) could be secured to the benches with 3D printed Clips that would snap into the oval/slot shaped hand holds at each corner.

Client's Thoughts
" Making bench covers that fit together to be a sling an interesting idea, I would not mind them if they fit the seats well"
"I think that much wood could be quite heavy and handle holds cut in line with the wood grain may not be that strong depending on the wood"



Figure 38

Rescue In Progress Render

Halves lifting force needed by acting as a parbuckle similar to other slings tested but has much improved rigidity due to being solid rather than fabric. This prevents it twisting in the water or flapping in the wind.

Con: Does not use methods of providing mechanical advantage for lifting built into the boat such as windlass winches . (F3)



Seat Configuration

Figure 39

Design Idea 2 – Folding Stretcher

This design Idea is a flat stretcher with accordion style sides that are stored concertinaed then extended during and propped open by length ways aluminium poles that hold it open during use.
This creates a 'bed' that the MOB can be moved onto and lifted by the halyard from the top of the mast.
The concertina mechanism works by having different shaped quadrilateral panels which unfold into the correct shape.
The design was loosely inspired by origami shapes creating 3D objects from flat panels to save storage space.



Figure 43

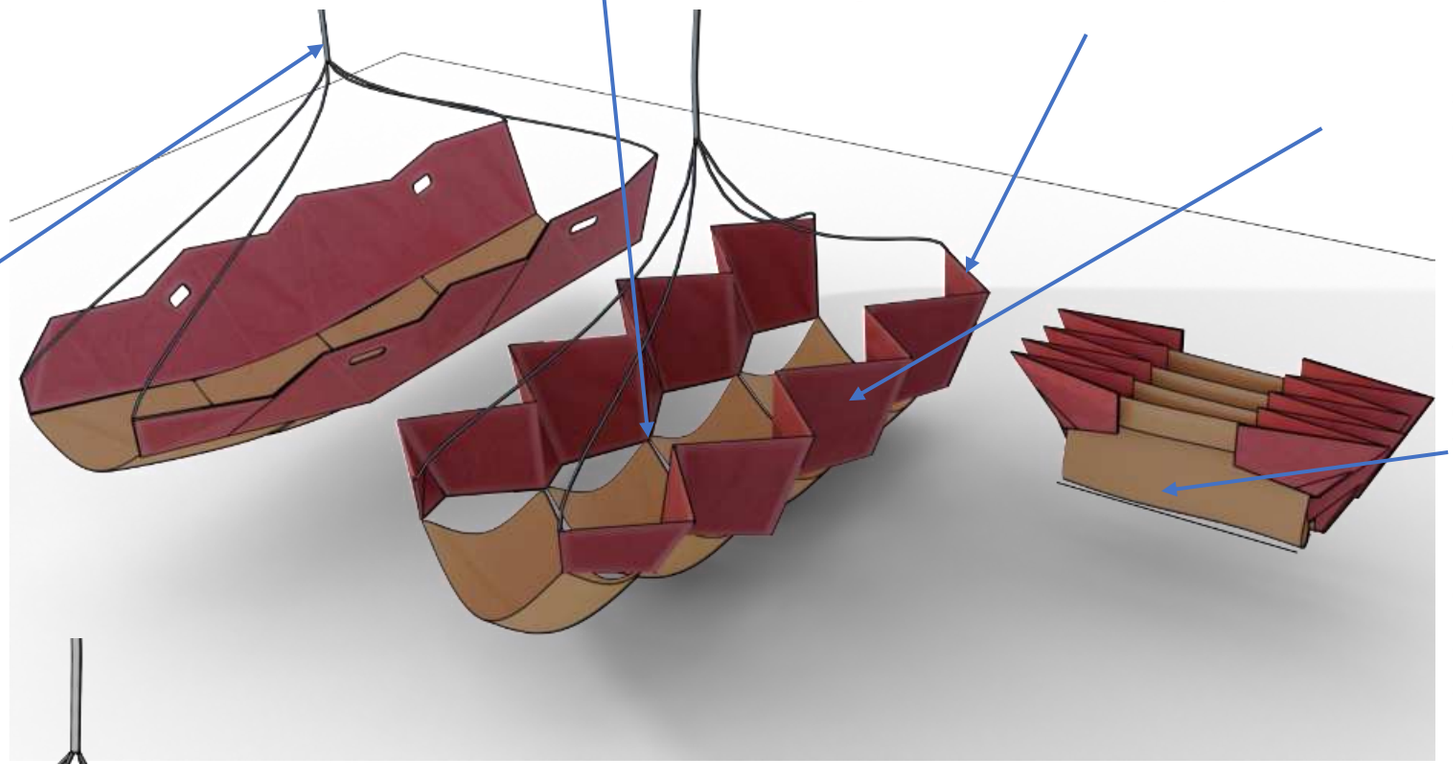
Since the only attachment point is the halyard, there is no limit to how far MOB can be lifted unlike other design ideas. This gives it adaptability to be used on different boats with a larger topside. This would also allow for lifting higher, over the guardrail, compared to slings with one side attached to the deck edge (F1).
Disadvantages of this are that to support each corner of the stretcher the halyard must split into four smaller ropes which creates a tangling hazard and would make it more difficult to get the MOB in.
No mechanical advantage (F3) with lifting is created as it is not a parbuckle design.

Another issue could be meeting the strength requirement (F1) of 105kg, the bed area fabric loops only being attached at the corners creates many weak points where this design idea could fail.

The main disadvantage with this design idea is that it does not help with securing the MOB (F2) - only lifting them out of the water. My clients only option to grab hold of a MOB would by hand, risking being pulled overboard

Client's Thoughts
"I like the idea of it being compatible with basically any boat I sail on because I sometimes charter boats abroad and would be able to bring it with me" "I'm not sure about how well that folding would work" "The jagged sides could catch on and scratch the hull of the boat as its lifted up"

Sides (coloured red (A2)) produced by laser cutting correctly shaped panels from wood or plastic then creating joints between them by fully encapsulating panels in a layer of heavy waterproof fabric that also acts as hinges between panels.
This makes it more flexible and corrosion resistant than exposed metal hinges, while also reducing complexity.
Con: Even though folding sides allow for easy storage and could be setup in a time efficient way by pulling in a single direction, their visual complexity could confuse new users during an emergency.



Panels shapes could be exported from CAD and laser cut (P4) in either Acrylic plastic or plywood sheet, so that their shape is accurate and the device folds as expected.
Con: Plywood panels would be absorbent if the outer fabric leaked. Acrylic panels are brittle and could shatter if dropped. This would reduce the durability (Su1)(M2) of the device - a key need of my client.

The textile used to encapsulate the panels could be woven CORDURA Nylon Fabric with a polyurethane coating to provide a waterproof barrier. This would completely surround rigid panels and held together with stitching and glued seams. This is the fabric used to make drysuits and high-end waterproof backpacks, so it is flexible but rugged enough to resist abrasion, tearing and repeated bending at "hinge" points between panels.
Con: A large amount will be needed, and it has a high cost per M^2.
Working with, cutting and sewing this material with standard equipment will be a challenge,(Co2).

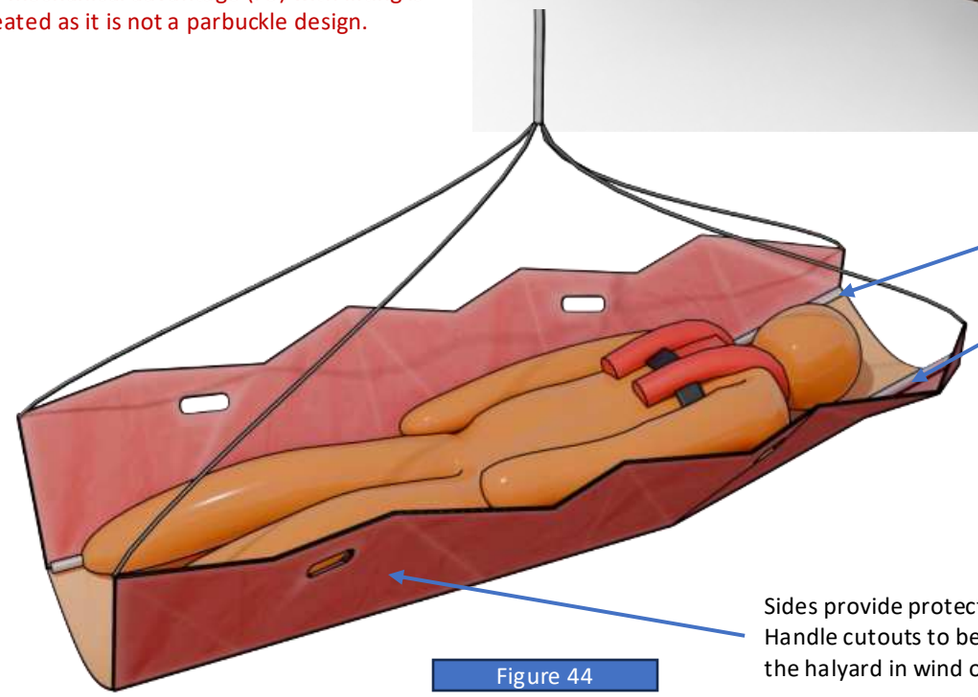


Figure 44

Rescue In Progress Render

Aluminium Poles running the length of the stretcher would hook in at each end to hold it taught and prevent it from collapsing. These would have a slight bend to lift the head, so the MOB is not lying perfectly flat in the stretcher, which could help prop their head above the water and provide some splash protection (S2).

Sides provide protection against splash and spray and prevent the MOB falling.
Handle cutouts to be used by rescuer to control the device and prevent swinging from the halyard in wind or waves.

The size of the stretcher is reduced for storage by removing the aluminum poles and pushing each end towards each other (Si1).
This would cause side panels to create a zigzag as they collapse and stack on top of each other. With the fabric bed area folded between them.

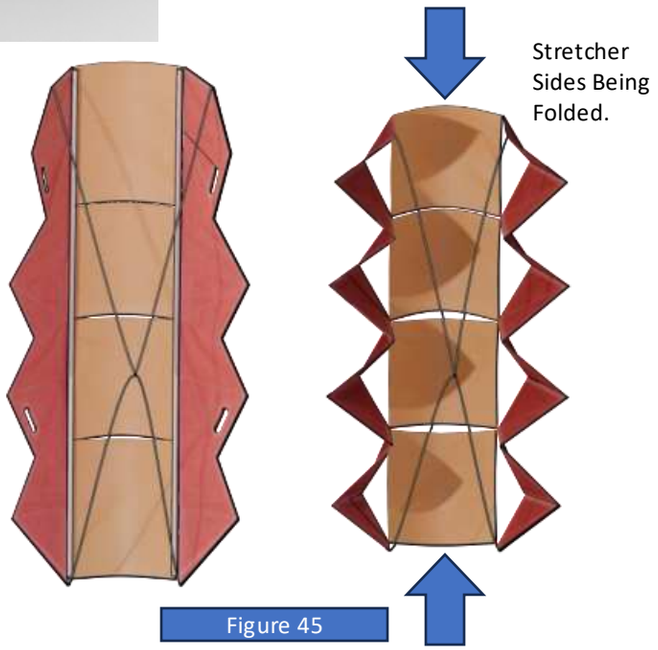


Figure 45

Design Idea 3 – Scoop Catcher



Figure 46

Design based on improving the C-Hero MOB VR-12 Rescue Pole (page 9).
Incorporates a similar square frame of aluminium tubing attached to a long arm for reaching MOB, this combines the functions of securing and lifting the MOB into a single device, potentially saving time (S3).

Con: This design idea would be able to secure an MOB at a long distance effectively but does not provide much help with lifting. Ropes could be tied to each corner of the square frame then the MOB hosted up seated with the halyard, but this would be time consuming and unstable as the MOB and chair part could swing or twist.

Figure 47

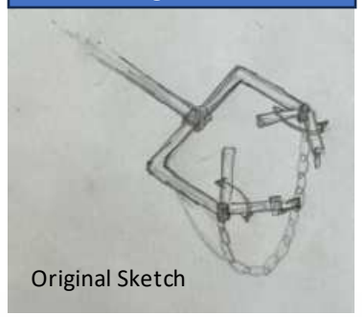




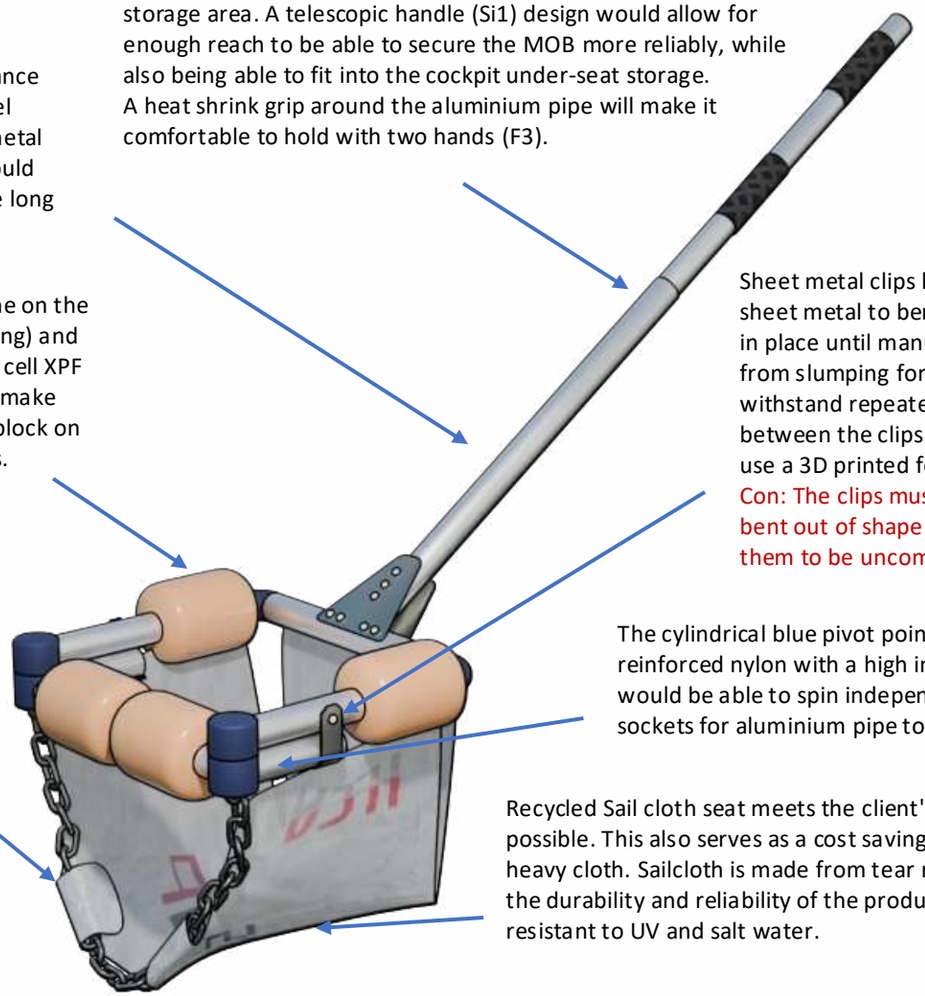
Figure 48

Collapsable telescopic handle to allow it to fit into under seat storage area. A telescopic handle (Si1) design would allow for enough reach to be able to secure the MOB more reliably, while also being able to fit into the cockpit under-seat storage.
A heat shrink grip around the aluminium pipe will make it comfortable to hold with two hands (F3).

Lightweight aluminium pipe construction would offer good corrosion resistance and decent strength. The only area where reinforcement with stainless steel plates could be needed is where the handle meets the square frame. Two metal plates with bolts though combined with tig welding the aluminium pipes would create a strong joint. This is required because of the leverage created by the long handle and the weight of the MOB on either side (F3).

Hot wire cut foam floats provide buoyancy (which would help keep the frame on the surface of the water, helping with positioning and reducing fatigue from lifting) and padding (Cl1)(S2) to avoid causing injury. These would be made from closed cell XPF foam, because it is non-absorbent and inexpensive. It would be possible to make them using hot wire cutting with the correct profile, or gradually sanding a block on a lathe then coring out the centre with a hole was so they fit on to the pipes.

A recycled sailcloth 'chair' (Su1) weighted to sink under the MOB with a chain at the front. This gives improved support while lifting upwards with a halyard, lifting knees higher to prevent to reduce shock (S4), compared to the chest loop on the C-Hero. This allows the device to be used to scoop up the MOB more quickly, safely and comfortably.



Sheet metal clips lock the front closed around MOB. They work using the flexibility of the sheet metal to bend enough to allow the pipes labelled orange to push into them but be held in place until manually released. This increases safety while lifting preventing the casualty from slumping forward and falling out of the chair. Stainless steel would be required to withstand repeated bending. To avoid galvanic corrosion, a plastic washer would be used between the clips and aluminium pipe. It would be possible to cut out the flat shape out then use a 3D printed former to get both clips the same shape so they work reliably.
Con: The clips must be thin and flexible enough to bend and snap around pipes so could be bent out of shape easily. Also being made from a thin sheet of stainless steel would cause them to be uncomfortable if they pressed against an MOB's torso.

The cylindrical blue pivot points would be 3D printed from a high-performance plastic such as CF reinforced nylon with a high infill and thick walls. The pivot comes from two stacked cylinders which would be able to spin independently, due to a print in place axle (Figure 49). Cylinders would have sockets for aluminium pipe to fit into.

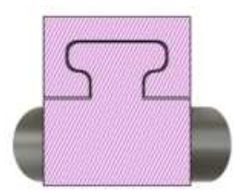
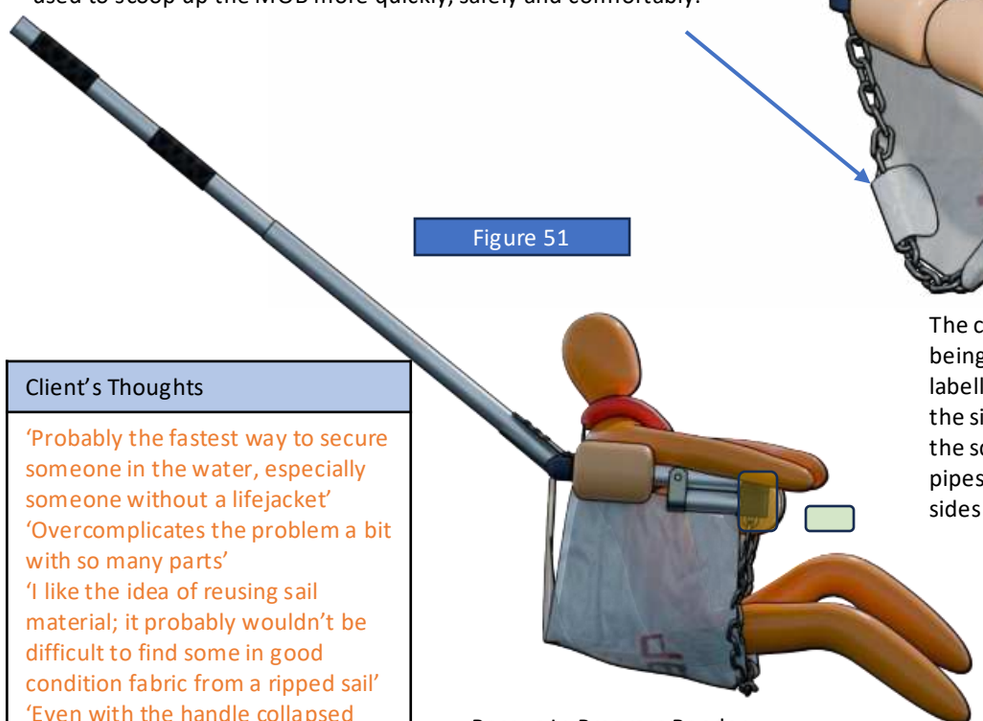


Figure 49

Recycled Sail cloth seat meets the client's goals of making sustainable choices where possible. This also serves as a cost saving measure (Co1) as it is a free source of durable heavy cloth. Sailcloth is made from tear resistant fabrics and has frequent seams increasing the durability and reliability of the product. Sailcloth will also have been designed to be resistant to UV and salt water.

Figure 51



Rescue In Progress Render
Shows how the MOB would sit in the sailcloth chair.

The closing/locking mechanism for the front side works by the area labelled in orange being pushed against the MOB, this causes the L shaped gates to pivot around the area labelled in green (On Figure 50 and 51). As the orange area is pushed backwards and too the side the perpendicular parts with floats on them rotate inwards to close the front of the square frame around the person. To hold the front side closed the orange labelled pipes are pressed into flexible clips (labelled in yellow)which attach them to the frame sides preventing movement in the attached front arms.

Con: This design idea has many complicated moving parts (Co2)(F4), the pivoting gates and telescopic handle make it a complicated system. This could compromise some of my clients key values of reliability and simplicity. Having many small mechanical components increases the possibility of something failing and also could make using the device unintuitive.

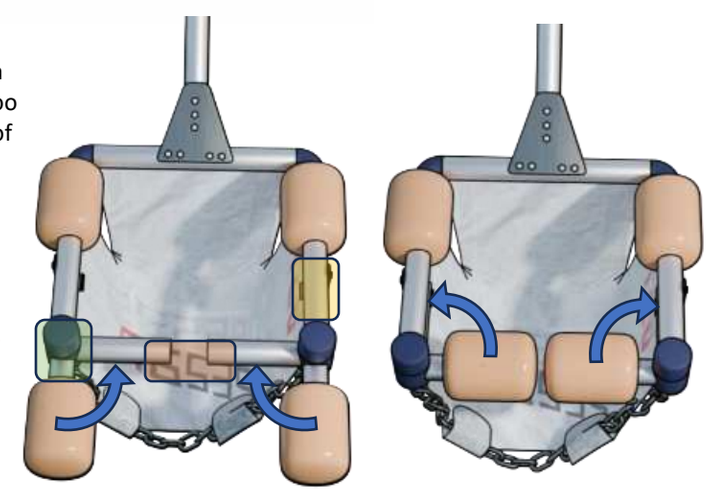


Figure 50

Client's Thoughts

'Probably the fastest way to secure someone in the water, especially someone without a lifejacket'
'Overcomplicates the problem a bit with so many parts'
'I like the idea of reusing sail material; it probably wouldn't be difficult to find some in good condition fabric from a ripped sail'
'Even with the handle collapsed down it would take up quite a lot of space'

Design Idea 4 – Sling and Rope Threader

The sling would be made from a high visibility mesh fabric (A2) to reduce its weight and allow water to drain out the bottom of the sling as it is lifted out of the water. A bonded sailcloth type white fabric could be used to cover the edges, prevent fraying and provide extra strength.
Con: Use of modern materials like rubber, metal handles and yellow fabric fulfils the goal of the device having a reassuring look but does not feature any wooden parts to match the aesthetic of my client's boat (A3).

Adding small aluminium (for corrosion resistance) handles made from bent pipe with plastic or wooden circles where they interface with the fabric to keep them pointed outwards, so they are easy to grab. These handles light enough to not be a hazard hitting people and not prevent the sling being rolled. Grip could be improved with a rubber sleeve or tape. Con: These handles would only be supported by fabric so could cause ripping if the join between fabric and handles does not distribute the weight effectively.
The handles provide an alternative to the halyard allowing rescue by hand with multiple people pulling the handles (F3) up and changing their grip between the handles in series. This would allow it to be deployed and used quickly without having to spend time taking down a sail and attaching a halyard. Handles could also be used as a ladder by an uninjured MOB.

Con: natural rubber floats in water so small steel weights would be needed to cause it to sink below an MOB. These could cause uneven sinking, and they would need to be securely attached to the sling with bolts or hooks (F4).



Figure 52

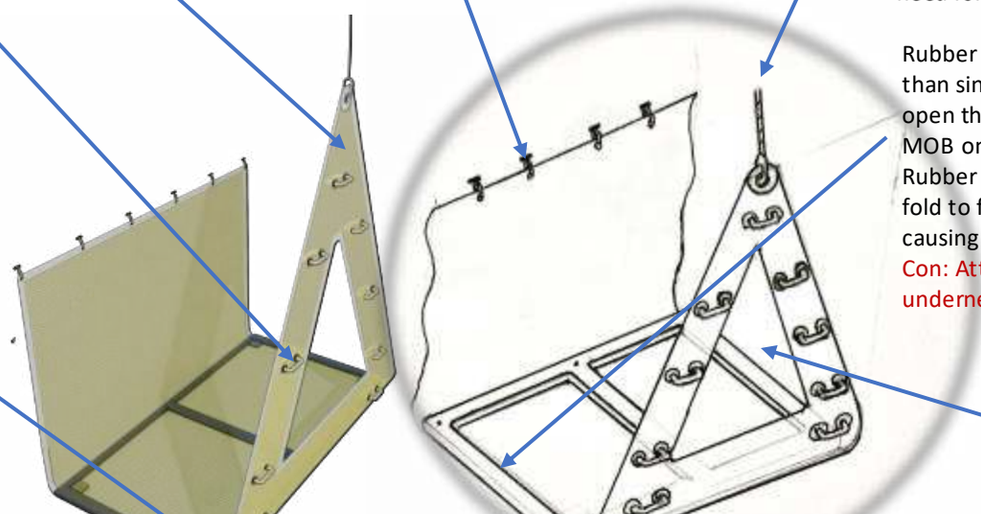
The Image above is of a Bouy Catcher, the device which inspired the threader mechanism. These are pushed against the top of a buoy and thread a rope though an eye for mooring. My mechanism is similar but angled at 90 degrees to the handle/pole so it can be slotted sideways underneath a lifejacket strap or into a lifting loop without having to push against anything.

Understanding how the threading mechanism works is difficult without seeing it working, the link below is to a video of a buoy catcher being used.

https://www.youtube.com/watch?v=g_eKeDgCPgU

Toggles (similar to design idea 1) are the fastest method of attaching sling edge to deck edge rail.
Toggles could be machined from aluminium bar or a strong plastic, with a slot milled though the middle to allow a strap of nylon/polyester webbing to be looped through.
Con: Toggle length cannot be adjusted to create tension to keep the sling edge taught, this could create a gap between the sling and deck edge (Si3).

Rescue Sling – Lifting MOB



Simple design without unnecessary deviation from products already on the market, makes small improvements to the function of a sling based on the problems found in my research.
Being a parbuckle sling with one side fixed fulfils the mechanical advantage criteria in the specification (F3). Also, optional use of the halyard with onboard winches would eliminate the need for strength to lift the MOB.

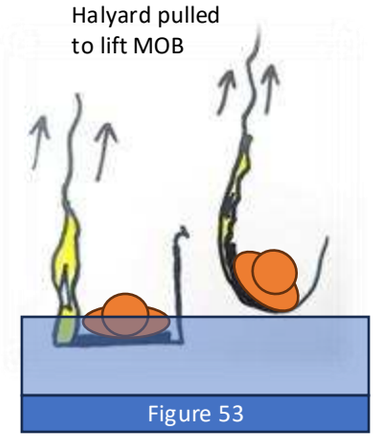


Figure 53

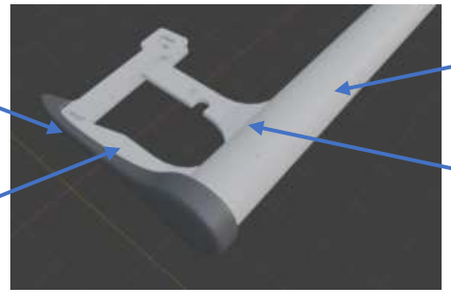
Rubber strips would reinforce some parts of the sling to create a flat "bed" area than sinks under the MOB (S4). This would hold the bottom of the sling much more open than the products I researched, creating a larger target (Si2) to position the MOB onto.
Rubber is needed as opposed to a more rigid material as the device must be able to fold to fit into available storage spaces (Si1). Also, soft rubber has less chance of causing injury to the MOB or damaging the boat if it is bounced around on waves.
Con: Attaching rubber strips on top of the sling could cause the MOB to get caught underneath them.

Triangular cutout in the furthest out side prevents fabric being blown back up over the side of the boat (F4) by reducing the vertical area. This should make lowering the sling into the water easier and prevent the MOB being covered by the sling side. The cutout could also improve visibility from the deck acting as a window so that the sling cannot block a MOB from view.

Threader Mechanism – Securing MOB

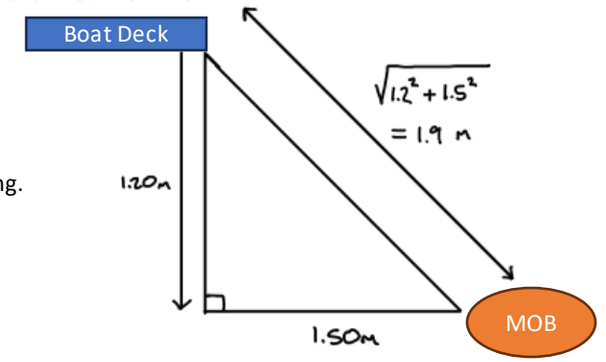
This design idea splits up the tasks of securing and lifting the MOB, to create a more modular adaptable (F4) system and avoid creating a single overcomplicated device. The MOB would be first secured by attaching a rope to them then lifted with the Sling.
Con: having to locate two components of a system in an emergency could take more time than one (S3).

A 3D printed TPU (flexible filament) rubber buffer/cap on the end of the device could act as padding to reduce injury to MOB from impacts from the device compared to bare aluminium (S2).
Mechanism made from 3mm thin sheet aluminium so parts can slide over and around each other.



Corrosion resistant aluminium broom handle could be used to utilise reused parts.

Exactly how the thin mechanism joins to the handle to be found during development.



The diagram above was used to work out the diagonal length the handle would need to be to meet the specification requirement of being able to secure a MOB who is 1.5m from the side of the boat (F2) horizontally. Pythagoras theorem and the height between the boat deck and waterline were used.

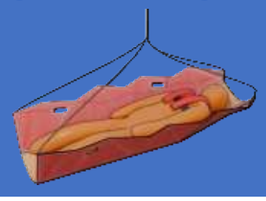
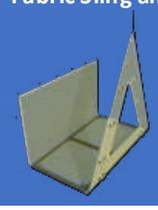
The sequence of images I produced below show the stages of how a Rope would be threaded behind a buoyancy aid strap. First the mechanism is pushed to the left so the strap is guided into a gate, pushing past the latch which rotates around the first pivot point. Next the mechanism is pulled right and backwards so the strap puts pressure on the latch in such a way so that it rotates around the second pivot point. The rope and mechanism are then retracted back towards the user once the rope has been looped around the strap.
Con: this method of securing only works if the MOB is wearing a lifejacket with easily accessible straps/loops



Client's thoughts

" Separating it into a two-part system seems like a good idea so that each component could be used individually"
"Getting that threading mechanism working reliably could be very difficult given its size with moving parts"
"This sling design seems to be the easiest to store, because it would completely fold up"

Evaluation Of Design Ideas

Criteria	Design Idea 1 - Wooden Sling	Design Idea 2 – Folding Stretcher	Design Idea 3 – Scoop Catcher	Design Idea 4 – Fabric Sling and Rope Threader
				
Key Advantages	Being stored as seat covers would mean it is instantly accessible. Very good rigidity and strength. Allows multiple people to help lift MOB. Aprox size of 1.5m x 0.7m fits space envelope.	Compatible with any boat and can lift an unlimited height. Compromise between fabric and solid construction provides cushioning and strength. High sides have good splash protection. Aprox size of 1.8m x 0.54m fits space envelope.	Best at securing an MOB at a the furthest from the boat distance without needing a lifejacket. Makes use of sustainable reused materials.	Lightweight and easy to fold up to pack away. Triangular cut-out reduces adverse effect of wind and waves. Toggles allow it to be setup and attached quickly. Modular two-part system, with Rope Threader. Aprox size of 1.5m x 0.65m fits space envelope
Specification Points Not Met	(S2) – Wooden construction gives no padding (Co1) – Hardwood too expensive unless reclaimed wood used. (M2) – Wood can rot if left damp (M4) – Wood is flammable (Cl1) – Would be uncomfortable to lie during training with the device. (Su1) – Some hardwood not sustainably grown.	(F4) – In strong winds the stretcher could swing or spin being difficult to control. (F4) – Not intuitive/easy to operate with many folded different shaped panels. (F1) – May not meet strength requirement due to flimsy joints between parts.	(Si1) – even with handle collapsed would take up significant space (F4) – Not intuitive/easy to operate (S1) – Rescuer could become off balanced holding heavy weight of square frame far from deck edge. (M1) – most foams are not temperature and UV resistant. (Co1) – Very complex to build within 20 hrs	(F1) – rope threader only works if wearing lifejacket. (A1) – No traditional wooden elements. (M4) – Polyester Fabrics are less flame resistant than some alternatives.
Immediate Design Changes Necessary (Development)	• Lowering the cost of the device by using an alternative imitation of Teak wood. • Reducing the weight of the device to make quick setup possible. This could be done by making hand holds larger, so more wood is removed. • Add a surface finish to protect the wood when it is immersed in water. A thick coat of resin or varnish could be enough to prevent rot. • Allowing boat winches and halyards to be attached for mechanical advantage.	• Adding handles at the head and foot to make it clearer which direction to pull and unfold the device. • Adding vertical fabric covered wooden panel side wall blocking the head to protect MOB against splash and waves travelling parallel to the boat. • Some way of preventing the MOB being tangled in the ropes that split from the halyard to secure each corner.	• Adding some built-in attachment points for ropes to make lifting the 'chair' easier. • Replacing bent sheet metal clips with another method of holding the front side of the square frame closed. Greater durability needed as sheet metal clips bending are likely to fatigue after many uses. • Reducing the overall complexity of the design idea, too many individual moving parts. • The posture of the MOB reclined more by tilting chair backwards.	• Removing handles built into the fabric, upon further consideration even though they provide another option for lifting without the halyard, they would most likely tear the fabric they are anchored to. • Thicker rubber strips to provide rigidity, thin ones on drawing would not be stiff enough. • Using a type of rubber with a density greater than water to avoid needing steel weights reducing part count. • Sandwich the rubber between layers of fabric.
Approximate Cost	40mm Diameter Aluminium Pipe 1.5m: £10.00 Stainless Steel Hinges x 14: £38.00 FSC Certified Teak Planks x 16: £640.00 Toggle Plastic and Webbing: £5.00 Heat shrink textured grips: £2.00 Total: £693.00 (much greater than £150 goal)	Plywood Pannels to be encapsulated in fabric one standard sized sheet should provide enough area. 9mm Plywood: £20.00 Waterproof Cordua Nylon Fabric 8m^2 : £140.00 Waterproof Adhesive and Thread: £5.00 Total: £165.00 (slightly greater than £150 goal)	40mm Diameter Aluminium Pipe 4m: £40.00 XPF Foam: £15.00 M8 Nuts & Bolts x4 : £4.00 Heat shrink textured grips: £2.00 Stainless Steel Plates x2 0.2m squared: £25.00 Nylon CF High Strength Filament: £12.00 Sail Cloth and Weight Chain Free if repurposed. Total: £96.00 (less than £150 goal)	Yellow Mesh Fabric 5m x 1.5m : £65.00 White Edge Tape 10m: £15.00 Webbing 3m (Strength Rated): £3.00 Toggle Plastic/Metal: Approximately £5.00 Aluminium Sheet 3mm thick : £5.00 Rubber Strip 15 cm wide: £55.00 Handle Free if repurposed from a Broom Total: £140.00 (less than £150 goal)
My Opinion **(Clients opinion on respective pages)**	Lowering the cost enough to be feasible by finding cheap wood would be a significant compromise. Wood could be cumbersome and heavy to setup in the required time. Some maintenance of numerous hinge joints would be needed.	Tries to compromise between soft and hard options, neither optimal. I don't think I have proficiency to sew panels inside fabric with complex shape to produce a product with satisfactory quality.	Too focused on securing MOB, while making lifting difficult and unergonomic. I thought it was a promising concept, just being able to push the square frame against any MOB and have it lock around them, but execution complicated.	Addressing Lifting and Securing with separate components allows both to be done well without compromises. The threading mechanism will need significant development, testing and precision manufacturing to work reliably.
Rank + Justification	2nd – Inventive solution but too heavy and markedly over budget. Too many changes needed to make a useable design.	3rd – Does not integrate with my client's boat well and would be too difficult to manufacture. My client would prefer a more standard sling design. Does not really help secure a MOB	4th - Not feasible to be produced within the timeframe available. Overcomplicated design with too many failure points. Slick securing, but poor lifting function	1st – Best all-rounder for all aspects of a rescue. Fewest number of specification points not met. (A1) failing not a top priority for my client. Only Minor changes needed to make a useable design.

Conclusion: Based on evaluation of each of the design Ideas the **Fabric Sling and Rope Threader** was selected. Immediate design changes suggested have been added to the design as implemented part of initial development. Over the next few slides each component of the systems will be iteratively improved before the materials and design are finalised.

Development 1 – Threader Mechanism Shape

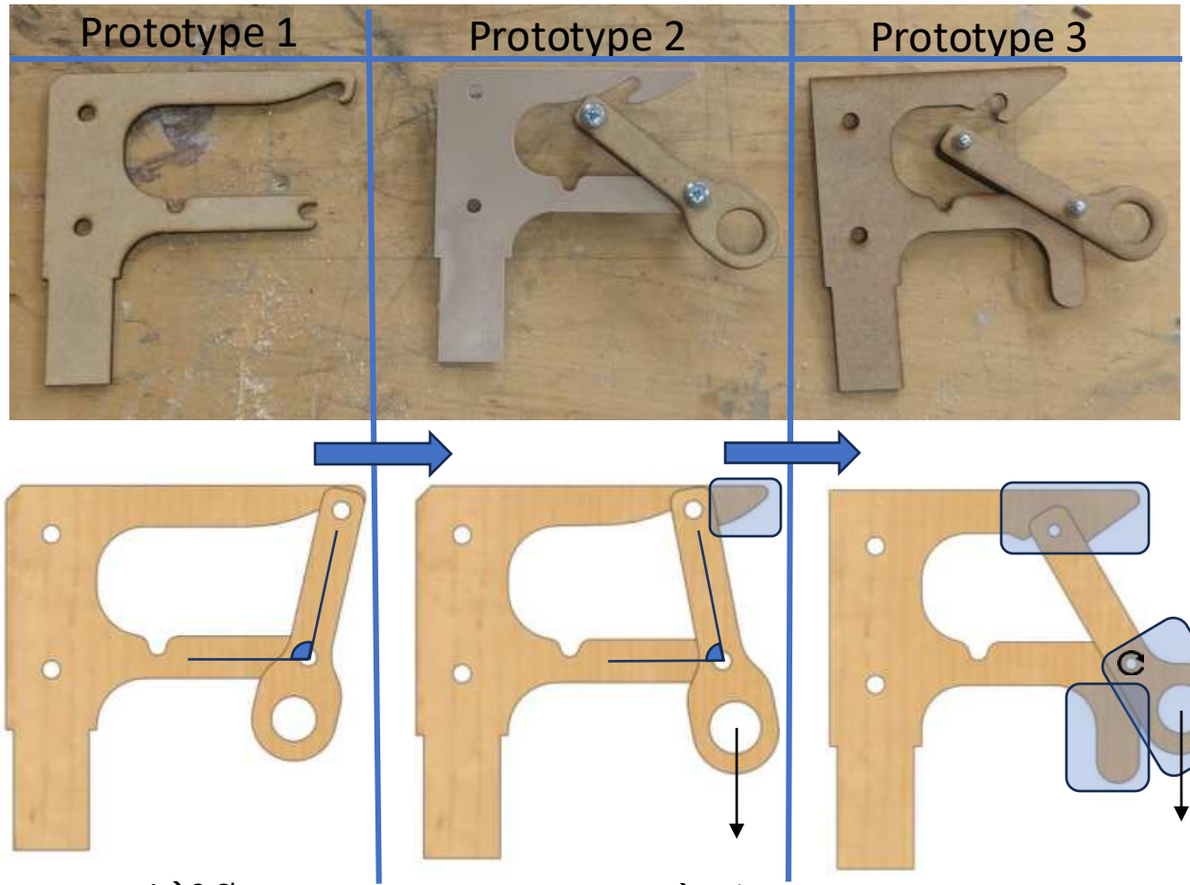
Prototyping Process

Laser cutting (Figure 53) was used to cut the 2D shape of the threading mechanism out of 3mm MDF. This allowed its operation to be physically tested, and changes quickly made for the next iteration. Judging how the latch would move between two pivots and the clearances needed between the gate and latch were difficult to judge in CAD. Making prototypes from a weak material also allowed me to see where the final aluminium design would also be likely to snap.

Threader Mechanism Background

The changes made during prototyping improved the reliability of the threader mechanism. The initial design took direct inspiration from buoy catchers that thread rope in a similar way but was rotated by 90 degrees so that instead of pushing against a solid buoy it could be slotted sideways underneath a life jacket strap. The design idea's mechanism was more suited to the angle found in buoy catchers with the latch perpendicular to the pole. The changes made between prototype iterations listed below are mostly tweaks to get the mechanism to work in the required orientation.

The only major change was the addition of a hook shape. This was to be able to move an MOB about in the water after they have been secured by attaching a rope to their lifejacket. This also gives an alternative if there is no easily accessible place to attach a rope to the MOB,



1→2 Changes

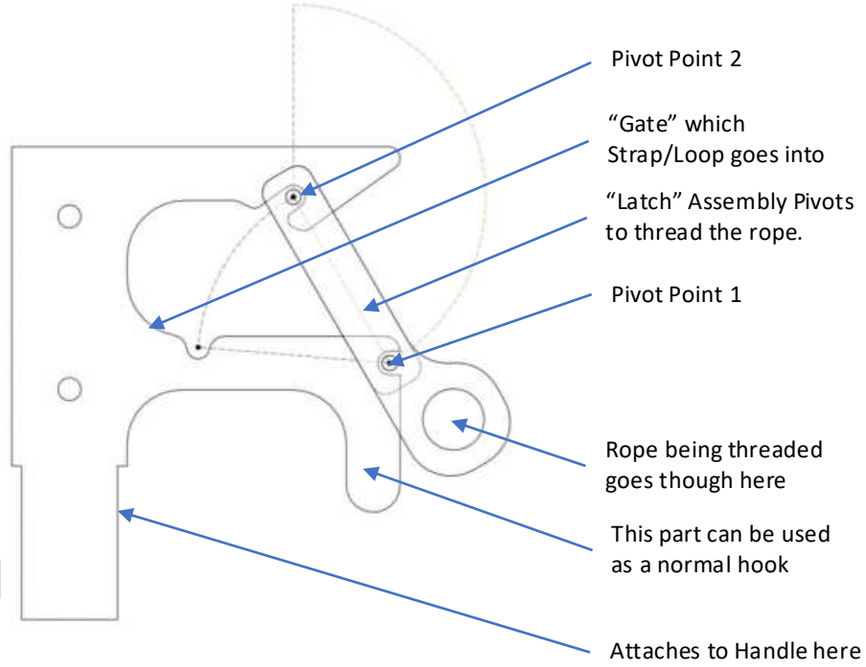
- The angle that the "latch" rests at in the closed position (when tension is kept on the rope) changed so that the amount of rotation between positions is reduced. This helps increase reliability as it does not have to be pushed as far by the strap that is being fed into the gate.
- A wedge shape added to the top side of the gate so that it can be slotted under a lifejacket strap or into lifting loop more easily.

2→3 Changes

- A large conventional hook shape added to the lower side of the gate mechanism to be used as a backup to move an MOB or secure them if threading/attaching rope is unsuccessful.
- Large fillets on hook end eliminate sharp corners and reduce chance of causing injury to MOB.
- The shape the upper side of the gate modified be thicker as this area snapped on prototype 2. Complex curved shape also slightly simplified for easier manufacturing.
- Overall length of gate sides decreased, and width increased for better strength and durability.
- The shape of the latch modified to move the hole which rope passes though further to the right of the pivot point so that it pulls the latch closed more effectively and is out of the way of the hook part added.



Figure 53



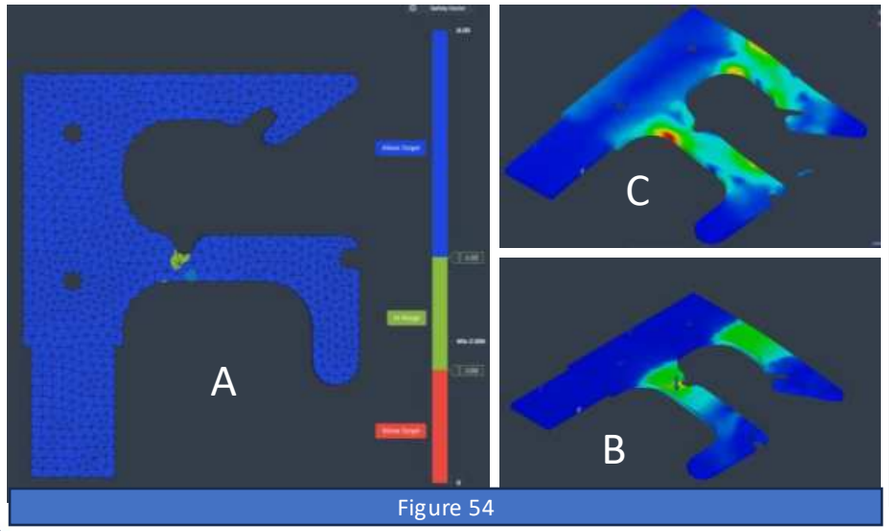


Figure 54

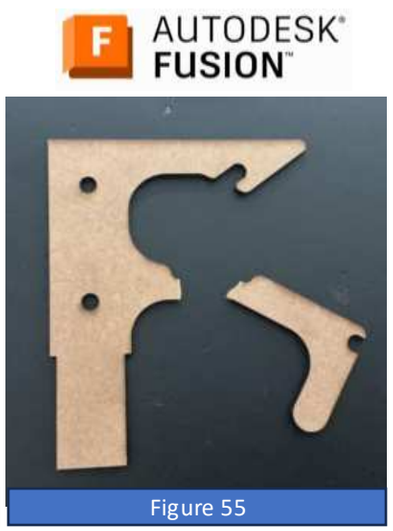


Figure 55

Finite Element Analysis

Autodesk Fusion's native FEA tools used to simulate applying loads in various directions to the part (Figure 54).

Simulations B and C show where stress concentrations occur on the part when forces are applied downwards or as if the hook is dragging an MOB. These show me the weakest areas of the part and where bending may occur. The red and yellow areas informed me that the gate sides were weak points. My prototype broke (Figure 55) during testing accidentally in the same location highlighted in the simulations, showing that they are an accurate representation of how the part would behave.

Simulation A was helpful in aiding with my choice of material. It shows that 3mm thick aluminium should have the strength to withstand a 300N (30kg) upwards force on the hook with minimal deformation even in weaker areas and a large safety factor. 300N was used as an estimation of the expected load on the hook moving a person though water without lifting them. This was very helpful as I was unsure if stainless steel or greater thicknesses would be necessary. Having no deformation in this direction is critical because it would move the three notches that the latch moves between, potentially jamming the mechanism.

Development 2 – Sling Materials

Rubber Selection

While designing the sling I considered a variety of rubber/flexible plastic materials. As mentioned in the design idea thick rubber sheet/strips are to be sandwiched between two layers of fabric with the purpose of stiffening the device to create a wide flat bed for the MOB to be slid onto. Also, the rubber must be heavy to prevent flapping in wind and to sink to below the water level.
Options I looked at were, neoprene, nitrile rubber and EPDM (Ethylene Propylene Diene Monomer). I ruled out neoprene due to its high cost and being too flexible and nitrile rubber because although it had the highest density it can crack and harden when outdoors. EPDM seemed to be the most well suited to my application as it is commonly used in roof seals.





Technical Specification

Thickness	1.5mm, 3mm, 6mm
Compound	EPDM
Density (Specific gravity)	1.13
Hardness (degrees Shore A)	70 +/-5
Tensile Strength (Mpa)	11.0
Elongation at break %	400
Minimum temperature	-40c
Maximum temperature	+100c
Maximum temperature for potable water	+23c
Colour/surface texture	Black/Cloth Mark finish
Compression Set (70c for 22 hours)	8

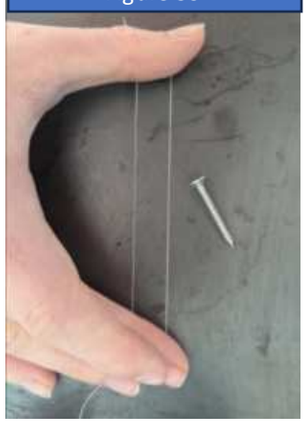
Figure 56

The technical specification of the 6mm thick rubber I purchased allowed me to be sure the EPDM had the necessary properties to meet points in my specification.

- Specific gravity (how dense it is compared to water) - having a specific gravity greater than 1 told me that it would definitely sink in water.
- Temperature range of -40 – 100 C is greater than the expected operating conditions of -20 to 60 C , so the rubber will not melt or become brittle when it is needed. Excellent UV resistance properties.
- Shore A hardness test value of 70 suggests it is firm but still flexible it is a similar value to a car tire or shoe sole. This should allow the rubber to lay flat but bend to conform as the MOB is rolled up the side of the hull.

This gave me the confidence to order a 1m x 1.5m x 6mm 6kg sheet for £40.00 to be able to cut down any shape to the required sizes.

Figure 59



Fabric Selection

During my research and producing the sample piece (Figure 61) I learned that the limiting factor for strength would probably be the thread holding the webbing to the fabric.
To maximise the weight the sling could lift I planned to:

- Use the thickest gauge of thread that can be used on the school sewing machines
- For the most important sections (attaching webbing) used a sewing machine designed for sail repairs at the yacht club, allowing thicker thread (Figure 59) to be used.
- Used polyester rather than cotton thread (Figure 59).
- Made multiple runs of stitching (one close to each edge) along the Dacron edging tape.



25mm wide red polyester webbing with a 440kg breaking strain. Easily visible against yellow and strong enough to carry load. To be used for attaching toggles and metal loops.





Figure 60

5 oz/m 100mm wide Dacron edging tape, designed for sail repairs, high strength bonded fabric. These properties will protect the edges from fraying and add strength to the sling.

Heavy duty Airtex mesh polyester fabric in fluorescent yellow was selected as it is similar to the fabric used in the sling I tested and a good compromise between strength, weight, and UV resistance. Holes are an important feature, to allow for water to drain out of the sling reducing its weight. It is also highly visible to alert other vessels of an emergency situation. Perforations also reduce suffocation risk.

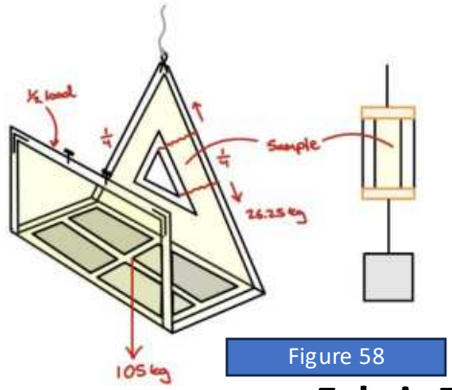


Figure 58

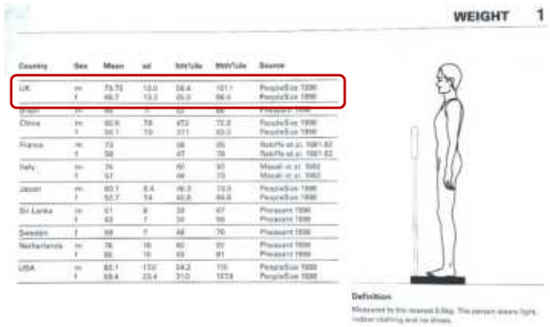


Figure 57

Fabric Testing

Producing a sample rectangular section of the sling design allowed me to develop techniques for working with the materials, improve my ability to sew straight on the machines in the textiles department and test that the strength of the edge tape and mesh would support an MOB.
Things that I learned while making the sample:

- That seam allowances would need to be incorporated into the design.
- That the strength/weight of thread used is the main factor that will affect the strength.
- Using double sided transparent 'basting' tape along with clips/pins is necessary to keep long lengths of edge tape in position while sewing.
- Paper templates could be used for non-rectangular parts to improve accuracy.

The sample was tested by being clamped between wooden blocks then weight added to create tension. The sample which is the same width as the thinnest section of the sling beside the triangular cutout would need to withstand ¼ the weight of a person + maximum 5kg of wet clothing. As seen in the anthropometric data (Figure 57) a 95th percentile UK male has a weight of approximately 100kg. As the sample is representative of the area shown in Figure 58 it will only take one quarter of the load 105kg / 4 = 26.25 kg. The sample could easily have 40kg of weight hung bellow it and I could even be suspended below it without it tearing. This confirmed that the strength of textile components was more than sufficient, with a large safety factor to account for uneven weight distribution, material fatigue or larger forces due to jerking from waves.



Figure 61



Figure 62



Figure 63



Figure 64

Development 3 – Design For Manufacture

While designing the hook/threader I was unsure what manufacturing method would be feasible. Sandcasting, (see below) while achievable in the school workshop and capable of producing complex parts, may lack the accuracy required to produce a reliable mechanism. The shape and position of notches which the latch pivots around is critical. Another option would be to have the part CNC machined to precisely the required shape, though this requires expert machinists and expensive technology only available in a commercial environment. Luckily, I was put in contact with a local company (Whitehouse Machine Tools Ltd) who supply a wide range of CNC machine tools and very kindly agreed to help me produce some parts.

Through extensive email communication and a zoom call, I was given advice on how to amend my design to be more suited towards CNC machining and to fix an issue that I had with constant tension being required on the rope that pulls the latch closed to prevent the design falling a part.

Changes that were made:

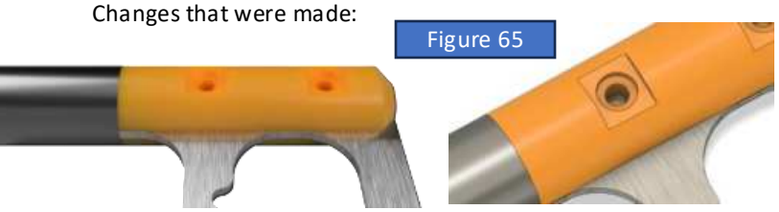


Figure 65

3D printed plastic spacers on each side of the aluminium gate to allow a flat part to be attached to a round pipe handle.

Holes in spacers made larger to fit standard M5 bolt.

Counterbore added so boltheads are flush with surface so less likely to catch on things.

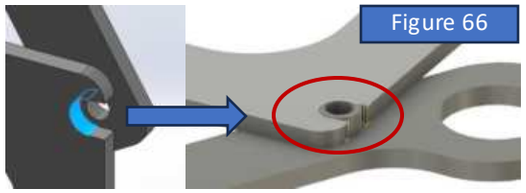


Figure 66

Subtle protrusions on each side of the pivot point notch so that a TPU 3D printed spacer is gripped and be better held in them but still be removed. This also better controls the distance between the two parts of the latch so they fit around the gate.

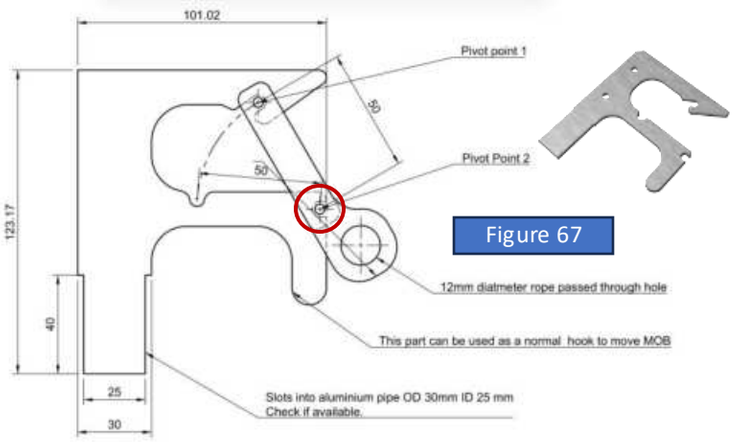
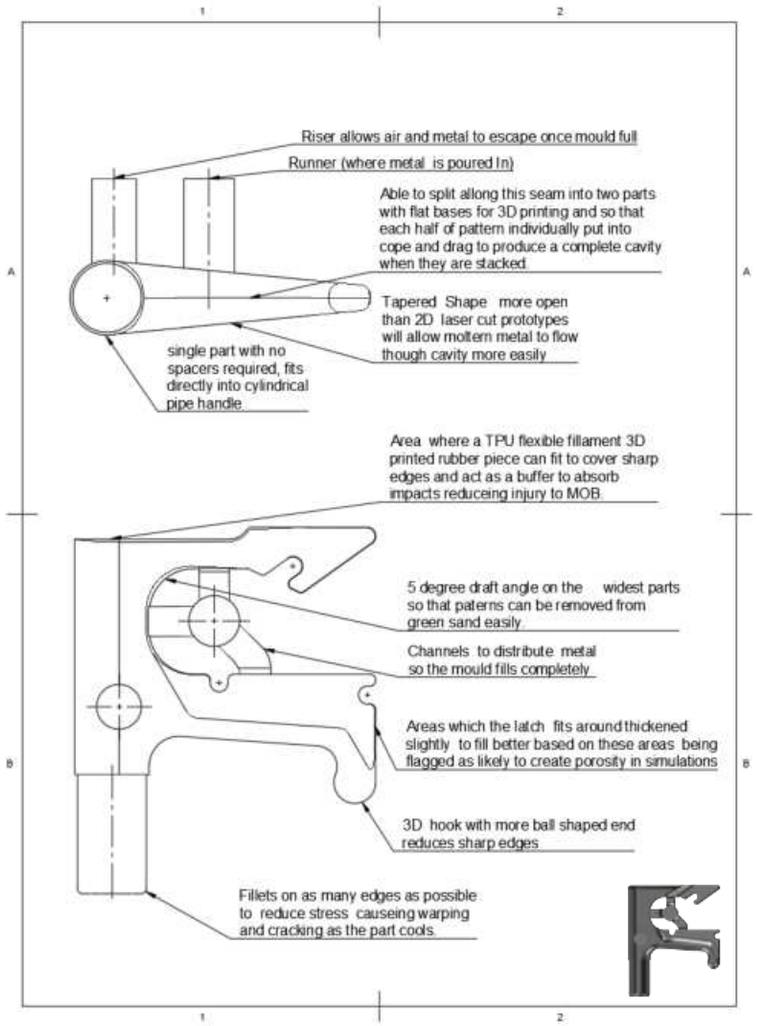


Figure 67

WHITEHOUSE
ADVANCED | ENGINEERING | SOLUTIONS
FOR WORLD LEADING INTELLIGENT PRECISION ENGINEERING SOLUTIONS
www.wmtcnc.com
AVEREX brother Biglia SPINNER WELE

Sandcasting Design Changes



Above is a labelled diagram of the updated design. The labels highlight features that were modified to suit sandcasting and justify changes. The gate shape and position of pivot points is largely unchanged, so that it still matches the laser cut prototypes that were proven to work.

Sand casting is a method of production available to me that would allow for 3D parts with complex geometry to be created from solid aluminium. The process involves compacting petrobond sand around a wooden or plastic pattern matching the shape of desired final component. Once the pattern is removed, it leaves a cavity in the sand mould. Molten aluminium is then poured in and allowed to solidify.

Though sand casting is a quite slow and labour-intensive process it is suited to making one off prototypes due to its low cost. By using sand casting, It should be possible for me to create a hook/threader mechanism that has fewer parts due to having a more optimised shape, with the structural characteristics necessary secure a person in the water and meet other specification requirements for corrosion resistance and durability.

I plan to use a 3D printer and CAD (Computer Aided Design) to produce a plastic pattern with greater accuracy than traditional carved wooden methods.

Figure 68



Problems I May have:

- Aesthetic defects such as pitting or small voids caused by imperfections in the mould
- Thin parts necessary for the latch mechanism to work not filling.
- Completing the casting within a reasonable timeframe.
- Warping or distortion/mould collapse of features important for the mechanism to function.
- Difficulty removing runner and riser from the casting without damaging it.

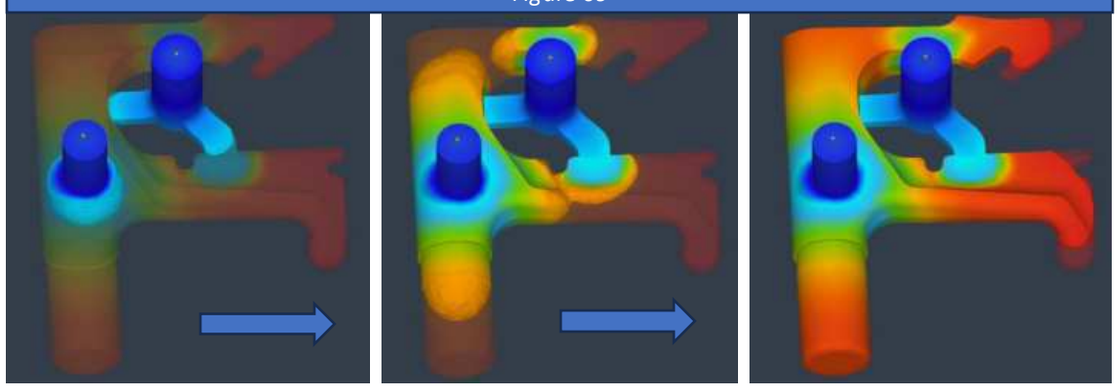




Once cast the curved shape designed to sandcast effectively has additional aesthetic appeal, once polished it should have a modern and metallic finish like objects of the streamline movement.

Another benefit of sandcasting is that it produces much less waste than machining as all material purchased is used, benefiting the environment and saving costs.

Figure 69



The CFD (Computational Fluid Dynamics) analysis of the shape of the mould cavity that my pattern would produce is shown above.

The three images in the first row (Figure 69) are screenshots of an animation that reveals how metal would gradually fill the mould, by moving outwards from the runner and riser cylinders where it is poured in. Time after pouring increases from the left most image and the temperature of the aluminium is shown by colour. Figure 70 shows which sprue each part of the mould will be filled from.

This allowed me to see which areas of the mould may not fill and assess if the three delivery channels I created would supply molten metal to all of the part.

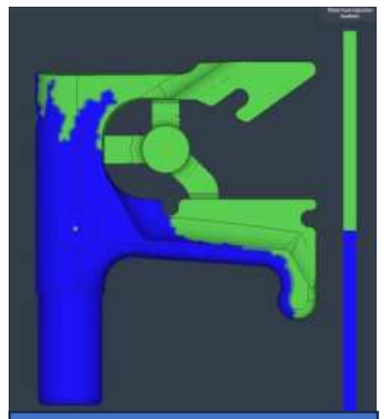


Figure 70

Material	Amount	Cost
Aluminium Ingots (LM25 alloy for casting)	Minium of 275g + spillage	£6.00
Aluminium Sheet 3mm	100 x 150mm (tessellate shapes)	£6.50
6mm round bar Aluminium	Minimum of 50mm length	£2.00
Handle	Aluminium with Polymer Grip	£6.00
Total Cost Of Sub Assembly:		**£20.50**

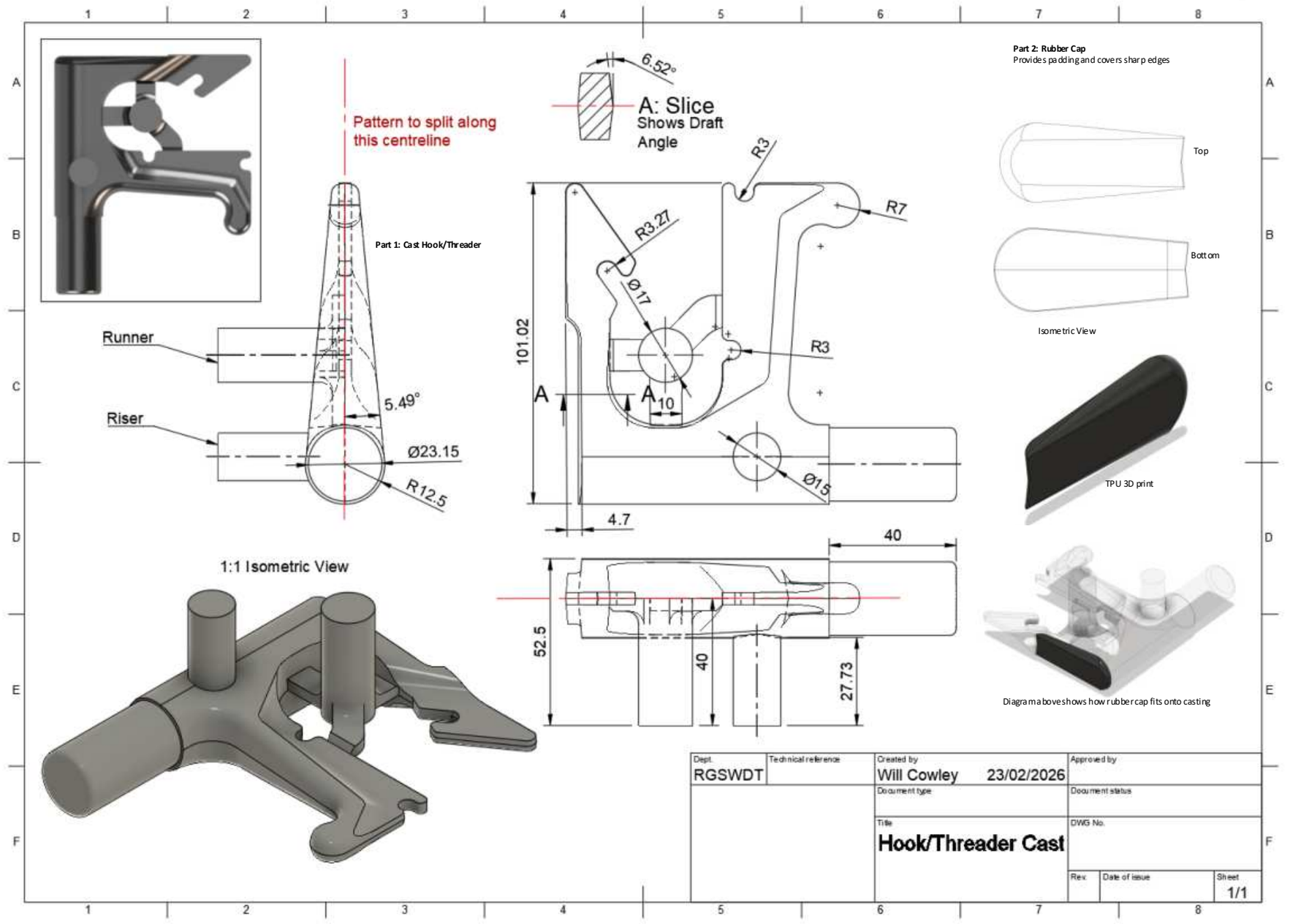


Figure 71

Final Design 1
Cast Hook/Threader



Figure 72



Please note that because these parts are designed to be 3D printed to produce the pattern only key dimensions are labelled, since for a third party to manufacture it the CAD files would be shared to be used directly for CAM. The main purpose of these drawings is to show the general size of parts and how they fit together.
The drawings include the runner, riser and channels that allow metal to flow into the mould during casting. These will be removed during the production process.

Client Comments - **"Casting may be less accurate, but I like the look of the 3D shape that can be produced and think it will look very good once it has been polished"**
"It will be interesting to see if the thin sections of the mould fill completely"
"The rubber buffer should help if your hit by end, but there's not much you can do about the wedge to slot under the strap"

Material	Amount	Cost
ABS Filament (Orange)	Minium of 275g + spillage	£1.00
TPU Filament (White)	1-2g to be used	£0.20
Nuts & Bolts	2 x M5 Nut+Bolt 2 x M4 Nut+Bolt	£0.50
Aluminium Sheet 3mm	100 x 150mm (tessellate shapes)	£6.50
Total Cost Of Sub Assembly:		**£8.20**

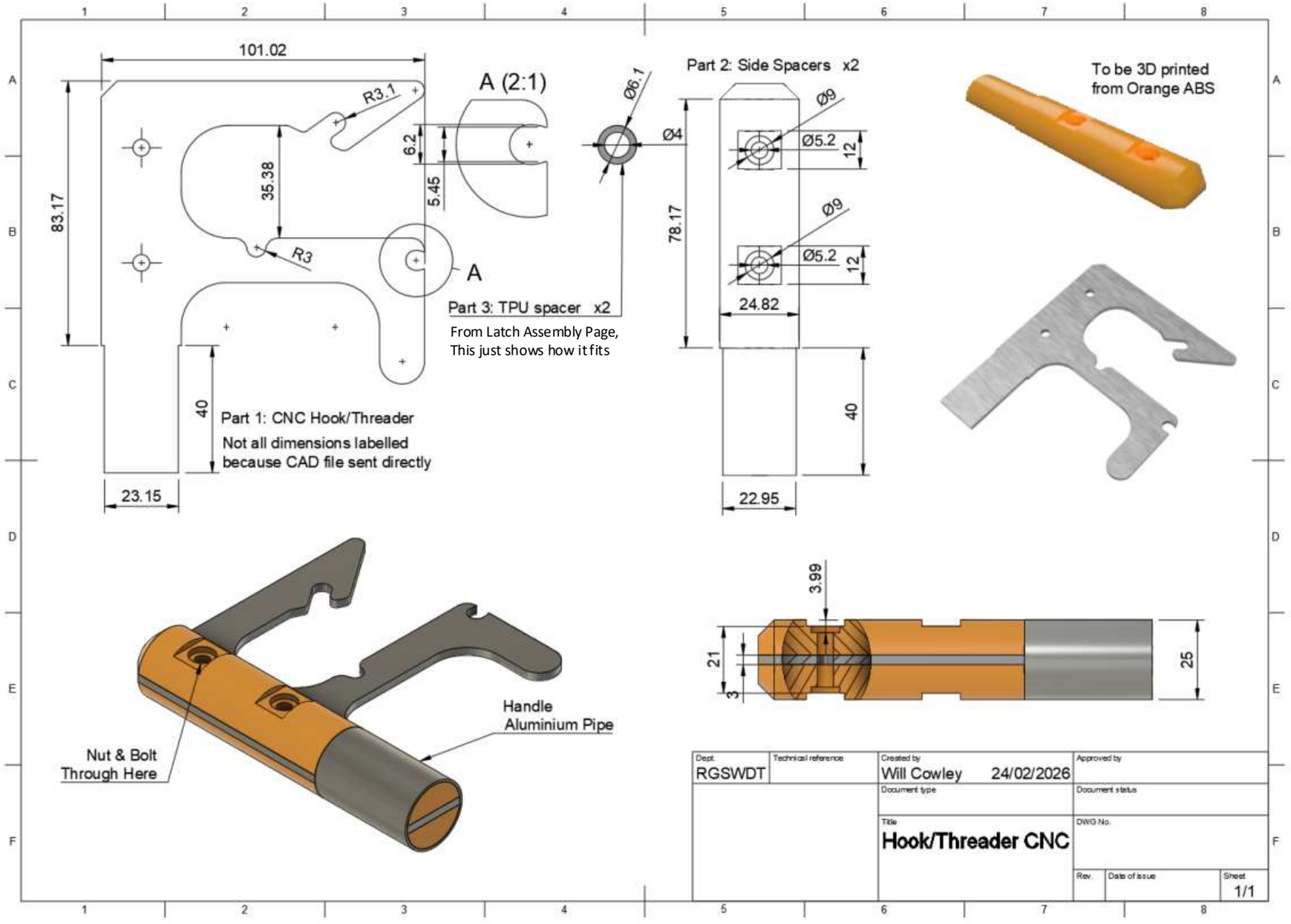
Final Design 2
CNC Hook/Threader



Figure 73



Figure 74



Please note that because these parts are designed to be 3D printed and CNC machined only key dimensions are labelled, since for a third party to manufacture it the CAD files would be shared to be used directly for CAM. The main purpose of these drawings is to show the general size of parts and how they fit together.

Client Comments - **" We will be able to test both versions and and see which works better, this one will probably be more accurate but the 3D shape of the hook part on the cast one might help it be stronger"**
"Using bolts instead of glue should allow the plastic pieces to be replaced easily if they ever crack or degrade"

Final Design 3 – Latch Assembly

Latch 1 and Latch 2 will need to be produced twice one for each Hook/Threader Option (Cast or CNC). I will be testing two different types of spacers that hold latch 1 & 2 apart. The spacers labelled on the render below use a nut and bolt with a TPU 3D printed washer (white) to separate parts. This was suggested by Whitehouse engineering as a method to stop the latch falling off, as rubbery washers are gripped by the CNC aluminium part. I will also be producing the aluminium spacers on the lathe for one of the latches I make because they are have a lower profile sitting which may reduce snagging.

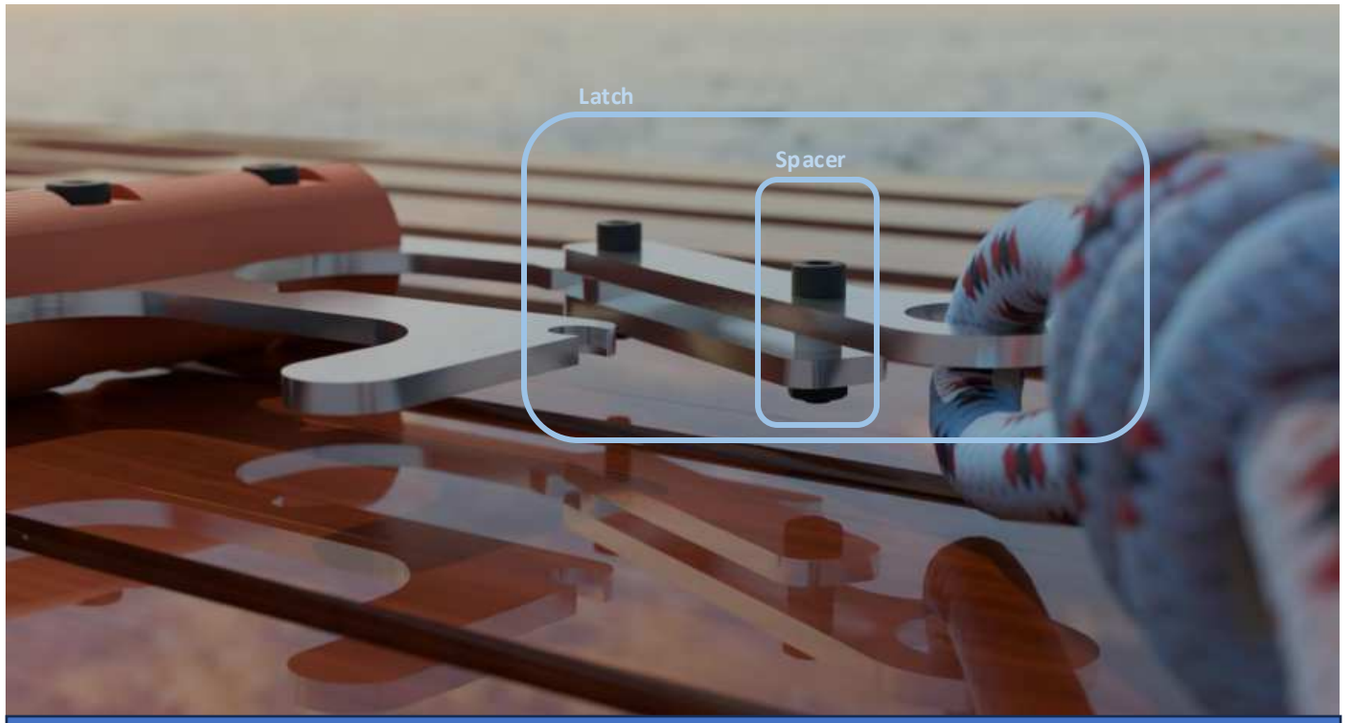
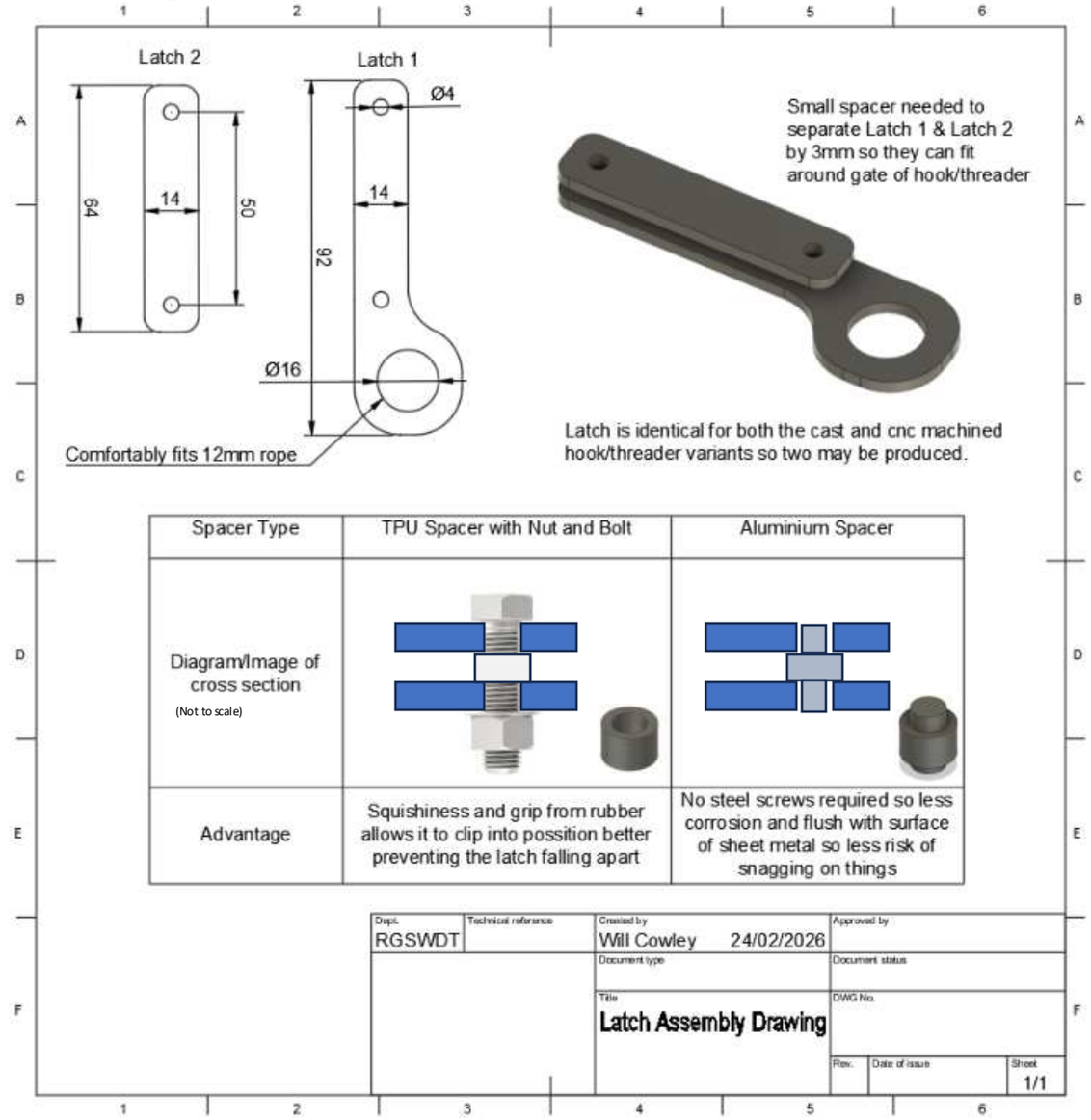


Figure 75



Latch Function

The Latch is a key part of both hook/threader mechanisms that has two pivot points that move freely between divots in the gate of the threader mechanism to pass a rope behind and though a loop to attach an MOB.

Joining Spacers To Latch 1 & 2

TPU spacers act like washers being held in place by a Nut and Bolt and Loctite . Aluminium spacers will be TIG welded to Latch 1 and 2, this will create a permanent join that is resistant to vibration and does not require an adhesive. Welding the spacers will be done at a low current setting to avoid melting the spacers.

Justifying Making Both Hook/Threader Options

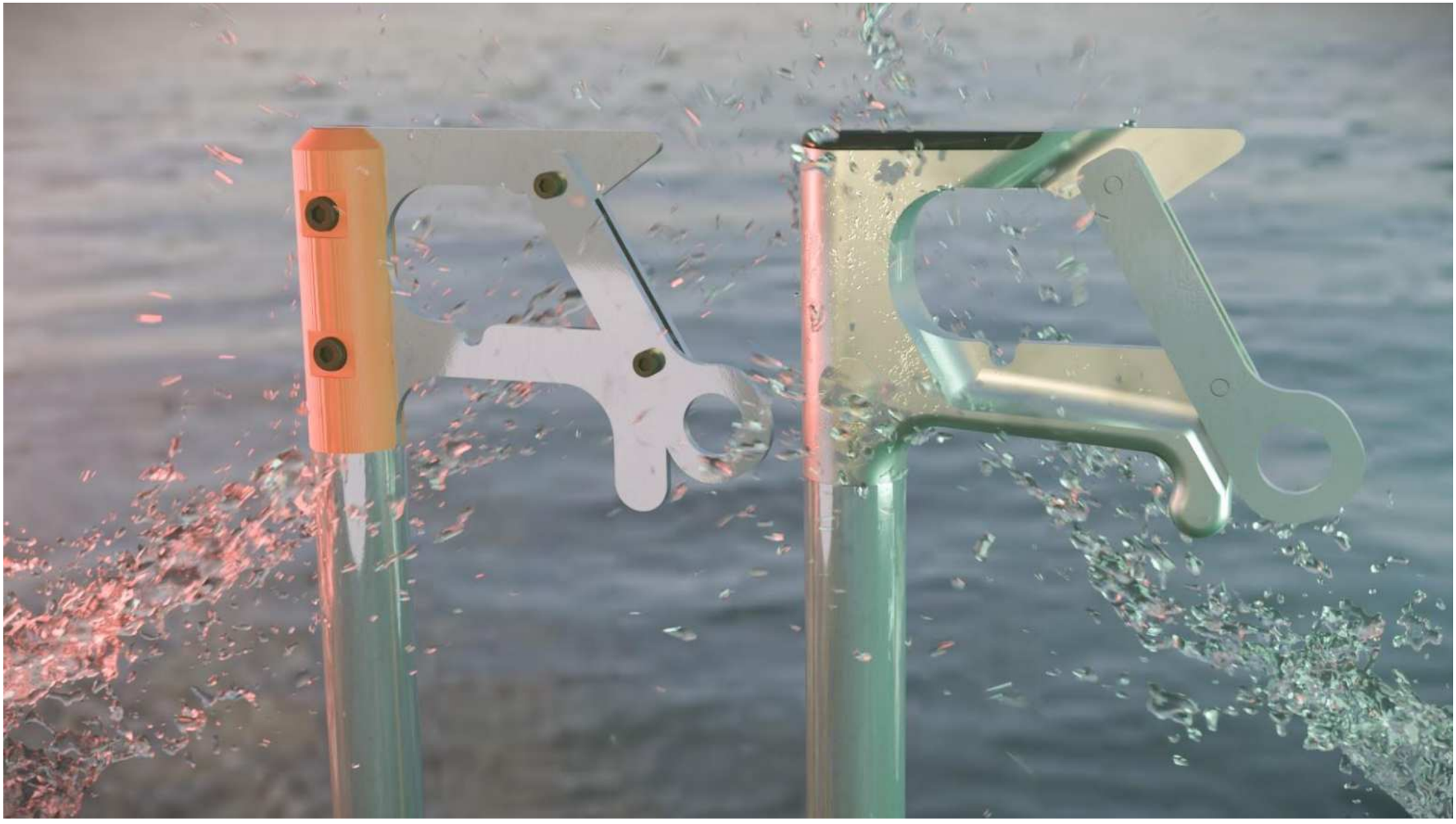
The table to the right directly compares the two manufacturing methods for the final design.
As you can see each design effectively meets specification in most areas but have different strengths and weakness in addressing individual requirements.

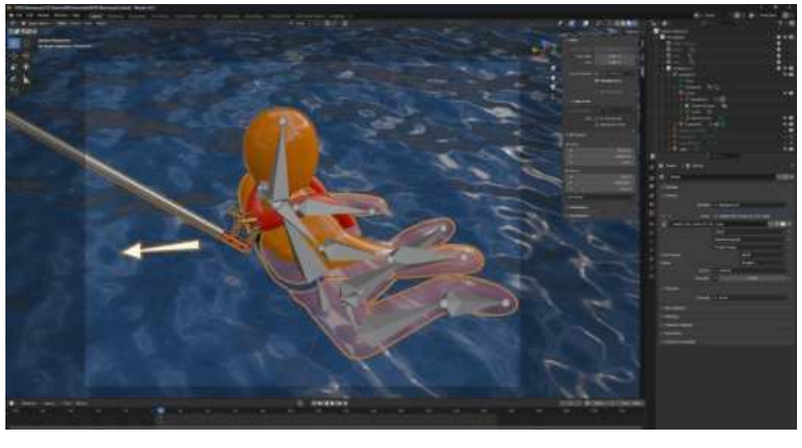
The fact that both design have a similar number of pros and cons has encouraged me to attempt making both options.

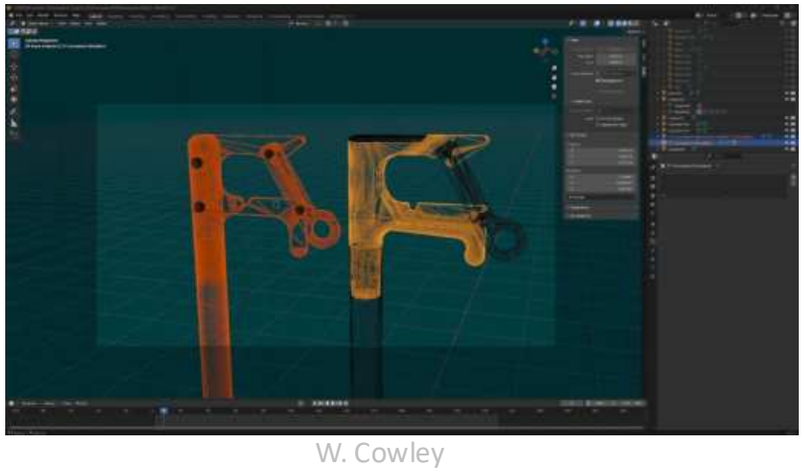
Client Comments - **"Choosing to produce both designs will allow them to be compared during testing, to get a more general sense of weather the mechanism works" "The CNC version seems to be the more mass-produceable"**

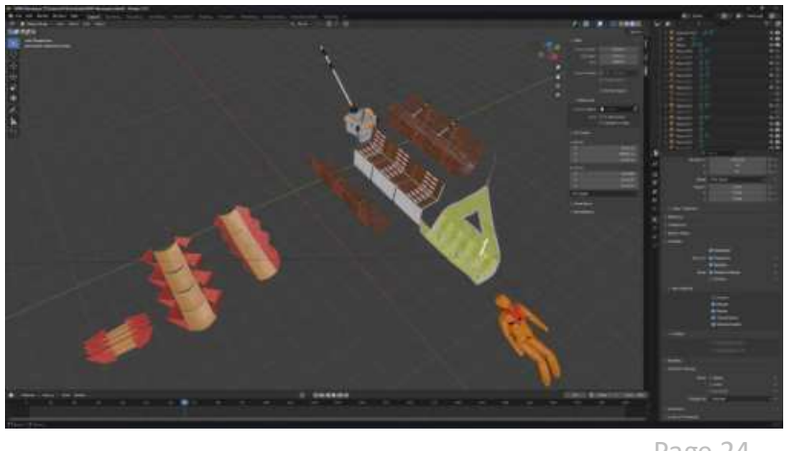
Feature	Cast Hook/Threader	CNC Hook/Threader
Manufacture	• Simplifies assembly by making a part with a complex shape which can do the job of multiple parts of the other design. • Slow labour-intensive process only feasible for a one-off prototype.	• Fast automated process which could allow production to be scaled up more easily. • 3D printed orange spacers needed so more parts must be produced.
Function	• Chance of warping or shrinkage distorting the shape. • May be brittle from casting or have internal defects that cause it to break unexpectedly. • No stainless steel bolts needed so entire assembly aluminium eliminating risk of galvanic corrosion. • TPU rubber cap will reduce risk of causing injury to MOB.	• Greater chance of functioning reliably due to higher accuracy and repeatability. • Stainless steel bolts to hold spacers on will be painted but could wear through paint, so greater risk of rusting.
Aesthetics	• Curves needed for casting give a futuristic look.	• Orange 3D printed spacers increase visibility for safety.
Cost	• Achievable in a school workshop environment. • Higher material cost as weight of aluminium to be used is higher.	• Expensive facilities needed, only available commercially. • Lower material cost as weight of aluminium used is less.
Sustainability	• No material is wasted during production. • Significant energy (gas combustion) required to melt aluminium.	• Subtractive process so material removed is wasted. • Aluminium and ABS plastic are both recyclable.

CNC Hook/Threader and Cast Hook/Threader Preview Render









Final Design 4 – Rescue Sling

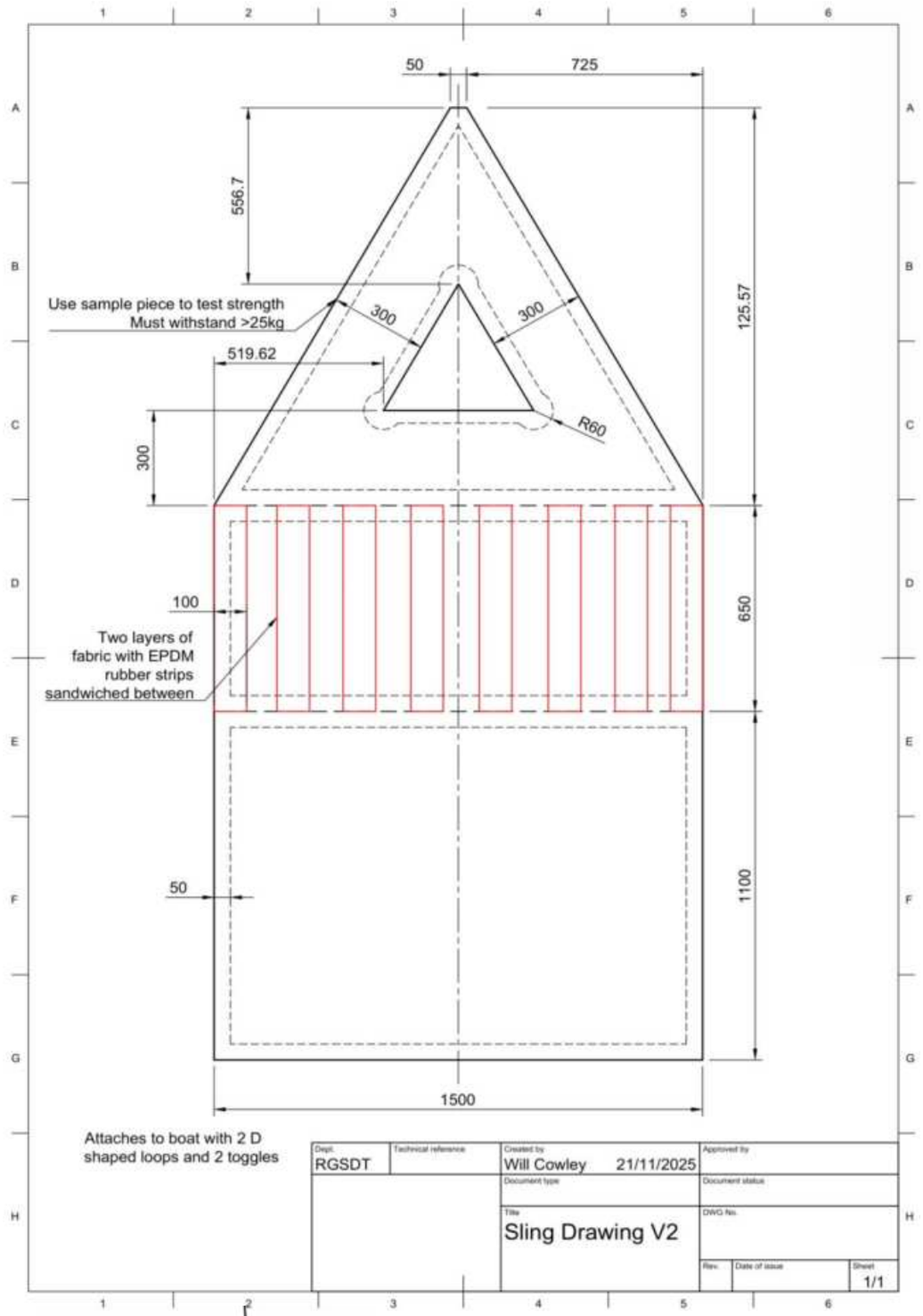
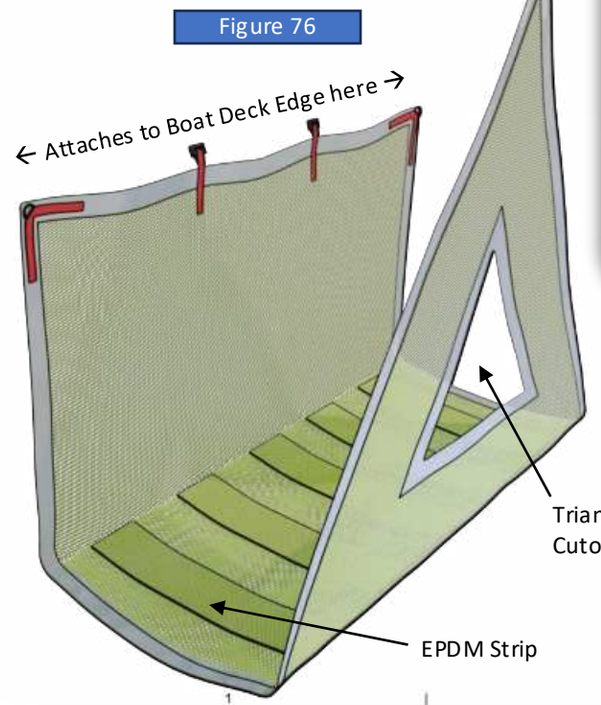


Figure 76



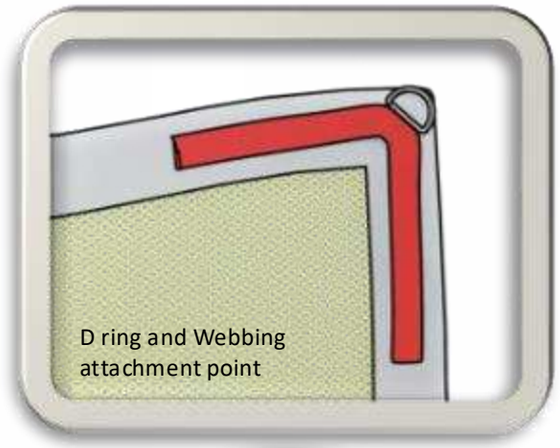
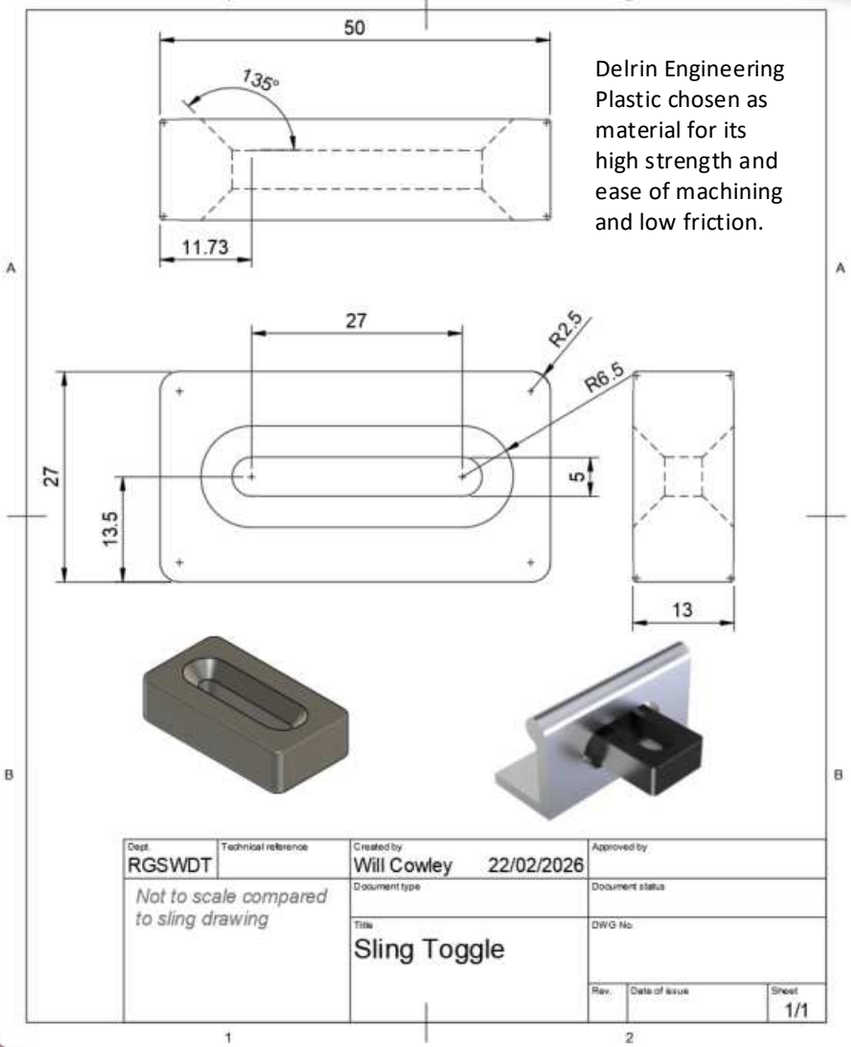






Figure 77



Delrin Engineering Plastic chosen as material for its high strength and ease of machining and low friction.

Material	Amount	Cost
Airtex polyester mesh fabric (yellow)	1.5 x 5m	£35.00
EPDM 6mm Sheet	1.5 x 1m	£40.00
Dacron Edging Tape 100mm	12m minimum	£18.00
Polyester Webbing 25mm (Red)	2m	£5.00
Stainless steel rings	2 x D shape 1 x O shape	£4.00
Delrin Plastic (For toggles)	12 x 80 x 30 mm	£1.00
Total Cost Of Sub Assembly:		**£103.00**

Figure 78

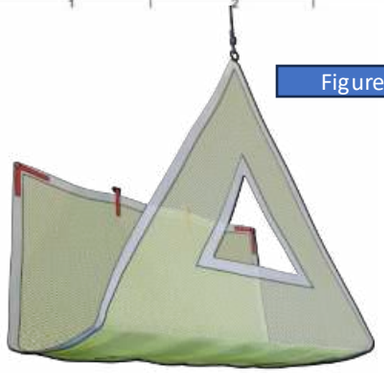
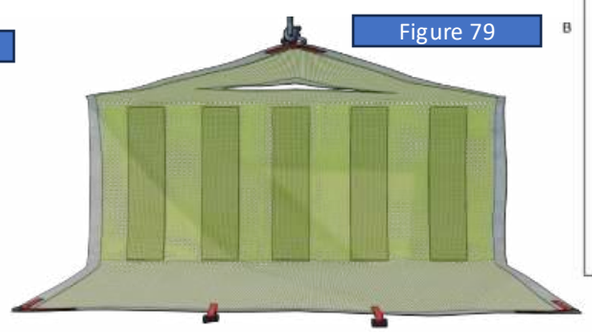


Figure 79

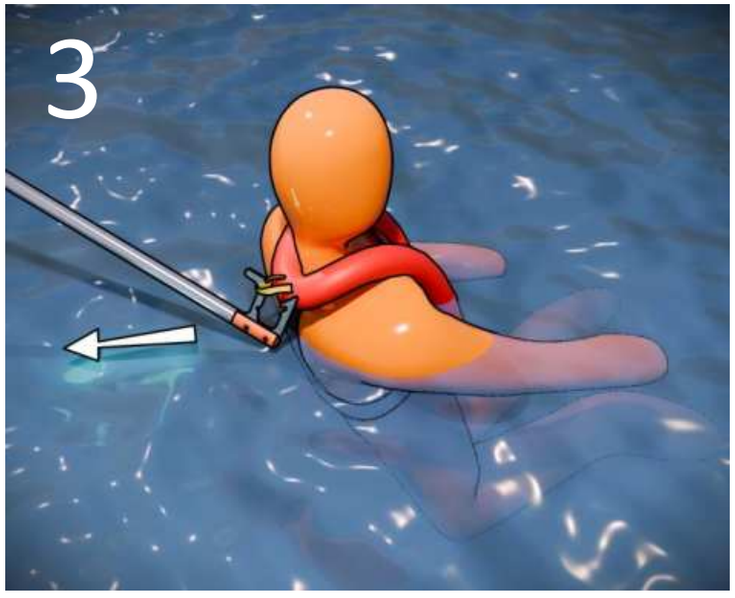


Client Comments - **"Adding the rubber strips and cutout window should significantly improve it over slings available on the market"**
"It seems like an effective method of lifting a man overboard in a controlled way"

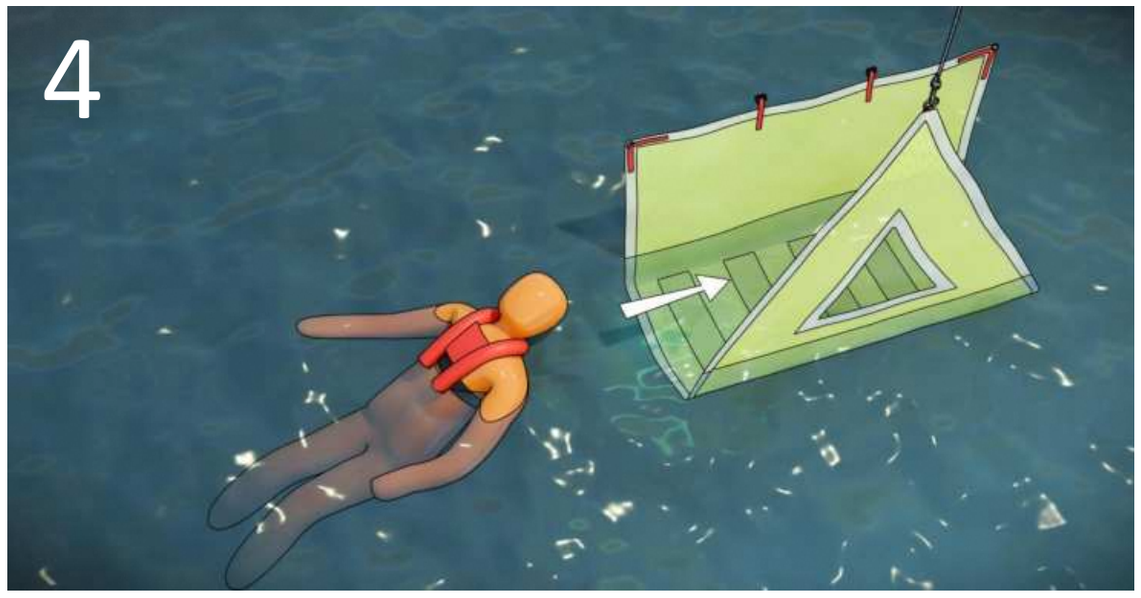
Pictographic Rescue Instructions





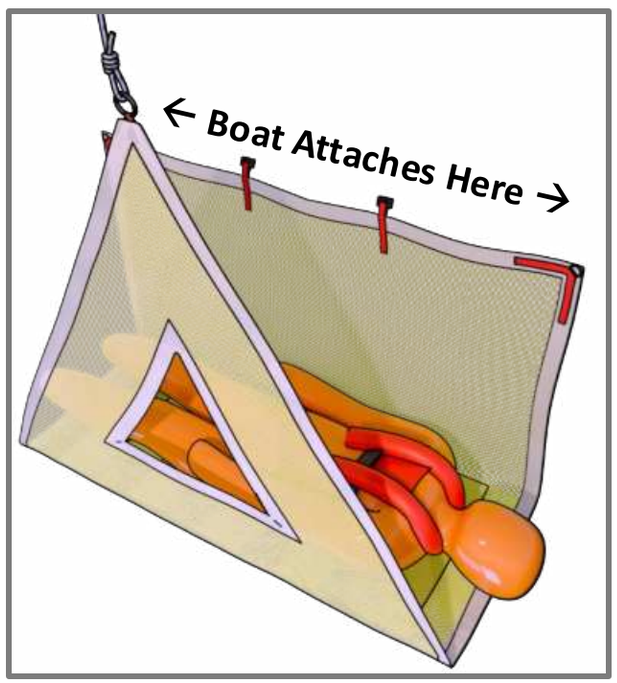


Use the Threader Mechanism to attach a rope to the Man Overboard' s lifejacket, to safely secure them. Next use the hook part to latch onto any part of their lifejacket or clothing.

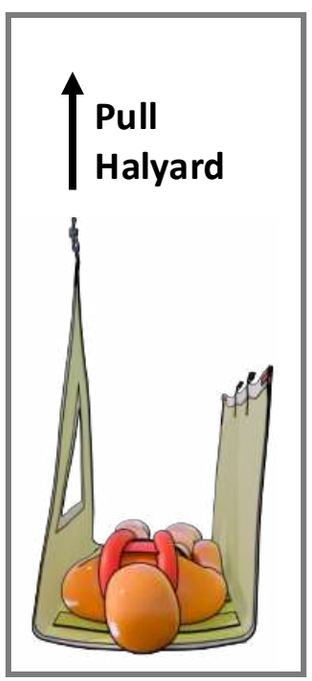


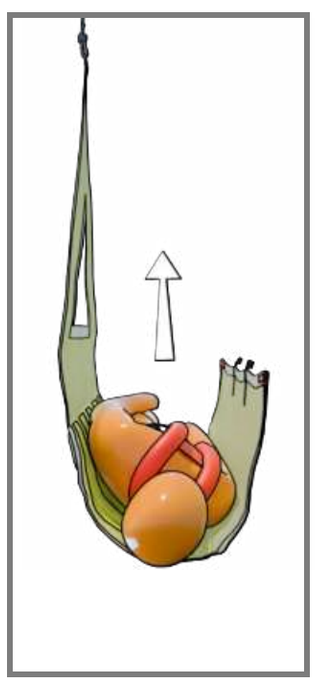
Use the Hook to move the Man Overboard in the water and pull them closer to the boat and backwards into position.

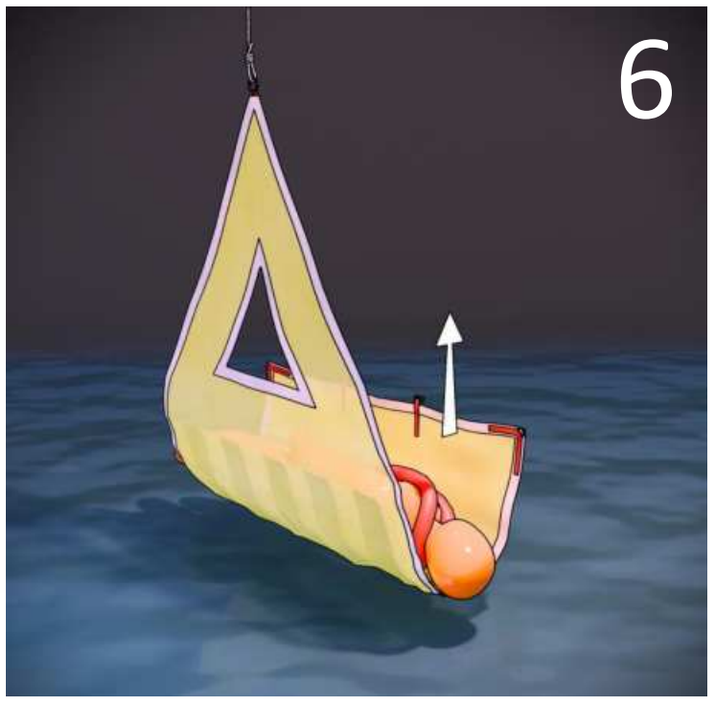




Pull Man Overboard onto the sling once it is setup as shown, with Toggles attached to Deck Edge Rail.







Use Windlass Winch to reel in Halyard tied to the sling. This will gradually lift the Man Overboard until they are level with the deck.

An A3 laminated print-out of these Images could be stored with the sling to act as step-by-step pictographic instructions for its setup and use, meeting the specification requirement for being intuitive for new users. Large text and brief description on this page are supposed to help it be read quickly.

Breakdown of components

CNC Hook/Threader

Parts: CNC Hook/Threader Gate, 2 x M4 Black Oxide Coated Nut & Bolt, 2x Side Spacers
Comments:

- As CNC machining some of the Aluminum parts has been outsourced to Whitehouse Engineering ltd to my own CAD design, I will continue our email exchange as I build other components, to stay up to date on progress and discuss any further changes that may need to be made.
- A high tolerance will be required so that parts in this component fit together and the threader mechanism functions reliably. Use of Computer Aided Manufacture techniques like 3D printing and CNC machining should allow parts to be created more accurately than is possible by hand.

Rescue Sling

Parts: Toggles x2 , Fabric Sections, EPDM Pannels, Round Metal Ring, 2x D Metal Rings
Function: Lifting a MOB out of the water.
Comments:

- Producing this component is likely to be the most time consuming. I have very little experience with sewing machines and initially I expect to spend time unpicking seams with possibly multiple attempts to be needed to get a secure joint between fabrics.
- The points where the red webbing attaches to the sling will experience the most stress because they have the smallest cross section so a solid connection between webbing and fabric must be created. This can be done by going over them repeatedly with zig-zags or crosses of stitching.
- The number and width of EPDM strips added is flexible (may not be exactly as shown on drawing) because it is not really possible to judge how much stiffness they will add before having received the material.
- Delrin toggles produced are extremely chunky for strength and are perfectly sized to the slots in the deck edge rail, To make sure they fit after facing off the sides of the Delrin blocks test that they fit though the 3D printed replica of the slots.

Cast Hook/Threader

Parts: Cast Hook/Threader Gate, TPU Rubber Cap
Comments:

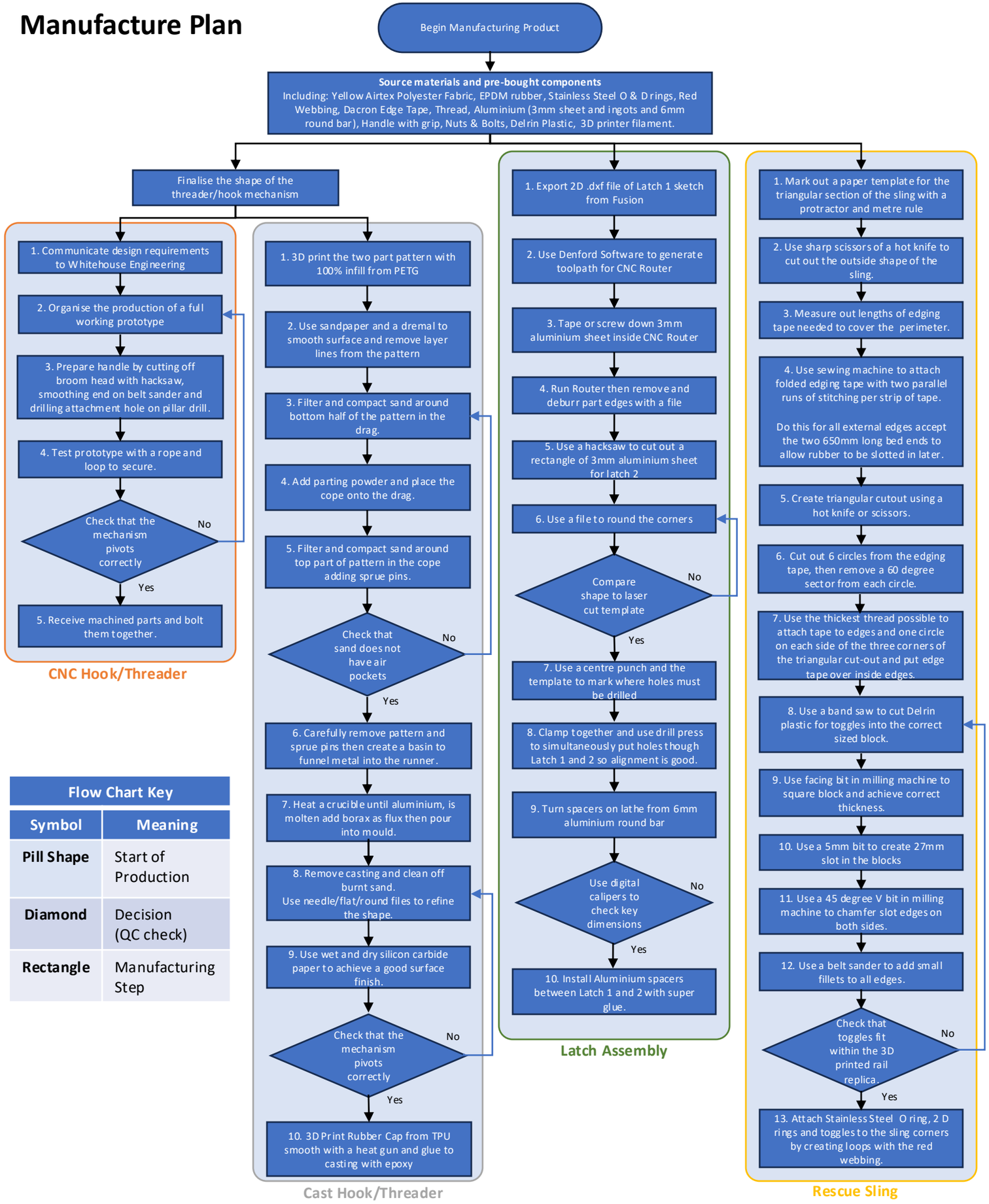
- Since CAD was used to make the modifications to the hook/threader design so it would suit sand casting, models for the pattern can be easily exported into a slicer for 3D printing.
- Sanding the plastic 3D printed pattern to create a mould without layer lines imprinted will be easier and less time consuming than removing the same imperfections from the metal part, so care should be taken preparing the pattern.
- Producing a good mould in as few attempts as possible will be important to stay on schedule, fully compacting green sand around the pattern in layers without rushing will ensure that the part produced has a smooth surface finish.

Latch Assembly

Parts: Latch 1, Latch 2, 2 x TPU Spacer or 2x Aluminium Spacer
Comments:

- Two complete versions of Latch 1 and Latch 2 will be made, one completely by me using the school's CNC router and hand tools the other will be accurately machined externally. This will allow each Hook/Threader to have its own latch assembly for testing. This also means both types of spacers can be permanently attached to their own latch assembly if necessary.
- I chose to make two types of spacers, Aluminium ones turned on a lathe (my original design) and TPU 3D printed washer, the advantages of each are listed on the final design page.

Manufacture Plan



Flow Chart Key	
Symbol	**Meaning**
Pill Shape	Start of Production
Diamond	Decision (QC check)
Rectangle	Manufacturing Step

Gantt Chart Production Schedule

Producing a Gantt chart for each component with estimations for how long each step will take, allows me to allocate my time well and have a target for what needs to be completed each lesson. Each of the numbers increasing horizontally represents a 1hr workshop session available to me for manufacturing. The steps being scheduled are written in full on the previous page in the Manufacture Plan flow chart.
After finishing making the device, I added the actual timings of when each step was completed. Generally, I stuck to the plan quite well, but some steps took longer than expected or had to be redone causing delays with a knock-on effect.
Causes of delays:
Rescue Sling Step 3 – Textiles Dept closed so no access to sewing machine.
Rescue Sling Step 10 – Many small passes needed to be made to cut slot, so it took one more lesson than expected.
Latch Assembly Steps 5,6,7 – Latch 2 was completely remade after failed attempt at welding it to spacers, second version was glued to spacers.

Gantt Chart Key	
Symbol	**Meaning**
Coloured Box	Proposed Timing of a step
X	Actual Timing of a step
X	Significantly Off Schedule
→	Overran Schedule

Component	Process	Step	1	2	3	4	5	6	7	8	9	10
Rescue Sling	Sling Fabric Work	1	x									
		2	x	x	x	x						
		3						x	x			
		4						x	x			
		5							x			
		6							x			
		7							x	x		
	Producing Delrin Toggles	8								x		
		9								x		
		10								x	x	x
		11									x	x
		12									x	x
	Attach Webbing	13										→

Component	Process	Step	11	12	13	14	15	16	17	18	19	20
Latch Assembly	Machining Latch 1 (School CNC)	1	x									
		2	x									
		3	x									
		4		x								
	Shaping Latch 2	5			x					x	x	
		6			x					x	x	
	Making then fitting spacers	7				x		x				
		8				x						
		9					x	x				

Component	Process	Step										
Cast Hook/Threader	Producing Pattern	1							x			
		2							x			
	Making Mould and Casting	3							x	x		
		4								x		
		5								x	x	
		6									x	
		7									x	
	Finishing the Casting	8										x
		9										→
		10										→

Component	Process	Step										
CNC Hook/Threader	Produce Prototype	1	x		x	x			x			
		2	x									
		3								x		
	Handle Prep	4		x								
	Assembly	5										x

Quality Assurance

Quality Assurance techniques are to be followed throughout manufacture to make sure parts are being produced accurately up to a high standard. Basic good working practice standards, such as checking key dimensions and inspecting parts for defects will be followed.

Using Template/Stencil

- Paper hand drawn or wooden laser cut templates of the correct shape will be used where possible.
- This Is more accurate than marking out directly onto a part because grid paper can be used to get the perfect angle and mistakes disposed off.
- Using templates also improves the aesthetics of the part, giving a cleaner look because parts do not get written on or scratched with a scriber.

- A template will definitely be used to precisely cut out the shape and angles of the triangular cutout in the rescue sling.
- A laser cut template of the intricate latch assembly parts will be held next to the aluminium so their shape can be exactly matched.

3D Printed Gauge

- A 3D Printed replica of a section of the deck edge rail (Photo'd Below) will be used similar to a go/no-go gauge to check that the toggles will fit though the actual slots present on my client's boat. This will allow more material to be removed immediately if they do not fit.



Working Prototype

- I plan to 3D print a full working replica of the CNC Hook/Threader and Latch Assembly so the function of the mechanism can be tested threading actual rope though a loop.
- This will allow fits and clearances between parts to be perfected before the actual parts are produced.
- This reduces the risk of discovering fit/tolerance problems at a later stage after resources have been dedicated to make the real part.
- Physically being able to put together the whole mechanism could expose flaws that were not noticeable in CAD where objects can phase though each other.

Tolerances

- 3D printing, milling, and CNC machining allows a high degree of accuracy in production, well within the range required for this product.
- Smaller Features or Parts such as Spacers, notches in gate parts, holes in Latch 1 and 2 will need to be sized and positioned correctly to a very high tolerance of around +- 0.2 mm so that the threader mechanism works.
- Less care needs to be taken with much larger dimensions such as the handle length or sling width where a +- 20mm deviation would have no effect on function and would be a smaller percentage of the overall length.

Risk Assessment For Each Process

In order to operate each tool, I use in a safe manner a risk assessment was done to identify risks associated with using each machine and plan control measures I will follow during manufacturing to mitigate them.

Sling Fabric Work

Tool: Sewing Machine
Risks:
- Needle puncture Injuries
- Entanglement In moving Parts

Control Measures:
- Use a correctly adjusted needle guard
- Remove foot from treadle when threading and changing needles.
- Wear Apron and tuck in loose clothing.

Tool: Pins, Clips and Scissors
Risks:
- Cuts or puncture Injuries

Control Measures:
- Count Pins out of and back into their box to make sure all are removed.
- Carry scissors with blades closed and down.

Tool: Hot Knife
Risks:
- Burns, Fire or Cuts
- Tripping over cable.
- Inhalation of toxic fumes.

Control Measures:
- Power off between uses and allow to fully cool after use.
- Use a flat non-flammable cutting surface.
- Have a well cleared workspace.
- Work in a well-ventilated space or with Local Exhaust Ventilation.

Producing Delrin Toggles

Tool: Band Saw
Risks:
- Cuts from direct contact with blade.
- Entanglement
- Flying Objects
- Dust Inhalation

Control Measures:
- Wear PPE, Safety goggles and ear protection.
- Wear Apron and tuck in loose clothing.
- Use push sticks, keep fingers out of safety zone and lower guard.

Tool: Milling Machine
Risks:
- Entanglement, cutting/amputation
- Workpiece being ejected

Control Measures:
- Wear glasses or face shield, Ensure vice securely fastened
- Wait for machine to stop before removing part.
- Never operate without guard closed.

Tool: Belt Sander
Risks:
- Abrasion if skin contacts belt.
- Dust Inhalation
- Workpiece being ejected

Control Measures:
- Wear PPE, safety goggles, do not wear gloves.
- Grip toggles well and keep firmly against table.
- Adjust guard height.

Machining Latch 1

Tool: CNC Router
Risks:
- Flying Debris/workpiece, Cuts and entanglement
- Burns from parts conducting heat, Cuts from on sheet edges

Control Measures:
- Do not leave machine running unattended.
- Only run the machine with the enclosure shut
- Initially run the program with a very slow feed rate.
- Secure the aluminium sheet effectively with double sided tape and many screws.
- Allow parts to cool fully before removing them.

Tool: Hand Files
Risks:
- Cuts and abrasions from file slipping, metal edges and burrs
- Eye discomfort from fine metal dust.

Control Measures:
- Secure workpiece in vice, wear glasses and wash hands after afterwards.

Shaping Latch 2

Tool: Centre Punch
Risks: Hitting fingers with hammer
Control Measures:
Take care to strike the centre punch in a controlled way.

Tool: Pillar Drill
Risks:
- Cuts and entanglement, workpiece being ejected

Control Measures:
- Clamp Part, wear goggles and lower guard.

Producing Pattern

Tool: FDM 3D Printer
Risks:
- Burns from bed or nozzle
- VOC inhalation

Control Measures:
- Print in a well-ventilated space with enclosure shut.
- Allow pattern to cool before removing.
- Print PLA or PETG to produce less fumes.

Tool: Dremel and Sandpaper
Risks:
- Grazes from abrasion
- Dust Inhalation or dust getting in eyes.

Control Measures:
- Wear goggles/glasses, work in a ventilated space.

Making Mould & Casting

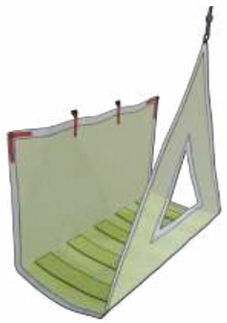
Tools: **Cope, Drag, Rammer and Green Sand**
Risks:
- Dropping heavy metal frames on feet.

Mitigation:
- Do not carry above feet or wear steel toe cap boots.

Tools: Furnace and Crucible
Risks:
- Burns, Molten metal spills, gasses released

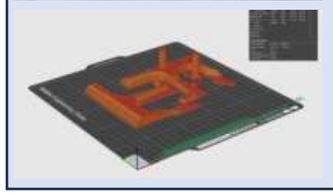
Control Measures:
- Local Exhaust Ventilation and wearing a respirator.
- Make sure mould is properly vented and free of any moisture.
- Wear PPE: face shield, leather gloves, apron and spats for leg/foot protection.
- Sand covering foundry floor to contain spills.
- Use long tongs to move crucible.

Manufacturing Specification

Component	Part	Part Type	Part Purpose	Material	Stock Dimensions (mm)	Performance Standard	Justifications For Material Choice	Tools Required
Rescue Sling 	EPDM panels x 4	Self Made	Add Rigidity and Weight to the sling so it stays flat under water.	EPDM Rubber sheet (6mm thick)	1 500 x 1000	Withstand Spec temperature range	Denser than water to sink below MOB Optimal stiffness to lay flat Durable enough for conditions.	Table Saw or Band Saw
	Toggles x2	Self Made	Quickly attach the Sling to the deck edge	Delrin Plastic	100 x 30 x 13	Withstand Spec temperature range	Easy to Machine Low friction so Toggles can be fitted easily	Milling Machine Belt Sander Band Saw
	Fabrics	Self Made	Support and Lift the MOB from the water to the level of the boat deck.	Airtex Mesh Polyester Fabric White Dacron Sail Edging Tape Red Polyester Webbing	1 500 x 3000 12 000 x 100 2000 x 25	Airtex Mesh & Edging Tape Self Tested Page 20 Webbing 1000kg Max Load	Contrasting colours red against yellow. Certified high tensile strength and tear resistance.	Sewing Machine Hot Knife Scissors
	Welded Round Ring	Bought In	Attachment Point For Halyard	A4 Marine Grade Stainless Steel (Polished)	OD 66 ID 50 (8mm thickness)	480kg Max Load	Corrosion Resistant High Strength	n/a
	Welded D Ring x 2	Bought In		A4 Marine Grade Stainless Steel (Polished)	20 x 25 (5mm thickness)	480kg Max Load	Corrosion Resistant High Strength	n/a
Cast Hook/Threader 	Casting	Self Made	Secure MOB by either attaching a rope or hooking onto their lifejacket	Aluminium Ingot (Likely an LM25 alloy for casting) PETG Filament for Pattern	275 g to fill mould cavity 110 g @ 100% infill		Able to be Cast Lightweight compared to steel Corrosion Resistant PETG strong enough for pressure during mould making	3D Printer & Sandpaper Dremel Cope & Drag + Rammer Furnace & Crucible File Set + Silicon Carbide Paper 80 – 1000 grit
	Rubber Cap	Self Made	Absorb hits to prevent injury	Black TPU Filament	6 g @ 100% infill		Flexible/Squishy to absorb impacts	3D Printer & Heat Gun
CNC Hook/Threader 	Side Spacers	Made Externally	Allow 2D Mechanism to attach to pipe handle.	Orange ABS Filament	45 g @ 100% infill	Withstand Spec temperature range	More UV stable than alternatives such as PLA. High visibility	3D Printer
	CNC Gate	Made Externally	Hook MOB and accommodate rope threader mechanism.	Sheet Aluminium (3mm thick)	130 x 100		Corrosion Resistant Easy to Machine	Large Industrial CNC Milling Machine.
	M4 Nuts and Bolts x2	Bought In	Hold Side Spacers either side of the Gate	Black oxide coated Stainless Steel.	30mm length		Surface coating prevents Galvanic Corrosion while contacting Aluminium	Allen Key
Latch Assembly x2 (One for each Hook/Threader Type) 	Latch 1	Self Made	Pivot in 2 places to thread a rope.	Sheet Aluminium (3mm thick)	92 x 20		Corrosion Resistant Easy to Machine	School CNC Router TIG Welder, Pillar Drill
	Latch 2	Self Made	Pivot in 2 places to thread a rope.	Sheet Aluminium (3mm thick)	64 x 14		Corrosion resistant	Hacksaw, Pillar Drill File Set
	TPU Spacer x2	Self Made	Separate Latch 1 from Latch 2 Act as a pivot point	White TPU Filament	1 g @ 100% infill	Withstand Spec temperature range	Flexible/Squishy to	3D Printer
	Aluminium Spacer x2	Self Made	Separate Latch 1 From Latch 2 Act as a pivot point	Aluminium Round Bar	6 OD		Corrosion Resistant Same Metal as Latch 1 & 2	Manual Lathe or CNC Lathe Facing tool, Parting off tool TIG Welder
	M4 Nuts and Bolts x2	Bought In	Join Latch 1 & 2 with spacers between.	Black oxide coated Stainless Steel.	12mm length		Surface coating prevents Galvanic Corrosion while contacting Aluminium	Allen Key

Manufacturing Evidence 1 – Cast Hook/Threader





Step 1 – Produce Pattern

CAD exported from Fusion as a .stl file, converted to gcode using Bambu Lab's Slicer and 3D printed with 100% infill.

The FDM 3D Print took 3hrs and 15 minutes and used 110g of PETG filament to accurately produce both halves of the pattern.

Comment: 100% infill was needed so that the pattern was completely solid, so features were not crushed by the pressure of having sand rammed around it.

Comment: Brim Ears were added in Bambu Lab Slicer to help with print bed adhesion at the corners. This kept the base of the parts flat and did not warp and peel up at the corners. Ensuring the accuracy of the shape





Step 2 – Pattern Surface Finish

Both sides of the pattern were lightly sanded to remove most of the layer lines which would otherwise be imprinted in the mould surface.

The pattern was then test fitted inside the handle pipe to verify that the final casting would fit.

Comment: Removal of the side of the pattern with the runner built in was quite difficult, as the sand gripped the sides of the tall cylindrical sections well. Multiple attempts were needed to produce this half of the mould without parts collapsing, cracking or being squished/distorted out of shape. If I were re-making the cast hook/threader I would put a greater draft angle in these areas.



Step 3 - First Half Of Mould

Petrobond Sand was sieved though a mesh into the drag and gradually compacted around the bottom half of the pattern. This was done with a wooden dowel rammer.

The mould half was upturned and coated in parting powder. The pattern half was replaced and the cope was put on top of the drag.



Step 4 - Second Half Of Mould

The top half of the pattern was placed onto the bottom half.

Petrobond Sand was sieved though a mesh into the cope and gradually compacted around the top half of the pattern. This was done with a wooden dowel rammer.

Sprue pins were aligned to the riser and runner modelled into the 3D print.



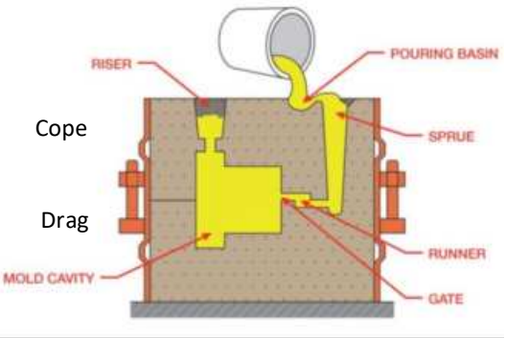


Step 5 - Forming Pouring Basin

Both halves of the pattern were removed and the quality of the mould cavity checked before assembling the complete mould.

A poring basin was dug out from the top of the mould and walls formed to guid molten metal into the runner and prevent spillage over the floor.

Sand Casting Diagram







Step 5 - Aluminium Casting

Aluminium heated with a gas furnace until molten in ceramic crucible and flux added to remove oxides.

Appropriate safety precautions (see Risk Assessment page) taken while pouring metal into the mould.

The part was allowed to cool for 4hrs before being removed from the mould.

Step 6 – Removing Riser/Runner

The channels where metal was poured into the mould were sawn off with a hacksaw.

The unburned mould sand and excess Aluminium can be reused.

A course File was used to hide where the riser and runner had attached to the casting.



Comment: Slippage of pattern halves over each other caused a slight misalignment between the two halves of the mould which created a step in the middle of the final casting. To correct for this, I could have used dowels to temporarily join the pattern halves together. The pattern halves could not be permanently joined together as a flat bottom surface is required for the third step when the drag is upturned. In the end most of the step could be hidden by with by filing it away but doing this and keeping profile of the cylindrical part circular where it attaches to the handle was difficult. Imperfection pictured right.





Step 7 – Filing The Casting

Files were used to clean up the shape of the casting and remove imperfections.

Round needle files were required to clear notches that the latch assembly pivots between.

Wet and dry silicon carbide paper was used to improve the shine of the aluminium.





Step 7 – 3D Printing Rubber Cap

The rubber cap was 3D printed from TPU flexible filament at 100% infill in 20 minutes.

This was then heated with a heat gun to smooth it's surface.

Two-part epoxy was used to glue the rubber cap onto aluminium casting.

Comment:
The TPU rubber cap was heated with a heat gun to partially melt and smooth its surface, something I only discovered was possible after printing it. The top blue part is a prototype to test the fit to the casting, the middle part had 15% infill to try to be more squishy but top layer sagged when heated, so I reprinted as solid and was able to achieve a smooth surface finish.

Manufacturing Evidence 2 – CNC Hook/Threader

Communication Timeline With Whitehouse Engineering

03/12/2025 - I sent an email introducing the project and asking if they would be able to help me CNC machine some parts. This included photos of the laser cut prototype. hook/threader mechanism.

05/12/2025 – At their request I shared my CAD design files as a 3D .step file containing all parts of the CNC Hook/Threader and 2D .dxf files of the shape of thin metal parts.

14/01/2026 – A Microsoft Teams call was organised where we discussed the technical requirements of the parts, considering their material and surface finish. Even after FEA simulation I was unsure if aluminium would be stiff enough to not bend.

27/01/2026 – Whitehouse Engineering 3D printed a full working prototype from PETG – HF and sent it to me for testing, so we could check that all the parts fitted together and the mechanism functioned correctly.

28/01/2026 - I received some recommendations of minor tweaks to the design: Adding more clearance for bolts, TPU spacers for latch with protrusions on notch edges to grip them and the idea of making the side spacers from Orange ABS.

12/02/2026 – The CNC machined parts arrived (Latch 1, Latch 2, Gate) and 3D printed parts arrived (Side Spacers x2) ready to be assembled.

Step 1 – Final Working Prototype

Once I sent the design files to Whitehouse Engineering, they were able to 3D print a PETG plastic version of all of the parts in the 3D model I sent.

This was sent to me, assembled with two M4 and two M5 bolts. This allowed me to fully test the threader mechanism cheaply before we committed to producing the aluminium parts.

I tested the fit of the side spacers and gate into the handle I planned to use for the actual device.

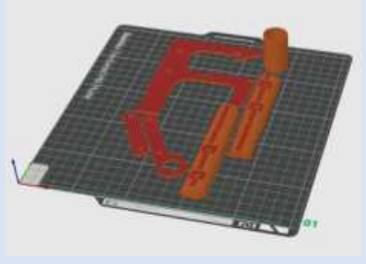












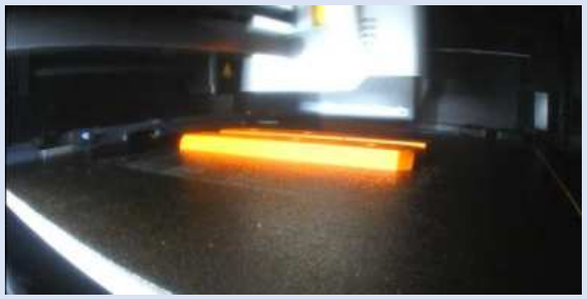
Step 2 – Wait For CNC Machined Parts To Arrive

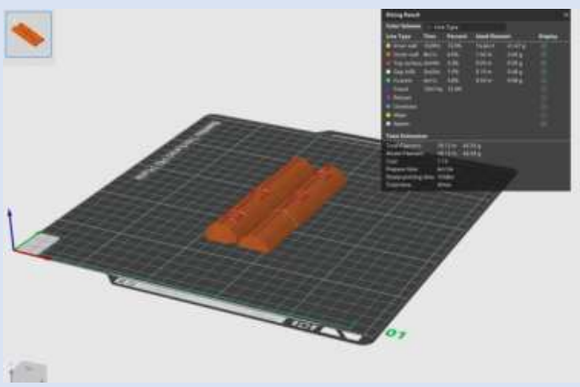
Once the final shape of the Hook/Threader Mechanism had tested and finalised I gave the go-ahead for the aluminium parts to be machined.

Orange side spacers were 3D printed with 100% infill out of Orange ABS Plastic to provide the maximum weight and durability.

The parts were sent to me and assembled, with two M5 bolts attaching the side spacers to the aluminium gate and two M4 bolts attaching Latch 1 and Latch 2, with the TPU them.

The handle was prepared by sawing off the broom connector then straightening the cut by pushing the end against the belt sander. The CNC Hook/Threader component was clamped in the milling machine vice inside the pipe handle so that the mounting hole could be drilled.







Manufacturing Evidence 3 – Latch Assembly

Step 1 – Program CNC

A .dxf file of the shape of latch 1 was exported from Fusion.

The Denford Router Software was used to generate a toolpath to program the CNC router to cut out the shape.

Comment: Tabs and Double-sided tape kept the part from moving once it was separated from the stock material.

Comment: CNC machining instead of free hand cutting was required to produce the shape of Latch 1 accurately so that pivot points are in the exact locations they were planned to be.







Step 2 – Cutting out Latch 1

A 3mm Aluminium sheet was screwed to a block of wood held securely in a vice to hold the workpiece.

The CNC router was run at the correct feed rate and spindle speed for aluminium. A 6mm Endmill was used to cut out the shape in multiple passes.

The Part was then removed from the CNC machine.



Step 3 – Deburring & Finishing Latch 1

Files were used to take off sharp edges left by the CNC Router. A round file was used to smooth the 16mm hole edges, so they do not slice the rope that will pass through It.

Silicon Carbide Wet and Dry paper was used to remove scratches and shine the aluminium part, before it was joined to latch 2 so all areas were accessible.



Step 4 – Cutting out Latch 2

Latch 2 has a much simpler rectangular shape so was cut out manually with a hacksaw.

The edges were then squared with a facing bit on a milling machine to create straight, flat and parallel sides.



Step 5 – Rounding Corners

To reduce injury if the part contacted MOB and improve aesthetics corners were rounded.

This was done repeatably by using the laser cut prototype as a template, clamping it in the vice next to the aluminium rectangle. This allowed the profile of the corners to be matched with a hand file.



Step 6 – Drilling Holes

A hammer and centre punch were used to create a divot to guide the drill bit. 4mm Diameter holes were drilled with a pillar drill. Latch 2 was held in a clamp so that holes could be positioned perfectly.
Once Latch 2 was done the holes for Latch 1 were marked through the holes in Latch 2 so the parts would be held aligned once joined with spacers between holes.





Step 7 – Turning Aluminium Spacers

6mm Aluminium Bar was turned into two spacers, spacers had shoulders where diameter increases to hold Latch 1 and 2 a fixed distance apart.

First the stock was faced off then a parting off tool was used to reduce the diameter to 4mm on either side then the spacer was parted off from the stock. This was done for the first time on a manual lathe.







Step 8 – Welding Spacers

My original plan was to Use a very small low current TIG welded tack to join the aluminium spacers to latch 1 and 2 permanently without adhesives. This was attempted, but controlling the power delivered to get parts to fuse together without warping them was not possible and ruined the aesthetics and dimensional accuracy of the parts.









Step 9 – Remaking Latch 2 and Spacers

Because I was not happy with how the welding attempt looked or performed, I decided to quickly remake Latch 2 (Steps 4-7) This threw off my manufacturing schedule, but time was saved by making the new spacers with a CNC lathe. This took about 2 minutes per spacer compared to the 25 minutes It took to do it manually. The new spacers were also made 1mm longer to account for design change to the casting thickness. Instead of welding a drop of super glue was used to join the parts.

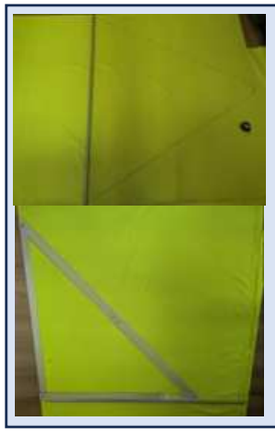


Manufacturing Evidence 4 – Rescue Sling



Step 1 – Paper Template

A 1:1 scale template accurately marked out on grid paper with the dimensions and angles for the triangular section.



Step 2 – Marking Out Shape

An angle section was used to provide a straight edge and a tape measure used to measure the length of each edge.
A very large folding aluminium square was used to make sure lines for edges are completely perpendicular. Dry wipe marker and pencil used to draw lines for good viability on yellow while being removable after cutting.



Step 3 – Cutting Out Shape

An electrically heated hot knife was used to cut out shape and melt edges to minimise fraying. This was done with the window open to reduce inhalation of fumes.
The fabric needed to be pulled apart as it was cut to prevent it melting back together behind the knife.



Step 4 – Pining/Taping/Clipping

Strips 100 mm wide of edging tape were folded in the middle, so they fit around the edges of the yellow fabric with 50mm on each side.
They were then kept in position with double sided 'Basting' Tape, Pins and small Clips at regular intervals.
All edges had edging tape attached except for two left open to allow EPDM rubber to be slotted in later.



Comment: Industrial Sailmakers Sewing Machine used for some sections, such as the triangular cutout where tearing is more likely. This machine permitted much thicker thread, with greater strength to be used and had a stronger needle capable of penetrating red webbing. Standard Sewing machine was used for outside edges.





Step 5 – Cutout Corner Reinforcement

Extra reinforcement was added to the inside corners of the triangular cutout. This was a change to the design, originally there was no overlap between edging tape covering each side so the corner would have been a weak point, where tears could start. Circles were cutout from the edging tape Dacron material, then taped into position one on each side of the corners.

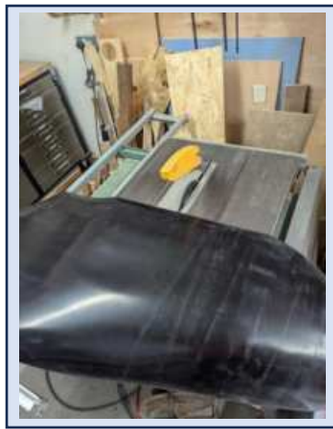


Step 6 – Sewing To Join Together

Each strip of edging tape was sewn on with two runs of stitching each 15 mm from the long edges of the tape, so they were securely bonded to the yellow airtex mesh fabric.
The outline of the circular cutouts was followed. The longest stitches possible were used to minimise the effect of the sewing machines slipping, struggling against the weight/thickness of the heavy fabric.

Comment: EPDM Rubber ordered as a 6mm thick sheet, not strips as in final design. This was much more economical than buying pre-cut strips.
EPDM was cut on a table saw, though if was quite difficult to get straight and parallel cuts because of its flexibility and it frequently getting jammed in the gap between the saw blade and table.
I ended up significantly modifying my final design which originally had many thin short strips of EPDM running widthways to one with four larger panels (see Changes During Manufacture page for more information)

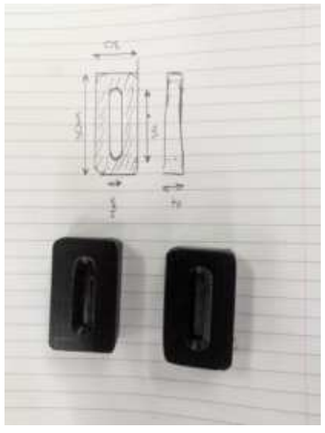




Step 7 – Cutting Out Rubber Pannels

A table saw was used to make two cuts in the sheet of EPDM to split it into four separate panels.









Step 8 – Making Toggles
Block of Delrin plastic roughly cut out with a bandsaw then face mill used to machine to accurate size and flatten faces.
4mm bit used to create a 27mm long slot. Multiple passes made lowering Z axis by 2mm each time and using digital readout to move the correct distance across.
A v bit was used to chamfer the edges of both sides of the slot so that they do not cause abrasion against webbing.
The outside vertical edges were also rounded using a belt sander.
The 3D printed Rail Section Replica was used throughout manufacture to check fit.



Step 8 – Attaching Toggles & Rings

EPDM rubber panels slotted into edge previously left open, to be encapsulated between two layers of yellow airtex fabric.

Loops of Red Polyester webbing attached by sewing a small rectangle then crossing diagonals to add more stitching and strength. Toggles and Round O Ring and two D rings threaded onto loops of webbing and attached to the sling.

Testing and Evaluation 1

Setup Test

My client and I were able to unpack and setup the sling (Figure 83). First both the hook/threader and sling components of the rescue system were removed from storage. Next the sling was laid across the boat while the toggles were attached and D rings tied to guardrail posts. Before pushing the sling off into the water the Main Sail Halyard was shackled to the Round ring at the top of the sling.

One issue that was noted was that the EPDM panels embedded in the sling to make it sink, add a substantial amount of weight (6kg) making carrying it quickly quite difficult, though the distance between its storage to the deck edge is only a few meters.

The time to setup the device was lowered by folding in a specific way so that all attachment points are visible while folded, allowing it to be oriented correctly more quickly. More practice, more people to help or better co-ordination could have reduced the setup time considerably.



Figure 83



Figure 80



Figure 81



Figure 82

Storage Test

Rescue Sling Storage

The design change during manufacturing to switch to a four panel EPDM layout allowed the sling to fold to perfectly fit the hatch size into the under-seat cockpit storage space (Figure 82). Being stored here offers excellent accessibility in an emergency, without having to go below deck or towards bow storage spaces that are more exposed to wind and waves. This storage area has pumps to remove any standing water but still gets quite damp, material choices were made to be resistant to moisture with to mitigate this.

Hook/Threader Storage

The hook/threader can be stored tied with elastic to the guardrail (Figure 80 & 81) , easily within reach from the cockpit. Elastic could be tied between the two sides of the gate area of the hook/threader to prevent it sliding off the end. The device would be secured even better if a channel for the elastic tie to sit in was added close to the handle end to prevent the elastic sliding off the end.

Being stored on the guardrail will lead to constant exposure to the elements. Use of UV stable ABS plastic for 3D printed parts and corrosion resistant aluminium for CNC or cast components will allow the hook/threader to withstand these conditions. The stainless-steel bolts selected are black oxide coated to provide high strength while having a thin layer separating them from contacting aluminium parts and causing galvanic corrosion in saltwater spray .



Figure 84

Ease & Strength Of Attachment

The combination of two toggles and two D rings to join the sling and deck edge was a good choice. The toggles (Figure 88) can be slotted in quickly and provide a secure connection that will not come undone. Even though tying the corners to guard rail posts (Figure 87) may take slightly longer, with rope the corners can be pulled taught (force in direction of arrows on Figure 86) reducing the gap between the boat and sling.

The Halyard used for lifting was attached to the round ring on the outboard side of the sling.



Figure 85

Electric Windless Turner

My client recently bought an electric windless turner (Figure 85) which was used during testing. This has a battery pack and high rpm electric motor to rapidly turn the winch.

Though using this was not planned, it Improves the experience of using the sling, removing all physical effort required and significantly increasing speed of lifting over manually turning the winch. A torque limit can be set to avoid causing injury to the MOB if a part of the sling got snagged. Also, given that it can be swapped for a manual winch handle if the electric turner fails, there is no reliability compromise.

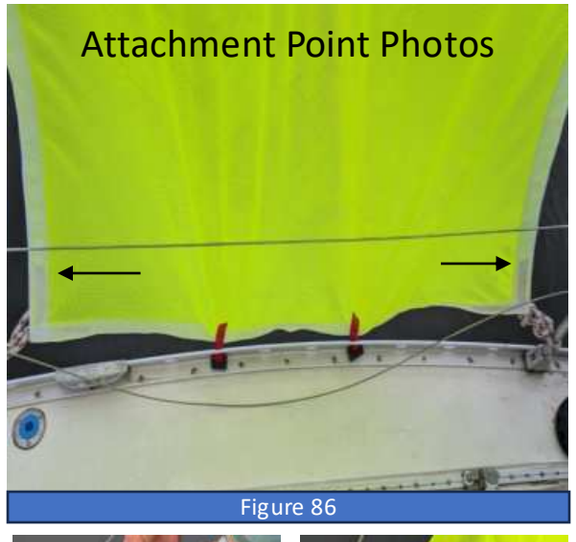


Figure 86

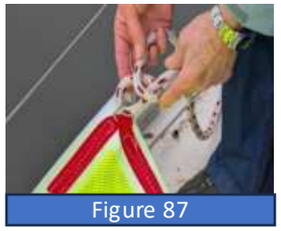


Figure 87



Figure 88

Land Testing Cast Hook/Threader



Figure 89

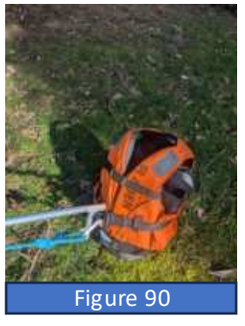


Figure 90



Figure 91

Due to a slight delay in producing the cast version of the Hook/Threader testing on the water was not possible.

Instead, I put a non-inflating foam child's lifejacket onto a backpack filled with weights. This functioned as an analogue for the torso of an MOB.

I was able to hook onto the lifting loop from a distance and drag the 30kg backpack, proving the hook part was strong enough to move someone though the water (Figure 89).

I also "secured" the backpack by attaching a rope though a strap on the front of the lifejacket with the threader mechanism (Figure 90 and 91). The cast version worked the same as the 3D printed prototype.

Using the Latch assembly with aluminium spacers meant that tension had to be kept on the rope to prevent the latch falling out of the gate, as was expected.

Testing and Evaluation 2

Physically testing a full rescue was a must to properly evaluate the functionality of the system. This was done safely on a calm day in a relatively sheltered area inside Cardiff bay, while the boat was stationary moored to a pontoon.
In reality the conditions where a MOB is likely to occur are in extremely strong winds when the sailboat is tilted and tall waves, so a real rescue would be more difficult.

Measurable Criteria From The Specification Tested

A Stopwatch and Tape measure were used to collect quantitative data during testing on the performance of the system.

Setup Time: 2 minutes and 30 seconds (less than 3 minute requirement)
Rescue Time: 4-5 minutes (In ideal conditions)
Hook/Threader Reach: 1.5m with arm extended, 1.2m without arm extended
Lifting Height: 1.2m (successfully lifted from water line to deck)
MOB weight: Tested up to 70kg (105kg weight limit target not reached)



Comment:
It was at this point I began to slightly regret making the device over the winter.



I jumped into the water wearing a wetsuit to stay warm and an auto-inflating lifejacket to simulate the scenario of a Man overboard. The lifejacket had a pressurised CO2 cylinder and chemical sensor that triggered upon contact with the water, filling the air bladders and p roviding buoyancy. This accurately mimics what would happen in a real MOB scenario. Throughout the test I tried not to provide any physical or verbal help to prove it is capable of rescuing an incapacitated MOB, the user the had access to the pictographic instructions as shown on slide 26.





Comment:
The threader mechanism was quite difficult to get working and to position precisely at the end of a long pole with quite a small lifting loop as a target. My client suggested enlarging the mechanism by 25% could fix this issue.
Comment:
Showing the exact motion required to use the threader was difficult to do with only photos.

Description: First the CNC hook/threader was used to attach a rope to me and prevent me drifting further away. The lifting loop (yellow and red) on this lifejacket popped out of the front, unlike the one depicted on page 25.

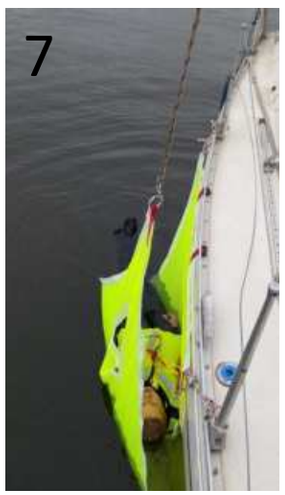


Once I had been secured by a rope the sling was pushed into the water and the hook/threader pole used to spread it flat to create a bed area below the water, where EPDM rubber sheets provided rigidity.

Comment:
Using the hook/threader to open the sling before sliding in the MOB was not something I had considered when designing it, but it seemed to be quite a useful feature to help prepare the sling.



The hook part on the CNC hook/threader was used to drag me backwards guiding me onto the sling, which sank in the water below me.









Once I was inside the loop of the sling the halyard attached was used to steadily lift me out of the water . This was done from inside the cockpit with an electric windlass winder pulling the rope in at the push of a button. Towards the top I was rolled over onto the deck as the sling became taught.

Comment:
The 0.65m width of the reinforced flat section was wide enough for me to be slid in easily even with the extra space taken up by the lifejacket.

Comment:
The guardrail wires did not have to be taken down as I fit underneath the lowest wire while lying flat. This saved time during the rescue.

Comment:
The experience of testing using the device was not uncomfortable, fabric and rubber provided some padding between me and the side of the boat.



I was successfully secured and lifted back into the boat in about 4-5 minutes this is much faster than the 11 minute maximum in cold water suggested by the UK's Marine Accident Investigation .

Evaluation Against The Specification 1

Over the next few pages I will evaluate the performance MOB rescue system against each of the specification point requirements. Each area has its own table with individual specification points referenced using their code. To get an overall sense of how well my prototype meets the needs of the client, I have assessed to what extend it met the criteria of each specification point (Fully/Partially/Not Met).

Purpose and Function

Specification Point	Evaluation	Fulfilled Requirement
(F1)	The Sling component proved capable in testing of easily lifting an MOB wearing an inflated life jacket the required height of 1.2m. We were unable to test the device lifting its maximum MOB weight of 105kg but given the lack of signs of being close to failure at lower weights was reassuring. The design of the device took into consideration the extra width taken up by a lifejacket so that the EPDM panels which cause it to lie flat under the water were made to be wide enough. The lifting loop present on lifejackets was determined to be the best way to secure the MOB with the hook/threader. By either threading a rope though the loop or hooking onto it the MOB could be moved onto the sling for lifting.	Fully
(F2)	The need to be able to secure an MOB 1.50m away from the topside (Outboard distance parallel to the water from the side of the boat). To be able to reach that far from a deck level of 1.20m, Pythagoras theorem (page 17) was used to work out that a 1.90m handle would be required. Reuse of an aluminium broom handle allowed me to save costs and manufacturing time, but the handle available was only 1.45m. The requirement can still be partially met by including the extra 0.45m of a rescuers arm in the reach calculation. Extra length would be preferable, a future design a telescopic handle would optimise this.	Partially
(F3)	The sling both creates its own mechanical advantage by using a parbuckle design to half the force required for lifting and takes advantage of windlass winches available and the main sail halyard to pull upwards. This made lifting easily manageable as winch used had a 30:1 ratio. Handles included in the original design idea that would have spread the load and provided grip were removed as I thought they may tear the fabric, but in theory the reinforced fabric edge or halyard attachment ring could still be held by multiple people and distribute the load without to help with lifting without needing to attach to the halyard.	Fully
(F4)	The chosen design idea split the two functions of lifting and securing to avoid over complicating a single device and causing confusion to an inexperienced rescuer. Picture instructions (Page 25) produced to be stored with the device so that anyone aware of stowage location can learn how to use it quickly. Clearly visible large toggles and metal rings that do not require fine motor co-ordination to attach so could be done with gloves on in cold/wet/panicked conditions. EPDM rubber panels provided enough weight along with the triangular cut-out lowering side area, to reduce flapping in the wind compared to the sling design I tested. Extra rigidity from EPDM panels also provides in a flat bed under the water that kept the sling open and prevented it being blown back over the boat, making it much easier to slide a person into it before lifting.	Fully

Materials

Specification Point	Evaluation	Fulfilled Requirement
(M1)	Dacron (brand name for a PET fibre) white edging tape specifically designed for reinforcing sails was used. This has excellent resistance to UV, as well as low absorbency to prevent mildew and is also resistant to stretching. Airtex Mesh Polyester Fabric is advertised as being usable for sunshades with very good lasting UV tolerance. Delrin (brand name for an Acetal Polymer) used for black toggles has stabilisers so should be adequately protected. It was mainly selected for its machinability, wear resistance, toughness and low friction, that make toggles durable and easy to fit. ABS plastic used in orange side 3D printed spacers on CNC hook/threader has acceptable UV resistance, much better than other 3D printable polymers like PLA. ABS parts will eventually degrade, but they are not strength critical components.	Fully
(M2)	A4 Marine Grade Stainless steel was chosen for 2 x D rings and 1 x round ring attachment points, offering good corrosion resistance while having strength to support the weight of a person. Aluminium was used for: the handle, cast hook/threader, CNC hook/threader and latches for each. This reduced weight compared to steel and made machining and casting possible. No direct contact between any Stainless Steel and aluminium parts, even black oxide coated bolts chosen to create a barrier and prevent galvanic corrosion.	Fully
(M3)	Delrin Temperature Range: -40 to 90 C Dacron Fabric Temperature Range: -60 to 130 C Airtex Mesh Polyester Fabric Temperature Range: -40 to +70 C ABS Temperature Range: -20 to 80 C All plastics are rated for a wider temperature range than the -20 to 60 C expected most extreme conditions.	Fully
(M4)	Finding information on the flammability of the specific fabrics was not possible, if they were treated with flame retardants was not included in their specifications. If the device were to be produced on a larger scale, suppliers could be asked directly. Polyester's which make up most of the project by weight (Webbing, Airtex Mesh and Dacron Edging Tape) are generally fine as they melt before burning and can shrink away from heat sources. Combustion of Polyester can produce carbon monoxide and some toxic VOC's, but these are lower risk than HCl, HCN or NOx gasses released by PVC or Nylon.	Partially
(M5)	Webbing rated to hold a 1000kg load which is probably a factor of ten higher than it will ever need to hold. Stainless steel rings have a breaking load of 480kg. These are exactly the type of large safety margin to create confidence that was requested by my client. I physically tested that the strength of the fabrics would be sufficient by producing a sample piece of the sling (page 20). FEA simulations were used to verify the thickness and material of the hook would be enough to drag an MOB though water, though I think the casting may be more brittle than expected.	Fully

Evaluation Against The Specification 2

Safety		
Specification Point	Evaluation	Fulfilled Requirement
(S1)	All lifting can be done by operating the winch that inside the cockpit area, so the chance of the rescuer falling in is low. Leaning over the edge during setup is not required as halyard attachment point is high enough to be level with the deck when the sling is fully lowered. The long handle of the hook/threader means reaching down towards the MOB is not necessary as they can be secured and manoeuvred at a distance.	Fully
(S2)	Both hook/threader prototypes have rounded corners and edges where possible to reduce their sharpness, if they were to contact the MOB during hooking or threading of rope though lifting loop, they are less likely to cause injury. The cast version of the hook/threader features a TPU rubber cap that would cushion some of the impact if it were to hit the MOB. The most outboard vertical side of the sling will provide some protection against spray since the mesh fabric does not allow splash through, but the triangular cutout may allow some water to splash over the top of the MOB. Also, if waves are perpendicular to the sling/boat no shielding is provided in this direction.	Partially
(S3)	During testing a stopwatch was used to determine unpacking and setup time. My client an I both helped setup the device and It may have taken longer with only a single rescuer. On the other hand, with more practice my client and his regular crew could probably reduce the setup time significantly.	Fully
(S4)	The device lifts the MOB in a face up position lying flat on their back, this provides the optimal support and orientation to provide comfort and reduce medical complications. The MOB does get rolled over slightly as they get close to deck level.	Fully

Cost		
Specification Point	Evaluation	Fulfilled Requirement
(Co1)	Cast Hook/Threader Sub Assembly: £20.50 CNC Hook/Threader Sub Assembly: £8.20 Rescue Sling Sub Assembly: £103.00 Total bill of materials: £131.70 Total material costs lower than target of £150 by 12%. If my device were produced commercially labour costs would also have to be factored in. This includes both hook/threader options produced, where only one is needed. Material costs were minimised by: • Ordering the EPDM rubber as a sheet not in strips, sheet form was cheaper as it is commonly used in rubber mats/flooring/roof seals. • Purchasing the minimum required amount of material for each part and avoiding wasting material. • Choosing more economical material options where possible, for example use of aluminium instead of stainless steel in hook/threader.	Fully
(Co2)	Gantt charts were used to schedule production, though there were some delays caused by factors outside of my control such as delivery time or tasks taking longer than expected. Overall, I spent roughly 22 hours manufacturing the prototype over 6 weeks. Though the 20 hr goal was not quite met I still was able to produce the device in the required timeframe. The device could be produced more quickly with less work hours to reduce labour costs by automating processes.	Fully

Client		
Specification Point	Evaluation	Fulfilled Requirement
(Cl1)	During testing I confirmed that the sling is not uncomfortable from the MOB's perspective. Choosing to produce from soft fabric and rubber unlike other wooden and metal options means that it conforms to fit the MOB well and provides even support. This will hopefully incentivise my client to practice using the device with new crew.	Fully
(Cl2)	No maintenance needed, apart from possibly a visual inspection after use to check for fabric tearing or metal bending of the threader gate. Use of moisture resistant non absorbent polyester lowers the time that the sling must be left to dry after use to only a few hours. Aluminium and stainless steel bolts do not need oiling or painting to keep their corrosion resistance.	Fully

Aesthetics		
Specification Point	Evaluation	Fulfilled Requirement
(A1)	Unfortunately, my client's preference for a dark wooden look was not possible to achieve while meeting safety requirements for visibility and would have negatively affected the sustainability, durability and cost of the device. Maybe a future iteration of the device could reach a better compromise by having a wooden handle for the hook/threader, though this would prevent it being stored outdoors attached to the guardrail.	Not Met
(A2)	Fluorescent yellow fabric is highly visible, making locating the device quickly possible and alerting other boats of the situation if the device is deployed. Red webbing contrasts against yellow making it easy to locate attachment points and orient the sling. Orange ABS was used to 3D print the side spacers for the CNC hook/threader so that on the end of the handle it's bright colour stood out against water so that it can be positioned accurately. In a commercially produced version, aluminium parts would be anodised orange to match the side spacers visibility and have even better durability.	Fully
(A3)	The precision of CNC cut Aluminium parts which were wet and dry sanded to a smooth surface finish creates a modern look. The casting was manufactured with only minor visual defects, so it has a fairly smooth 3D shape. The sling has a robust appearance due to the large size of toggles and extra reinforcement circular patches on the corners of the triangular cutout. These features should inspire confidence in the device, making it more likely to get used in an emergency MOB scenario.	Fully

Evaluation Against The Specification 3

Manufacturing Process		
Specification Point	Evaluation	Fulfilled Requirement
(P1)	The laser cut prototype cut from 3mm mdf was used as a template to guide filing while adding radii to the corners of latch 2 parts. This was achieved by clamping against the aluminium version in a vice, to match the profile of the corners. A paper template was also produced with the correct angles and size for the triangular cut-out off the sling for accuracy.	Fully
(P2)	Gantt charts used to break down processes into manageable steps and schedule what should be achieved in each of the manufacturing sessions available to me.	Fully
(P3)	A full working prototype was 3D printed by Whitehouse Engineering and sent to me for testing before I gave the go-ahead for the aluminium parts to be cut. This was attached to the handle and rope was threaded though a loop on my school bag to test if it could reliably secure a “person” at a distance.	Fully
(P4)	I forgot to compensate for shrinkage of the cast hook/threader which reduces an objects size by 1-2%, shrinkage measured was only 1.6 % and did not impact the function of the mechanism. Shrinkage calculation: (Pattern Length - Casting Length)/Pattern Length x 100 (119mm – 117mm)/119mm x 100 = 1.6% Lots of time was spent filing the cast hook/threader to remove the step produced where patterns were misaligned, to produce an accurate shape. Use of Computer Aided Manufacture, such as 3D printing orange side spacers and having some aluminium parts CNC machined created parts with high precision.	Fully

Size		
Specification Point	Evaluation	Fulfilled Requirement
(Si1)	The sling is able to fold with the four EPDM rubber panels being placed on top of each other, this makes it slightly smaller than the hatch into under seat cockpit storage. The sling easily fits this space without careful packing required. This storage spot is easily accessible from the cockpit area in an emergency. The hook/threader is thin enough that it can be tied to the guardrail with elastic without obstructing the view while sailing significantly.	Fully
(Si2)	Anthropometric data for weight, shoulder width and ankle to chest height were used so that the sling could be used by the 95th percentile of each. This means that most dimensions are oversized for the average user, but due to the flexibility of EPDM rubber sheet and Fabric the device is still comfortable/usable for smaller lighter people.	Fully
(Si3)	The size of the sling fits fully into the space envelope defined in my research. The length of the product was limited to 1.5m to be smaller than the 1.8m distance between guardrail posts, while also being a multiple of the 150mm gap between holes in the deck edge rail. The latch has a 16mm hole - larger than any of the 12mm ropes on board, so the hook/threader mechanism could be used with what was available.	Fully

Sustainability		
Specification Point	Evaluation	Fulfilled Requirement
(Su1)	Aluminium: Infinitely recyclable, using less energy than new refining. Polyester(Dacron): Does not biodegrade but, is the most recycled plastic globally, used to make rPET plastic soft drink’s bottles where quality of polymer is less important. Delrin: Thermoplastic so can be recycled, but quite uncommon so not usually accepted by normal recycling companies. Plant derived Delrin made without fossil fuels from sugar cane or corn is being developed. EPDM Rubber: Very difficult to remelt and recycle but can be shredded and used in construction for things like playground surfaces. ABS Plastic: Commonly used for toys and car interior panels, can be easily recycled but not accepted in standard residential recycling.	Partially
(Su2)	No adhesives was used in the CNC hook/threader with bolts used to secure the side spacers on and attach the mechanism to the handle. These could be undone quickly and easily. The Latch with Aluminium spacers use a very small amount of super glue to join the spacers to latch 1&2. Since the parts joined together are Aluminium, it could be recycled as one piece. The sling could be disassembled into individual materials by cutting out the EPDM panels and Stainless-Steel rings. Webbing and Airtex Yellow Fabric are both polyesters so could be recycled together.	Fully
(Su3)	Durable material choices were prioritised to increase the lifespan of the product so that a replacement will probably never be required. Reuse of a standard aluminium broom handle with an injection moulded grip as the handle for the Hook/Threader’s will have lowered the devices carbon footprint compared to manufacturing a new handle.	Fully

Assessment Of Specification Compliance

Area	Proportion of Specification point requirements fully met
Purpose and Function	3/4 (Partially Met F2 – To reach 1.5m arm length must be included)
Materials	4/5 (Partially Met M4 – Information on flammability not available)
Safety	3/4 (Partially Met S2 – Does not provide great splash protection)
Cost	2/2
Client	2/2
Aesthetics	2/3 (Not Met A1 – Does not match teak boat interior style)
Manufacturing Process	4/4
Size	3/3
Sustainability	2/3 (Partially Met Su1 - Some materials not easily recyclable)

Total	**25/30 Fully Met**	**4/30 Partially Met**	**1/30 Not Met**

Changes During Manufacture

Triangular Cut-out Corner Reinforcement

Disks of white Dacron with a 60 degree sector removed on inside corners of cut-out added on both sides sandwiching the yellow airless fabric. Originally the gap between edging tape strips could have acted as a weak point for tears to start, ripping the sling in half. The circles add extra tear resistant material and act as a bridge between the strips of edge tape. A compass was used to draw the 100mm diameter circle then scissors used to cut the shape out from the same Dacron strip/tape as was used for the edges, but without folding it. Circles were then slotted underneath the edging tape so that when I went over both with the sewing machine they were bonded together and to the fabric. This change was added to the final design drawing and flowchart of the sling but not renders.

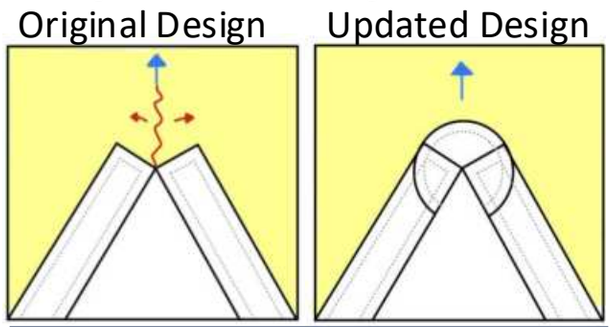


Figure 92



Figure 93


Figure 94

EPDM Rubber Layout

A different layout (Figure 95) for the EPDM rubber reinforcement in the flat "bed" area of the sling with four larger panels instead of the widthways strips planned originally .

The change was made for several reasons:

- Cutting the rubber safely was difficult as it got jammed in the saw frequently and produced a lot of potentially toxic fumes as the edges melted. The final layout of rubber reduced the number of cuts needed. If I were to do this again, It may have been possible to score the rubber with a utility knife and cut it by making multiple passes instead of using a table saw.
- The EPDM rubber was also slightly stiffer than I expected so I decided a lengthways cut would be required so that the four panels could fold on top of each other to better fit the storage area available since the new design can fold in two directions.

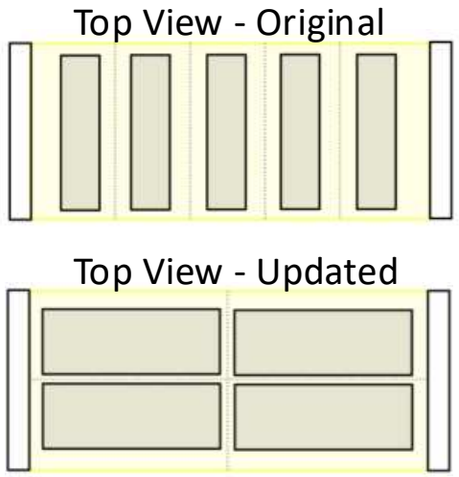


Figure 95

Casting Thickness

Late in production, I decided to print a second version of the casting mould pattern with 4 mm (Figure 96) in the thinnest areas instead of 3mm. This is because I wanted to be sure that these areas of the mould would fill, to avoid wasting time remaking the mould if the casting failed. Originally the thinnest parts of the cast hook/threader were the same thickness as the 3mm sheet aluminium used for the CNC Hook/Threader, so the latches with different spacers could be interchangeable.

This modification led to the need for a second set of aluminium spacers between latch 1 and latch 2. These needed to be 1mm longer than the original. To save time, they were produced on a CNC lathe instead of manually (Figure 97 and 98).


Figure 96


Figure 97


Figure 98

Future Improvements

Based on the areas of the specification that were partially or not met and direct feedback from my client, there are a few minor possible improvements to the design that I would make if another iteration was to be attempted in the future. Most of these changes are based on knowledge gleaned during testing and evaluation, unknown at the design/development phase.

Wooden Grip for Handle (Relates To A1)

One of my client's requests during our interview was for the device to match the aesthetic of his boat. His boat is produced from white GRP with a wooden teak interior. I was unable to match any aspect of the device to the hardwood interior.

Turning a cylindrical hardwood grip on a wood lathe would address this and could improve the aesthetics of the device giving it a more premium feel. In my final design I chose to prioritise durability, with the injection moulded plastic grip from the broom handle fitted.

On reflection, wood may be able to withstand the elements with an appropriate surface finish, such as being dipped in epoxy/polyester resin or having a thick layer of varnish applied to completely seal it from water.

My grip design could be similar to Figure 99 and 100 pictures sourced online, with a curved shape and ridges to provide grip.


Figure 99


Figure 100

Telescopic Extendable Handle (Relates to F2)

The target reach of 1.50m from the side of the boat at the waterline was only achieved when 0.45m of the rescuer's arm length was included in the total reach calculation. To be able to secure a MOB at the maximum distance, the rescuer must fully extend their arm while holding the grip with one hand. This posture makes it difficult to adequately support the handle and accurately position the Hook/Threader.

Increasing the handle length from 1.45m to more than 1.90m would enable the rescuer to use both hands to support the handle while reaching at the maximum required distance, improving control.

To be able to increase handle length without creating storage issues a telescopic mechanism could be incorporated, to allow for the handle to extend during use and retract for storage. The simplest possible design would consist of a smaller tube sliding within a larger tube, secured by a detent pin to lock the handle at any length.


Figure 101



Figure 102

Larger Hook/Threader Mechanism (From client feedback after testing)

My client suggested increasing the size of the hook/threader mechanism to improve accessibility . The current hook/threader mechanism looks quite small from a distance when it is on the end of the handle . Making it larger would improve visibility allowing the hook and gate part of threader mechanism to be positioned and oriented more easily.

Recyclable EPDM Alternative (Relates to Su1)

EPDM Rubber was picked to add rigidity to the sing for its specific properties well suited towards the role. I required a durable, inexpensive material with the correct stiffness that is also denser than water so that it sinks. EPDM makes up a significant proportion of the device by mass and is the only material chosen that cannot be easily recycled, apart from being shredded to use in construction. EPDM is not a thermoplastic so cannot be melted down and remoulded. Finding a more sustainable elastomer with similar properties would reduce the environmental impact of the device.

Making The Hook/Threader Float In Water

This was not a feature requested by my client so was not considered as a requirement. If either the cast or CNC machined versions were dropped overboard during a real rescue or training, I believe they would likely sink immediately. Being made of either solid aluminium or a combination of aluminium and ABS plastic means they have very little buoyancy. I think the buoyancy could be improved by Injecting the inside of the aluminium pipe handle with expanding spray foam. This would displace the water that would otherwise fill the handle with a material full of air pockets.

Client Feedback

Overall, my client was very happy with the final rescue system produced. Its solid construction and simple operation fulfilled his key functionality, aesthetics and safety goals of the device. Some areas of the prototype could be improved but it meets all most all of the specification requirements we agreed upon.

Direct Feedback:

"I'm really pleased with how the design turned out, and am looking forward to permanently installing it on the boat as soon as you have finished testing it and written up the evaluation"

"much better than anything I could have bought"

"We proved that it works, so I would feel confident relying on it in an emergency as a last resort along with the other equipment I have"

"Athletically I think I prefer the shiny cast version, but appreciate the extra reliability of the machined hook/threader"

"Lifting with the device required no effort because the electric windlass winder was used to reel in the main halyard"

"Being a one-off device it has been sized to fit the features and storage of my boat"

"Training using the system should be quite easy and fun, it seemed very comfortable to be lifted in"

"Definitely very visible, you would struggle to lose it, it would probably alert other boats quite quickly if it were used on open water"

"Obviously there some room for improvement in a couple of areas, but It's only a prototype and it was a difficult problem to solve"

"The threader mechanism could do with being slightly larger so it can be seen at the end of the pole"

Social, Moral & Ethical Impact Of The Prototype

- The low cost of the device compared to similar products makes it affordable for recreational sailors giving most a new capability to rescue incapacitated MOB, making them less reliant on external professional help which often takes too long to arrive.
- Morally reducing the risk of worsening the situation, causing more harm by putting the rescuer at risk was an important consideration. The prototype is designed so that lifting can be done without leaning over the edge of the boat, once the MOB has been guided into the sling with the hook.
- Safety standards such as ISO 15085 and SOLAS Chapter III were considered and adhered to where possible, to create a reliable system with approved materials while following advice on the posture/orientation of an MOB during lifting to reduce the effects of hypothermia.
- Ethical materials resistant to degradation were chosen to maximise product lifespan and lower microplastic pollution released.
- Use of mechanical advantage in lifting removes some of the need for physical strength to rescue someone from the water, potentially allowing older sailors like my client to feel more comfortable continuing their hobby.
- Inclusive design by oversizing length, width and weight limits will allow nearly anyone to fit into the sling easily.
- Focusing on making the device simple to use and producing an easy-to-read pictographic instruction card that is stored with the device, makes learning to use the device possible for people with poor eyesight.

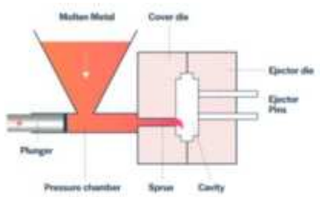


Level Of Production - Mass Production Feasibility

Having made a bespoke prototype, the following comments on its suitability for mass production, with changes needed to scale up production.

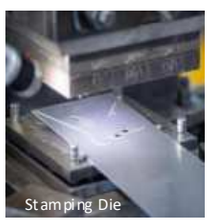
Sandcasting → Pressure Die Casting

Pressure Die casting could be used to mass manufacture the cast version of the Hook/Threader. In PDC a precision machined metal mould can be reused thousands of times unlike the single use sand mould I made. The process can also be entirely automated with parts being produced and ejected autonomously. The Hook/Threader could also be made with thinner as metal is forced into the mould cavity, not just flowing in under gravity. A better surface finish without porosity could be achieved.



CNC Machining → Metal Stamping or CNC Plasma Cutting

Even though CNC machining is extremely precise it is excessive for the 2D metal shapes used in the CNC Hook/Threader. These shapes could be stamped instantly using a hydraulic punch to cut out the required shape. If some design flexibility is needed Plasma Cutting could cut out aluminium shapes with less setup time than a CNC mill. Plasma cutting uses a jet of electrically charged gas to cut conductive materials.


Stamping Die


Plasma Cutter

3D Printing → Injection Moulding

Parts such as the ABS side spacers and TPU spacer between Latch 1 & 2 could be injection moulded. This would allow them to be produced in a few seconds instead of taking 5 minutes and 2 hrs 15 minutes respectively to 3D print. Injection moulding would create a smooth polished surface finish without layer lines. This however would have a very high setup cost to produce the moulds.


Injection Moulded Part

Life Cycle Assessment

Sustainability of the device was partially assessed in the Evaluation Against The Specification, this section details the environmental impact, lifespan and the sustainability of materials chosen.

Sourcing Materials and Environmental Impact

Materials would be sourced locally where possible to reduce carbon footprint in transport. Although the robust materials necessary for marine equipment limits scope for recycling for some parts (as they necessarily need to be capable of withstanding constant exposure to UV, salt water and temperature extremes), the long lifespan expected offsets this negative.

Impact of Metal Production

Metals Used: Aluminum, Stainless Steel

Source: Metals are refined from impure ores that are mined. This is an energy intensive process that often causes chemical pollution. Aluminum is infinitely recyclable so waste such as cans can be melted down and reused without being affecting the properties of the metal produced.

Impact of Polymer Production

Polymers Used 1: ABS, PETG, EPDM, TPU, Delrin

Source: Polymers above derived from crude oil a non-renewable finite resource. Or produced by the reprocessing of plastic waste.

Polymers Used 2: Polyester (Dacron)

Source: Can be produced as a Biopolymer from corn starch. Can be sourced by recycling polyester clothing or PET bottles.

Lifespan

The product has had durable components chosen, necessary to survive harsh marine environment. Plastics are UV resistant, with non-corroding metals chosen. I anticipate that this product, when stored carefully and maintained, to last the life of the boat.

Disposal/Recycling

Design, and method of manufacture has carefully considered how recyclable components can be separated and individually recycled. Where components are not recyclable (for instance where longevity in harsh marine environment necessitates this), these components have been kept to a minimum to fulfil brief. Instructions on dismantling, recycling, and disposal would be included in the packaging, as well as clear labelling on the device (once packaging has been disposed of)

Images of Final Product

CNC Hook/Threader Images







Cast Hook/Threader Images





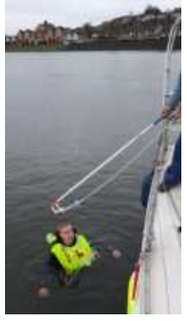
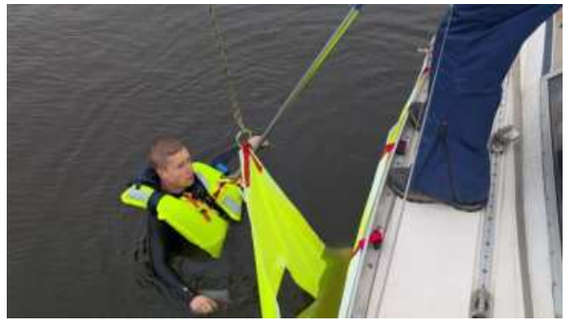





Closeup, Use and Storage Images






Rescue Sling Images







